EXPLICIT HAMILTONIAN BALL PACKINGS IN EVERY DIMENSION

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Abstract. Biran's packing stability theorem gives nearly full packings of the four-ball by any $k \geq 9$ equal balls. We construct the motion. For every $k \geq 10$ and $\varepsilon \in (0,1)$, one construction defines a smooth compactly supported Hamiltonian $H_{k,\varepsilon}$ on $\mathbb{R} \times \mathbb{R}^4$ whose time-one map carries the closed unit balls centred at $(3i-3,0,0,0)$ into the ball of radius $R = (k/(1-\varepsilon/4))^{1/4}$, filling the fraction $1-\varepsilon/4$. This gives a mathematical solution to the construction problem posed by Epoch AI. The two main ingredients are a linear comparison function for which the increasing rearrangement of the profile defect is exactly linear, in place of a contraction fixed point, and a return from the Fermat cubic disc-bundle model to the ball, in which a cube-root symmetry proves avoidance of the line at infinity. The construction develops the twisted disc-bundle mechanism of a 2026 preprint of OpenAI and proves four-ball packing stability without J-holomorphic curves or Seiberg–Witten theory. In dimension $2n \geq 6$ it realises every packing allowed by the volume and two-ball inequalities, and in dimension four every packing with $\sum_i c_i^2 < C^2$ and $3c_i < C$. We also prove robustness under perturbation with an explicit margin and give a terminating algorithm whose output is a rational polynomial times a fixed cutoff. The main theorems are verified in Lean 4.

1. Introduction

1.1. **The problem.** Points of $\mathbb{R}^{2n} = \mathbb{C}^n$ are $z = (z_1, \dots, z_n)$, $z_j = x_j + iy_j$, the symplectic form is $\omega_0 = \sum_j dx_j \wedge dy_j$, and the Hamiltonian vector field of a function $H(t,z)$ is defined by $\iota_{X_H}\omega_0 = dH$, i.e. $X_H = (\partial_{y_1}H, \dots, \partial_{y_n}H, -\partial_{x_1}H, \dots, -\partial_{x_n}H)$ in the coordinates (x,y). An integral curve of X_H is a differentiable γ with $\gamma'(t) = X_H(t,\gamma(t))$, and ϕ_H^t denotes the flow of X_H from time 0 to time t. We write $B(c)$ for the closed ball of capacity c, that is $\{\pi|z|^2 \leq c\}$, $B^\circ(c)$ for its interior, and $B(s,c) = s + B(c)$ for $s \in \mathbb{R}^{2n}$.

A *symplectic packing* of a ball by finitely many balls is a family of symplectic embeddings of the balls into the ball with pairwise disjoint images. Since $\operatorname{vol} B(c) = c^n/n!$, balls of capacities $c_1, \dots, c_k$ can be packed into $B^\circ(C)$ only if $\sum_i c_i^n < C^n$, and Gromov's two-ball theorem [Gro85] gives the second obstruction $c_i + c_j < C$ for $i \neq j$. In dimension four there are further obstructions: five balls of capacity $\frac{11}{25}$ satisfy both inequalities, but the fifth ECH capacity of their union is at least $\frac{11}{5}$, and that of $B^\circ(1)$ is 2 [Hut11].

In dimension four the existence question is settled. Biran [Bir97] showed that for every $k \geq 9$ and every c with $kc^2 < C^2$, k balls of capacity c embed in $B^\circ(C)$, and the work of McDuff–Polterovich, McDuff, Li–Liu and Li–Li [MP94, McD98, LL01, LL02] reduces every four-dimensional ball packing problem to a list of inequalities indexed by exceptional classes. These proofs use J-holomorphic curves and, through Taubes' theorem and inflation on blow-ups of $\mathbb{CP}^2$, Seiberg–Witten theory. They show that embeddings exist; they do not produce a Hamiltonian that moves given balls into the target. Explicit full packings for square k, and several other constructive methods, are discussed in §1.7. Here the goal is one Hamiltonian construction for every $k \geq 10$ and every positive error.

Epoch AI [Epo26] asked for such a motion: for every integer $k \geq 10$ and every $\varepsilon \in (0,1)$, an algorithm that, on input (k,ε), outputs an explicit smooth time-dependent Hamiltonian $H(t,z)$

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on $\mathbb{R}^4$ whose time-one map sends the closed unit balls centred at $(3i-3,0,0,0)$, $1 \le i \le k$, into a closed ball $B^4(R)$ of Euclidean radius R centred at the origin, with $k/R^4 > 1-\varepsilon$. Since the flow preserves volume, k/R^4 is the fraction of $B^4(R)$ filled by the images. The construction below supplies one Hamiltonian for each (k,ε), defined by the same formulas for all of them; Theorem 7.4 supplies a uniform algorithm with finite output. Remark 7.7 relates these results to Epoch's program submission format.

1.2. **The four-dimensional theorem.** For admissible data (Definition 1.2) we define in §6 one Hamiltonian $H[n,k,C,c,s]$ (Definition 6.16), by a construction that is the same for all data. The first theorem is its four-dimensional instance for the data of the Epoch question.

Theorem A (Equal balls in the four-ball)**.** *Let $k \ge 10$ be an integer and $\varepsilon \in (0,1)$. Put*

$$R = \left(\frac{k}{1-\varepsilon/4}\right)^{1/4}, \qquad C = \pi R^2, \qquad s_i = (3i-3,0,0,0) \in \mathbb{R}^4 \quad (1 \le i \le k),$$

and $H_{k,\varepsilon} = H[2,k,C,(\pi,\dots,\pi),(s_1,\dots,s_k)]$. Then $H_{k,\varepsilon}$ is C^∞ on $\mathbb{R}\times\mathbb{R}^4$, vanishes for $t \notin (0,1)$ and outside $\mathbb{R}\times\mathcal{S}$ for an explicit compact set $\mathcal{S}$, and satisfies $\mathrm{Pack}(2,k,C,(\pi,\dots,\pi),s)$ *(Definition 1.1): its flow is global, and its time-one map sends each closed unit ball $B(s_i,\pi)$ into $B^\circ(\pi R^2)$, the open Euclidean ball of radius R about the origin, with pairwise disjoint images. Moreover $k/R^4 = 1-\varepsilon/4 > 1-\varepsilon$.*
[Lean: full statement verified; `four_dim_theorem, main_time_support, main_support_radius, flow_images_disjoint`]

Proof. The centres are at distance at least 3, so the closed unit balls are disjoint. Since $R^4 = k/(1-\varepsilon/4) > k \ge 10$, we have $k\pi^2 < \pi^2R^4 = C^2$ and $3\pi < \pi R^2 = C$, so $(2,k,C,(\pi,\dots,\pi))$ is admissible and Theorem C(i) below applies. For these data the normalised capacity $\rho = \pi/C = R^{-2}$ satisfies $\rho\sqrt{k} = \sqrt{1-\varepsilon/4} > \sqrt{3}/2 > \frac{3}{4}$, which selects the linear comparison datum of Proposition 3.8 in Construction S20.1. □

The theorem leaves an explicit margin. The images lie in the closed ball of radius R, while the density requirement $k/R'^4 > 1-\varepsilon$ is already met by $R' = (k/(1-\varepsilon/2))^{1/4} > R$. The following estimate converts this margin, a closed formula in (k,ε), into a tolerance on the vector field.

Proposition B (Robustness)**.** *Let $k \ge 10$, $\varepsilon \in (0,1)$, and let $R < R'$ be as above. Let $\widehat{H}$ be a smooth function on $\mathbb{R}\times\mathbb{R}^4$ vanishing outside $\mathbb{R}\times K$ for a compact set K. Suppose that, for $t \in [0,1]$, the field $X_{\widehat{H}}(t,\cdot)$ is L-Lipschitz for some $L \ge 0$ and $|X_{\widehat{H}} - X_{H_{k,\varepsilon}}| \le \eta$ on $\mathbb{R}^4$, where $\eta e^L \le R' - R$. Then the time-one map of $\widehat{H}$ sends each closed unit ball $B(s_i,\pi)$ into the closed ball of radius R' about the origin, and $k/R'^4 = 1-\varepsilon/2 > 1-\varepsilon$.*
[Lean: full statement verified; `four_dim_stability`]

Proof. Let γ and $\hat\gamma$ be integral curves of $X_{H_{k,\varepsilon}}$ and $X_{\widehat{H}}$ starting at the same point of $B(s_i,\pi)$. Then $|\hat\gamma(1)-\gamma(1)| \le \eta(e^L-1)/L \le \eta e^L$ by Grönwall's inequality (Lemma 7.1, with the Lipschitz bound on $X_{\widehat{H}}$), and $|\gamma(1)| \le R$ by Theorem A. For $L = 0$ the Grönwall bound is η. □

Thus every smooth compactly supported approximation with vector-field error at most $(R'-R)e^{-L}$ still meets the Epoch requirement, with images in the ball of radius R'; the geometric margin $R'-R$ depends only on (k,ε). Theorem 7.4 gives one algorithm which, on input k and a name of ε, halts with $B \in \mathbb{N}$ and $P \in \mathbb{Q}[t,z]$ such that $\beta_B \cdot P$, a rational polynomial times a fixed smooth cutoff, satisfies the same packing specification. Each output is certified by a finite rational test, and the construction shows that the search terminates.

The packings of Theorem A exist by Biran's theorem. The proof here works with explicit maps, forms and flows only and uses none of the existence results quoted above. Through

Corollary E it gives a proof of Biran's packing stability for B^4 without J-holomorphic curves or Seiberg–Witten theory, which the earlier proofs [Bir97, Bir99, BH13, BHO16] use, directly or through the results they cite.

1.3. **The mechanism.** The construction follows a mechanism due to the OpenAI preprint [Ope26]. We describe it in dimension four, where the picture is classical. Give $\mathbb{CP}^2$ the Fubini–Study form for which a line has area 1, so that $\operatorname{vol}\mathbb{CP}^2 = \frac{1}{2}$ and the complement of a line is $B^\circ(1)$. Let $E \subset \mathbb{CP}^2$ be a smooth cubic, of area 3. Biran [Bir01] showed that the complement of an isotropic skeleton in $\mathbb{CP}^2$ is symplectomorphic to a disc bundle over E: the normal bundle, of degree $E \cdot E = 9$, with fibre discs of capacity $\frac{1}{3}$. With fibre moment $p = \pi|w|^2$, the base area of the circle bundle at level p is

$$A(p) = 3 - 9p = 3(1 - 3p) \qquad (0 \le p < \tfrac{1}{3}), \qquad \int_0^{1/3} A(p)\,dp = \frac{1}{2} = \operatorname{vol}\mathbb{CP}^2.$$

The cubic is the first smooth plane curve with the topology needed by this handle construction. Its closed base must carry a closed one-form β with nonzero period around the handle. A line or conic has genus zero, so $H^1(S^2;\mathbb{R}) = 0$ rules out such a form; a cubic has genus one. Thus the fibre bound $\frac{1}{3}$, and hence $3c_i < C$, reflects the choice of the smallest-degree smooth plane curve supporting this mechanism.

A ball $B(a) \subset \mathbb{C} \times \mathbb{C}$, sliced by the moment p of its second coordinate, is a family of discs of capacity $a - p$ for $p \le a$; its slice profile is $(a-p)_+$, and $\int_0^\infty (a-p)_+\,dp = a^2/2 = \operatorname{vol} B(a)$. Placing balls fibrewise in the bundle requires $a < \frac{1}{3}$: this is the source of the condition $3c_i < C$. If $\sum_i (a_i - p)_+ \le A(p)$ for every p, the base discs can be placed side by side over E. For nine balls of capacity $\frac{1}{3}$ this holds with equality. For $k \ge 10$ balls of capacity close to $1/\sqrt{k}$ the integrated inequality $\sum_i a_i^2/2 < \frac{1}{2}$ holds, but at $p = 0$ one needs $\sum_i a_i \le 3$, which fails: the difference $A(p) - \sum_i (a_i - p)_+$ has positive integral and is negative near $p = 0$ (Figure 1).

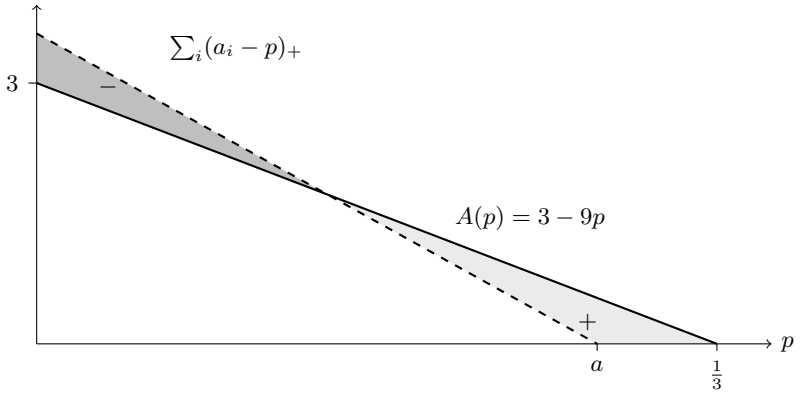


FIGURE 1. The area profile A of the cubic model (solid) and the slice profile of 13 balls of capacity $a = 0.99/\sqrt{13}$ (dashed). The deficit (dark) is smaller than the surplus (light), since $\int_0^h (A - \sum_i (a_i - p)_+)\,dp = \frac{1}{2}(1 - 13a^2 - (1-3h)^2) > 0$ for $h = 0.33$, but stacking needs $A \ge \sum_i (a_i - p)_+$ pointwise. The twist moves area from the surplus to the deficit.

The construction of [Ope26] pays the deficit with the surplus by twisting the bundle. Across an annulus in the base, the fibre over one boundary circle is identified with the fibre over the other by an area-preserving map q of the fibre disc instead of the identity. Paired with a function K of the moment, the twist changes the base area available above a fibre point w from

$A(p(w))$ to $A(p(w)) + K(p(q^{-1}w)) - K(p(w))$ (equation (3)). Since q preserves the fibre area, this redistributes area between moment levels without changing the total. The balls fit if, for some $\gamma > 0$ and $P = A - \sum_i (a_i - p)_+$,

$$(K - P)\big(p(q(w))\big) \le K\big(p(w)\big) - \gamma \qquad \text{for every fibre point } w.$$

This forces $\int P > 0$. Conversely, for concave P with positive mean that is positive outside a core, [Ope26] obtains K as the fixed point of a contraction built from increasing rearrangements, realises the rearrangements by isotopies, and turns the inequality into embeddings by Moser's method and a transfer to the target. For $k \ge 10$ equal balls with $\frac{3}{4} < \rho\sqrt{k} < 1$ a linear K suffices: $K - P$ is, up to a constant, a tent whose increasing rearrangement is linear, and the inequality becomes an identity up to the error of one planar map (Proposition 3.8).

1.4. **The main steps.** The four-dimensional contribution consists of four constructions, listed in Table 1: the linear comparison identity, which removes the fixed-point step for the Epoch family; an explicit planar rearrangement; a symmetry-controlled return from the Fermat cubic model, written in explicit cover coordinates, to the affine ball; and one global motion from the prescribed source balls. We first give the construction chain for Theorem A.

The construction chain. After the dilation by $1/\sqrt{C}$ the target is $B^\circ(1)$ and each ball has capacity ρ. Put $\eta_\star = 1 - \sqrt{1 - \varepsilon/4}$, $t_0 = 1 - \eta_\star/2$, $a = t_0/\sqrt{k}$, $h = t_0/3$ and $g = 3\eta_\star/2$. Then $H_{k,\varepsilon}$ is obtained in five steps, each determined by the data and by the previous steps.

(1) *Model.* The cubic disc-bundle model associated with Biran's decomposition, written on the cover $\mathbb{R}^2 \times \mathbb{C}$ of the Fermat torus with the explicit form Ω_0 of §5.1, cut at fibre moment $p \le h$.

(2) *Comparison.* The linear function $K(p) = (3\sqrt{k} - 9)p$ and the fibre isotopy (q_t) of Theorem S5.34, generated by an explicit planar Hamiltonian; its endpoint carries the centred disc of area I into the sublevel set $\{\Lambda \circ p \le I + 4e_q\}$ of the tent Λ of §3.4, with e_q as there. The comparison gap is $\frac{31}{32}g$ (Proposition 3.8).

(3) *Handle.* Over an annulus $\mathcal{A}$ in the base, the bundle chart $\chi_{\mathcal{A}}$ and the twist $\mathcal{D}$, which applies $q_{f(s)}$ to the fibre over the handle coordinate s for a fixed clock f; the Moser map Φ, the time-one map of three explicit fields; the layer maps E_i, given by an explicit strip map and unique roots (§4, Remark 4.7).

(4) *Return.* The map $\mathcal{T} = b_{\mathrm{ball}} \circ \psi_1^{-1} \circ \mathrm{pr} \circ \tau_1 \circ \iota \circ \mathsf{N}$ from the model into $B^\circ(1)$: it carries the model into $\mathbb{CP}^2$ minus the line $\{Z_0 = 0\}$, which the ball chart b_{ball} identifies with $B^\circ(1)$; τ_1 and ψ_1 are time-one maps of explicit ordinary differential equations (§S13). The embeddings are

$$f_i(u) = \sqrt{C}\,\big(\mathcal{T} \circ \Phi^{-1} \circ \chi_{\mathcal{A}} \circ \mathcal{D}^{-1} \circ E_i\big)(u/\sqrt{C})$$

(Remark S13.25).

(5) *Motion.* One isotopy $F_{i,t}$ from the translations $u \mapsto s_i + u$ to the f_i, with disjoint images at all times (Proposition 6.6), and $H_{k,\varepsilon}$, the pushforward along it of a radial primitive computed in the source balls (Definition 6.10).

The parameters t_0, a, h, g, the slope of K and the height $d_n = \frac{1}{3} - \frac{\eta_\star}{12}$ used by the return are closed-form expressions in $\sqrt{k}$ and ε (Remark S13.25); the maps are formulas, integrals, time-one maps of explicit vector fields, inverses of explicit injective maps and extrema over explicit compact sets.

Table 1 lists the four constructions with the statements that resolve them. The handle step (3) follows [Ope26] with explicit fields once the comparison datum is given.

The linear comparison identity and the symmetry argument for the return are the two simplifications particular to the four-dimensional construction. Together with the planar rearrangement

Step	What is needed, and the difficulty	Core statement
comparison	a finite datum (K, q) with a quantitative gap in place of the fixed point of a contraction; for equal balls in dimension four, a linear K for which the comparison is an identity up to the planar error	Prop. 3.8; Thm. 3.5
planar rearrangement	an area-preserving disc map carrying round discs into prescribed sublevel sets: a convex potential whose sublevel areas follow the increasing rearrangement, made quadratic near its minimum; a Morse chart with rotation and area normalisation; a Hamiltonian path on the whole disc, the identity near the boundary, ending at this map on a core	Thm. 3.4; Thms. S4.7, S4.21, S5.34
return to the ball	explicit coordinates on the cover of the Fermat cubic and explicit transport and return fields carrying the model form to the Fubini–Study form; proof that the images avoid the line at infinity $\{Z_0 = 0\}$, and are injective modulo deck maps, with a compact margin inside $B^\circ(1)$	Thm. 5.1; Thm. S13.21, Prop. S13.19
from the given balls	one isotopy from the prescribed balls to the model embeddings keeping the closed balls disjoint at all times (a route ordered by ranks, a path in $\mathrm{Sp}(2n)$, a shape stage, a common dilation), and a global Hamiltonian generating it on the balls	Props. 6.6, 6.15; Prop. S21.14

TABLE 1. The main steps of the construction and the statements that resolve them.

and the common isotopy, they turn the disc-bundle mechanism into an explicit motion of the given balls. The parameters are fixed uniformly in the data, and the margins pass quantitatively from step to step: the comparison gap $\frac{31}{32}g$ pays for the planar error and for the excess of the handle caps over the ball profiles, the return leaves a compact margin inside $B^\circ(1)$, and the isotopy uses explicit radii between the closed balls and the domains of the embeddings. Each map is defined on the whole domain on which the next one uses it: the planar isotopy on the whole disc, the Moser flows on the whole model, the Hamiltonian on $\mathbb{R} \times \mathbb{R}^{2n}$.

1.5. **All dimensions and unequal balls.** In dimension $2n \geq 6$ the volume and two-ball inequalities are the only obstructions to packing a ball by balls: this is Conjecture A of Siegel and Yao [SY25], proved in a 2026 preprint of OpenAI [Ope26]. The construction of Theorem A extends to these problems and to unequal balls in dimension four.

Definition 1.1 (Packing specification)**.** Let $n \geq 1$, $k \geq 0$, $C > 0$, $c = (c_1, \dots, c_k) \in (0, \infty)^k$ and $s = (s_1, \dots, s_k) \in (\mathbb{R}^{2n})^k$. A function $H \colon \mathbb{R} \times \mathbb{R}^{2n} \to \mathbb{R}$ satisfies $\mathrm{Pack}(n, k, C, c, s)$ if

(a) H is C^∞, and there is a compact set $\mathcal{S} \subset \mathbb{R}^{2n}$ with $H(t, z) = 0$ for all $t \in \mathbb{R}$ and all $z \notin \mathcal{S}$;
(b) for every i and every $z \in B(s_i, c_i)$ there is an integral curve $\gamma \colon \mathbb{R} \to \mathbb{R}^{2n}$ of X_H with $\gamma(0) = z$, and every integral curve $\gamma \colon [0, 1] \to \mathbb{R}^{2n}$ of X_H with $\gamma(0) = z$ satisfies $\gamma(1) \in B^\circ(C)$.

[Lean: full statement verified; `Problem.PackingSpec`]

For H satisfying (a) the flow ϕ_H^t is global and consists of symplectic diffeomorphisms, and (b) says $\phi_H^1(B(s_i, c_i)) \subset B^\circ(C)$ for every i.

Definition 1.2 (Admissible data)**.** Let $n \geq 2$, $k \geq 1$, $C > 0$ and $c \in (0,\infty)^k$. The data (n,k,C,c) are *admissible* if

$$\begin{aligned} n \geq 3: &\qquad \textstyle\sum_i c_i^n < C^n \quad \text{and} \quad c_i + c_j < C \ \text{ for all } i \neq j; \\ n = 2: &\qquad \textstyle\sum_i c_i^2 < C^2 \quad \text{and} \quad 3c_i < C \ \text{ for all } i. \end{aligned}$$

[Lean: full statement verified; `Problem.Admissible`]

For $n \geq 3$ admissibility is exactly the pair of necessary conditions above. For $n = 2$ the condition $3c_i < C$ is stronger than the two-ball inequality; admissible data include $k \geq 9$ equal balls with $kc^2 < C^2$ and the unequal capacity vectors of Corollary G. This is a sufficient small-capacity range, for any $k \geq 1$, rather than the full four-dimensional packing criterion.

Theorem C. *Let $n \geq 2$, $k \geq 1$, $C > 0$, $c_1, \dots, c_k > 0$ and $s_1, \dots, s_k \in \mathbb{R}^{2n}$ be such that the closed balls $B(s_i, c_i)$ are pairwise disjoint and (n,k,C,c) is admissible. Then:*

- *(i) the function $H = H[n,k,C,c,s]$ of Definition 6.16 is C^∞ on $\mathbb{R} \times \mathbb{R}^{2n}$, vanishes outside $\mathbb{R} \times \mathcal{S}$ for an explicit compact set $\mathcal{S}$ and for $t \notin (0,1)$, and satisfies* $\mathrm{Pack}(n,k,C,c,s)$*; in particular ϕ_H^1 maps each $B(s_i,c_i)$ into $B^\circ(C)$, and the images are pairwise disjoint;*
  [Lean: full statement verified; `main_theorem, main_time_support, main_support_radius, flow_images_disjoint`]
- *(ii)* (L1) *the definition of $H[n,k,C,c,s]$ is the same for all such data and uses only the operations of level L1 in Definition 1.3;*
- *(iii)* (L2, L3) *there is one oracle algorithm which, given n, k and names of the real inputs $C, c_1, \dots, c_k, s_1, \dots, s_k$ (a rational number is named by its constant sequence), halts on all such data and outputs an integer B and a polynomial $P \in \mathbb{Q}[t,z]$ such that $H_{\mathrm{fin}} = \beta_B \cdot P$ satisfies* $\mathrm{Pack}(n,k,C,c,s)$*. The output may depend on the names; for computable inputs, given by computable names, it is an ordinary algorithm.*
  [Lean: full statement verified; `original_name_algorithm`]

Parts (i) and (ii) are proved in §6.4, part (iii) is Theorem 7.4. Part (ii) is a statement about the definition: Table 5 lists, for each ingredient of $H[n,k,C,c,s]$, the operation of Definition 1.3 that produces it; none of them is a search over Hamiltonians, maps or packings.

For $2n \geq 6$ the fibre of the model is $\mathbb{C}^{n-1}$ and the base a compact piece of an affine curve, the diagonal curve $y_j^2 = t^2 - j$ for capacities below $\frac{1}{2}$; the two-ball inequality allows at most one ball of capacity at least $\frac{1}{2}$, which is reserved while the others are packed in a model of its complement (Table 2). The comparison datum is the finite rational iterate of Theorem 3.5. In dimension four, inputs other than those of Theorem A, including $k = 9$ and unequal capacities, use the scalar comparator of §S6; the handle construction for $n = 2$ is summarised in Remark 4.7 and carried out in §S10.9.

The packings of Theorem C also exist by known results. For $n \geq 3$ this is [Ope26]. For $n = 2$, by [MP94, McD98, LL01, LL02], closed balls of capacities c_i embed in $B^\circ(C)$ if $\sum_i c_i^2 < C^2$ and $\sum_i m_i c_i < dC$ for every exceptional class $(d; m_1, \dots, m_k)$ with $d \geq 1$; such a class has $m_i \geq 0$ and $\sum_i m_i = 3d - 1$, so $3c_i < C$ gives $\sum_i m_i c_i < (3d-1)C/3 < dC$. The threshold $k \geq 9$ of Corollary E is Biran's [Bir97].

Corollary D (The Siegel–Yao criterion by explicit flows)**.** *Let $n \geq 3$, $k \geq 1$, $C > 0$ and $c_i > 0$, and let $B(s_i,c_i) \subset \mathbb{R}^{2n}$, $1 \leq i \leq k$, be pairwise disjoint closed balls. There is a symplectic embedding of a neighbourhood of $\bigsqcup_i B(s_i,c_i)$ into $B^\circ(C)$ if and only if $\sum_i c_i^n < C^n$ and $c_i + c_j < C$ for all $i \neq j$. In that case ϕ_H^1 with $H = H[n,k,C,c,s]$ is such an embedding.*
[Lean: full statement verified; `neighbourhood_criterion, flow_neighbourhood_embedding`]

Necessity follows from volume and from Gromov's two-ball theorem, whose proof uses J-holomorphic lines. Sufficiency is the main theorem of [Ope26]; Corollary D reproves it by

explicit flows, without J-holomorphic curves, and realises every such packing by the time-one map of an explicit Hamiltonian.

Corollary E (Equal balls)**.** *Let $n \geq 2$, $\varepsilon \in (0,1)$, and $k \geq 2^n$ if $n \geq 3$, $k \geq 9$ if $n = 2$. Let $c > 0$, let $s_1, \dots, s_k$ be centres with $B(s_i, c)$ pairwise disjoint, and let $C > 0$ with $kc^n < C^n < kc^n/(1-\varepsilon)$. Then $H[n, k, C, (c, \dots, c), s]$ satisfies* $\mathrm{Pack}(n, k, C, (c, \dots, c), s)$*, and $kc^n/C^n > 1 - \varepsilon$.*
[Lean: full statement verified; `equal_balls`]

Since ϕ_H^1 preserves volume and the images are disjoint, they fill the fraction kc^n/C^n of the volume of $B(C)$. For $n \geq 3$ and $k \geq 2$ the bound $k \geq 2^n$ is forced by the two-ball inequality, since $2 \leq k < 2^n$ equal balls fill at most $k/2^n$ of a ball. For $n = 2$ the bound $k \geq 9$ comes from the condition $3c < C$; for $k \leq 8$ full packings do not exist except for $k = 1, 4$ [MP94].

Corollary F (The Epoch data)**.** *Let $k \geq 9$ be an integer, $\varepsilon \in (0,1)$, $s_i = (3i - 3, 0, 0, 0) \in \mathbb{R}^4$, $R = (k/(1 - \varepsilon/2))^{1/4}$ and $C = \pi R^2$. Then $H = H[2, k, C, (\pi, \dots, \pi), s]$ satisfies* $\mathrm{Pack}(2, k, C, (\pi, \dots, \pi), s)$*: its time-one map sends each closed unit ball $B(s_i, \pi)$ into $B^\circ(\pi R^2)$, the open Euclidean ball of radius R about the origin, and $k/R^4 = 1 - \varepsilon/2 > 1 - \varepsilon$.*
[Lean: full statement verified; `epoch_packing, epoch_theorem_ten`]

Theorem A is the case $k \geq 10$ of Corollary F at $\varepsilon/2$. At density $1 - \varepsilon/2$ the linear comparison datum applies for $k \geq 10$ and $\varepsilon < \frac{7}{8}$; the remaining cases use the scalar comparator. The corollary is the form used by the Epoch machine of Theorem 7.4: on input k and a name of ε it outputs (B, P) such that $\beta_B \cdot P$ satisfies the same specification.

Corollary G (A cone of unequal balls in dimension four)**.** *Let $k \geq 1$ and $c_1, \dots, c_k > 0$ with $\max_i c_i \leq \frac{1}{3}\big(\sum_i c_i^2\big)^{1/2}$, let $\varepsilon \in (0,1)$ and $C_\varepsilon = \big(\sum_i c_i^2/(1 - \varepsilon/2)\big)^{1/2}$. For all centres s_i with $B(s_i, c_i)$ pairwise disjoint, $H[2, k, C_\varepsilon, c, s]$ satisfies* $\mathrm{Pack}(2, k, C_\varepsilon, c, s)$*, and $\sum_i c_i^2/C_\varepsilon^2 = 1 - \varepsilon/2$; the images fill the fraction $1 - \varepsilon/2$ of the volume of $B(C_\varepsilon)$.*
[Lean: full statement verified; `unequal_cone`]

Proofs of Corollaries D–G. The sufficiency statements follow from Theorem C after checking admissibility; the necessity in Corollary D uses the classical obstructions. See §6.4. □

1.6. **Explicitness and the finite output.** The Hamiltonians of Theorems A and C are explicit in the following graded sense.

Definition 1.3 (Levels of explicitness)**.** (L1) An *explicit definitional construction* is a definition built from rational constants and the input by arithmetic and elementary functions, integrals over compact domains, limits of series with an explicit rate of convergence, maxima and minima of continuous functions over explicit compact sets, unique solutions of explicit ordinary differential equations, inverses of explicit injective maps, unique roots of explicit equations, case distinctions on propositions about the input and about the arguments of the function being defined, and least elements of searches over a fixed enumeration of finitely described objects (integers, rationals, and tuples of these and of rational polynomials) whose validity is decided by finitely many explicit inequalities; definite descriptions are used only after unique existence has been proved (Definition 1.4).

(L2) The *finite form* is $H_{\mathrm{fin}} = \beta_B \cdot P$ with $B \in \mathbb{N}$ and $P \in \mathbb{Q}[t, z]$, where $\beta_B(z) = \prod_{l=1}^{2n} \beta(\zeta_l/B)$ for the real coordinates $\zeta_1, \dots, \zeta_{2n}$ of z, and $\beta(x) = 1 - S\big((x^2 - \frac{1}{4})/\frac{5}{16}\big)$ with the flat step S of Definition S2.3; thus $\beta\colon \mathbb{R} \to [0,1]$ is smooth, equal to 1 on $[-\frac{1}{2}, \frac{1}{2}]$ and to 0 outside $(-\frac{3}{4}, \frac{3}{4})$.

(L3) *Computability* means that one oracle algorithm, given names of the input (a rational input is named by its constant sequence), halts on every admissible input and outputs some (B, P) as in (L2) for which H_{fin} satisfies the packing specification. The output

may depend on the names and not only on the numbers they name; this is necessary, as explained below.

Definition 1.4 (Definite description)**.** For a property $\mathcal{P}$ of elements of a set X and a proof h that exactly one $x \in X$ satisfies $\mathcal{P}(x)$, the(h) denotes that element. Phrases such as "the unique u with $F(u) = y$" denote this operation.

Level L1 describes the operations defining the geometric Hamiltonian; a metaprogram checks their formal counterparts (§8). Theorems A and C(i),(ii) specify this Hamiltonian through the model, comparison, return and isotopy maps, and prove directly that it packs. "Explicit" refers to the operations defining the maps; computation from names of the real input is the subject of level L3.

Level L3 is of a different kind. The algorithm of §7 accepts the first rational polynomial whose Euler trajectories from a finite grid, with an error bound valid on $[0, 1]$, stay where the cutoff equals one and end inside the target; every accepted output is correct by a soundness theorem. Termination is proved with $H[n, k, C, c, s]$, but any packing Hamiltonian with a positive margin would do. The statements marked [Lean: …], including Theorems A, C and 7.4, are verified in Lean 4 (§8).

The dependence on names in L3 is necessary for finite output. A computable single-valued map to the discrete set of polynomial codes is continuous, hence constant on $\varepsilon \in (0, 1)$ for fixed k. But the images of $k \geq 2$ disjoint closed balls under any fixed Hamiltonian diffeomorphism form a disconnected compact set. In their smallest enclosing ball the complement contains a nonempty open set, so their density is strictly below one. That fixed output therefore fails the requirement for sufficiently small ε.

1.7. **Relation to previous work.** McDuff and Polterovich [MP94] related packings of B^4 by k balls to the blow-up of $\mathbb{CP}^2$ at k points and determined the maximal packing densities of B^4 by at most nine equal balls; with [McD91] this reduces four-dimensional ball packing to the existence of symplectic forms on blow-ups. Biran [Bir99] extended packing stability to closed symplectic four-manifolds with rational cohomology class. Buse and Hind [BH11, BH13] proved packing stability for balls and for closed symplectic manifolds with rational class in higher dimensions, and Buse, Hind and Opshtein [BHO16] proved it for all closed symplectic four-manifolds, removing the rationality assumption. Edtmair [Edt25] proved packing stability for every compact connected symplectic four-manifold with smooth boundary, by filling its volume with a single ellipsoid, and related packing stability in the presence of boundary to the algebraic structure of Hamiltonian diffeomorphism groups; Cristofaro-Gardiner and Hind [CGH25] showed that it fails for some bounded open domains in $\mathbb{R}^4$. These are existence theorems for embeddings of a manifold; the present paper fixes the domain (the four-ball), the number k and the initial positions, and controls one specific Hamiltonian, its support, its margin and its dependence on (k, ε). Constructive methods in dimension four include almost toric fibrations [Sym03] and the toric mutations with which Casals and Vianna [CV22] produced volume-filling ellipsoid embeddings along the known staircases. Their input is an almost-toric base and a mutation sequence; our input is (k, ε) and the prescribed source balls, and our output is one Hamiltonian generating their motion. Siegel and Yao [SY25] stated Conjecture A and proved that taking the product with a closed surface does not change the answer to a four-dimensional ball packing problem, and a stabilised two-ball theorem in every dimension.

Explicit constructions go back to Traynor [Tra95], who built packings from toric pictures and explicit area-preserving maps of the plane, including full packings of B^4 by d^2 and of B^{2n} by d^n equal balls; further explicit constructions, including symplectic folding and wrapping, and a survey of the case $k \leq 9$ are in [Sch05b, Sch05a, Sch18]. Wieck's tunnelling constructions [Wie09]

include maximal seven- and eight-ball packings of B^4. Opshtein [Ops15, Theorem 4(ii), §5.4] constructs ball packings from symplectic curves with prescribed singularities, including explicit six-, seven- and eight-ball examples. His singular-polarisation theorem [Ops13, Theorem 3] constructs full ellipsoid packings from a supplied polarisation and its component areas. These results provide constructive geometric tools; here the comparison data, model maps and common Hamiltonian are fixed uniformly for every $k \geq 10$ and $\varepsilon \in (0,1)$. The common Hamiltonian also specifies the motion from the prescribed source balls. Biran's decomposition of $\mathbb{CP}^2$ along a cubic [Bir01] supplies our four-dimensional model; Witt Nyström [WN24] embedded large neighbourhoods of zero sections of normal bundles into Kähler manifolds.

Our construction follows the mechanism of [Ope26] and its division into three cases for $2n \geq 6$. The explicit replacements of its existential steps are those of Table 1, together with the explicit models of §5 and the compilation of §7. For $2n \geq 6$, Theorem C with Corollary D gives a second proof of the sufficiency theorem of [Ope26]. It builds on that work: it uses the rearrangement contraction of [Ope26, §2], and the formal proof uses lemmas of the Lean library accompanying [Ope26], whose proofs the kernel checks along with the new construction. The packing existence theorem of that preprint is not an input.

1.8. **Conventions.** Besides the notation of §1.1, we use the following. The dilation $z \mapsto \lambda z$ multiplies ω_0 and capacities by λ^2; conjugation by it normalises the target capacity to 1. In fibre coordinates w_j the moments are $p_j = \pi|w_j|^2$ and the angles are $\theta_j = \arg w_j/2\pi \in \mathbb{R}/\mathbb{Z}$, so that the fibre form is $\omega_V = \sum_j dp_j \wedge d\theta_j$. Fibre Hamiltonians Q follow the sign of [Ope26], $\iota_{X_Q}\omega_V = -dQ$, opposite to the global one. In affine coordinates we use $\omega_{\mathrm{FS}} = dd^c \log(1+|z|^2)$ with $d^c = i(\bar\partial - \partial)/(4\pi)$; the map $z \mapsto z/\sqrt{\pi(1+|z|^2)}$ identifies this chart symplectically with $B^\circ(1)$.

1.9. **Organisation.** Section 2 states the three inputs of the mechanism (model, comparison datum, transfer) and the packing theorem in a toric model. Section 3 constructs comparison data, Section 4 proves the packing theorem over a handle, and Section 5 states the packing theorems of the four models, proved in §S8, §S14 and §S16. Section 6 joins the embeddings to the prescribed balls by a common isotopy, defines $H[n,k,C,c,s]$ and proves Theorem C(i),(ii) and the corollaries. Section 7 proves Theorem C(iii), and Section 8 describes the formal verification. The supplement contains the complete proofs of the statements that the main text proves only in outline; each such statement points to the place of its proof. Numbers with the prefix S, as in §S6, refer to the supplement.

2. The mechanism

The construction follows the handle-packing scheme of [Ope26]. In dimension four the model is the disc bundle of degree 9 over a cubic curve $E \subset \mathbb{CP}^2$ from Biran's decomposition [Bir01] (§1.3): its fibres are discs of capacity $\frac{1}{3}$, the base area at fibre moment p is $3(1-3p)$, and the total volume is $\frac{1}{2} = \operatorname{vol}\mathbb{CP}^2$. The fibre capacity gives the condition $3c_i < C$, and the total volume gives the volume budget. The higher-dimensional models replace E by a compact piece of an affine curve and the fibre disc by $\mathbb{C}^{n-1}$. In each case the construction has three inputs: a model with a positive base-area profile, a comparison on the moment polytope, and a symplectic transfer to the target. The comparison moves area between fibre levels, and the handle turns the comparison inequality into disjoint ball embeddings. Throughout, $n \geq 2$, $m = n-1$, and the target capacity is normalised to one; the general target is recovered by the dilation of Definition 6.1.

2.1. **Toric balls and area profiles.** On $\mathbb{C}^m$ let $p_j = \pi|w_j|^2$, $|p| = \sum_j p_j$ and $\omega_V = \sum_j dx_j \wedge dy_j$. Away from the coordinate axes this is $\sum_j dp_j \wedge d\theta_j$, with each angle of period one. A downward polytope is

$$\Delta = \{p \in \mathbb{R}^m_{\geq 0} : \langle b_\nu, p\rangle \leq \ell_\nu,\ 1 \leq \nu \leq N\}, \qquad b_\nu \in \mathbb{R}^m_{\geq 0}, \quad \ell_\nu > 0.$$

Write $V_\Delta = \{w : p(w) \in \Delta\}$. The equalities $\langle b_\nu, p\rangle = \ell_\nu$ define its outer faces, and $\tau_\Delta\Delta$, $0 < \tau_\Delta < 1$, is the core used to keep the source caps away from those faces.

Definition 2.1 (Toric ball)**.** In $\mathbb{C}^n = \mathbb{C} \times \mathbb{C}^m$, put $\varpi = \pi|z_{\mathrm{b}}|^2$. The ball of capacity a is $B(a) = \{\varpi + |p| \leq a\}$. At a fibre point with $|p| \leq a$, its base slice is a disc of capacity $a - |p|$; its slice height is 0 for $|p| > a$. Thus the slice-height function is $(a - |p|)_+$.

Lemma 2.2 (The cap integral)**.** *For $a \geq 0$,*

$$\int_{\mathbb{R}^m_{\geq 0}} (a - |p|)_+ \, dp = \frac{a^{m+1}}{(m+1)!}.$$

Proof. The simplex $\{|p| \leq t\}$ has measure $t^m/m!$. Integrating its distribution function against $(a - t)_+$ gives the displayed identity; equivalently, integrate one coordinate at a time. □

For each higher-dimensional model, let Σ_ℓ be its compact affine core, let $\Gamma_{\mathrm{phys}}(p)$ be the horizontal primitive given in its model section, and put $\kappa(p) = d_\Sigma\Gamma_{\mathrm{phys}}(p)$. The physical form on the model is $\Omega = \omega_V + d\Gamma_{\mathrm{phys}}$. The base profile and its moment budget are

$$H_{\mathrm{b}}(p) = \int_{\Sigma_\ell} \kappa(p), \qquad \mathcal{B}_\Delta = \int_\Delta H_{\mathrm{b}}(p)\, dp. \tag{1}$$

The first integral is the intrinsic integral of the physical base curvature; the model sections give its formula, or a uniform bound for its distance from an explicit limiting profile. The packing proof uses this scalar budget only through the comparison inequality below.

The four-dimensional periodic model has profile $H_{\mathrm{b}}(p) = 3(1 - 3p)$ on $[0, h]$, $h < 1/3$, and

$$\mathcal{B}_{[0,h]} = 3h - \frac{9}{2}h^2 \longrightarrow \frac{1}{2} \qquad (h \uparrow 1/3).$$

Its periodic coordinates and deck maps are given in Section S8.

2.2. **Why stacking fails.** Choose auxiliary capacities a_i larger than the given capacities, with $\{|p| \leq a_i\} \subset \tau_\Delta\Delta$. Placing their base discs side by side would require the pointwise inequality $\sum_i (a_i - |p|)_+ < H_{\mathrm{b}}(p)$. Set

$$P_{\mathrm{b}}(p) = H_{\mathrm{b}}(p) - \sum_i (a_i - |p|)_+.$$

The cap integral gives the exact scalar balance

$$\int_\Delta P_{\mathrm{b}} = \mathcal{B}_\Delta - \sum_i \frac{a_i^n}{n!}. \tag{2}$$

A positive integral leaves room for a comparison, but does not give the pointwise inequality required by stacking. For example, in the periodic cubic model take $k \geq 10$, $t_0 \in (0, 1)$, $a_i = t_0/\sqrt{k}$ and $h = t_0/3$. Then $\int_0^h P_{\mathrm{b}} = t_0(1 - t_0) > 0$, whereas $P_{\mathrm{b}}(0) = 3 - \sqrt{k}\, t_0 < 0$ whenever $t_0 > 3/\sqrt{k}$ (Figure 1). The handle construction uses the positive integral without requiring P_{b} to be positive everywhere.

2.3. **The twist moves area.** Fix the model's annulus, with coordinates $(s, v) \in [0,1] \times \mathbb{R}/\mathbb{Z}$, and its compactly supported closed one-form $\beta = b(v)\,dv$, where $b \geq 0$ and $\int_0^1 b = 1$. Three further objects enter, each defined in Section 4. The handle data include a positive reference area form Λ on the base (Definition 4.1(d)). The comparison datum below carries, besides its gap γ, a reserve γ_* with $0 < \gamma_* \leq \gamma$; it is the part of the gap set aside for the normalisation. From these, (11) fixes a constant $e_\Lambda > 0$, and the first Moser path of §4.2 replaces the physical curvature, on the annulus, by the background $e_\Lambda\Lambda$ plus the base mass $H^*(p) \geq 0$ carried by β:

$$\omega_V + e_\Lambda\Lambda + d\big(f(s)H^*(p)\beta\big), \qquad f = S, \qquad H^*(p) \geq H_{\mathrm{b}}(p) - \gamma_*/16$$

(equations (13) and (14)). This reserve rule and the clock $f = S$ are those of dimension $2n \geq 6$. In dimension four the budget is set directly: the handle mass is $H^* = H_{\mathrm{b}} - e_h$ with $e_h = g/16$, the comparison gap is $\frac{31}{32}g$, the clock is $f(s) = S(2s - \frac{1}{2})$ in the handle coordinate s, and the residual stays above $\frac{7}{8}g$ (Lemma S10.15); this budget replaces the higher-dimensional reserve rule $\gamma_* \leq \gamma$.

Let q_t be the comparator's Hamiltonian isotopy of the fibres, $q = q_1$. The suspension $\mathcal{D}_\lambda(s, v, w) = (s, v, q_{\lambda f(s)}w)$ inserts this isotopy across the annulus. For a smooth function K of the moments, put $J = K - H^*$. Adding the globally defined form $d(J(p)\beta)$ changes only toric mixed terms away from the annulus. After the twist, the amount available across the annulus at w is

$$K\big(p(q^{-1}w)\big) - J\big(p(w)\big) = H^*\big(p(w)\big) + K\big(p(q^{-1}w)\big) - K\big(p(w)\big). \tag{3}$$

The comparison supplies, for a minorant $P \leq P_{\mathrm{b}}$,

$$(K - P)\big(p(qw)\big) \leq K\big(p(w)\big) - \gamma \qquad (w \in V_\Delta), \qquad 0 < \gamma_* \leq \gamma. \tag{4}$$

For $n \geq 3$, applied at $q^{-1}w$, together with $P \leq P_{\mathrm{b}}$ and $H^* \geq H_{\mathrm{b}} - \gamma_*/16$, it shows that the available amount exceeds $\sum_i (a_i - |p(w)|)_+$ by at least $\gamma - \gamma_*/16 > 0$. This amount is divided among the ball layers and a final residual layer, and the non-toric dependence on w is confined to the residual layer.

In dimension four the identity (3) can be read off directly. The fibre is a disc $\bar{D}(h) \subset \mathbb{C}$ with moment $p = \pi|w|^2$, and over the annulus every form in the construction has the shape $\omega_V + e_\Lambda\Lambda + d(G\beta)$ for a function $G(s, w)$ independent of v. By Lemma 4.2(b) its restriction to the annulus through a fixed fibre point w is $e_\Lambda\Lambda + \partial_s G\,ds \wedge \beta$, so, apart from the background $e_\Lambda\Lambda$, the area above w is

$$\int_0^1 \int_0^1 \partial_s G(s, w)\, b(v)\, dv\, ds = G(1, w) - G(0, w),$$

the increase of the β-coefficient across the annulus. After the first path $G = f(s)H^*(p(w))$, so the annulus carries $H^*(p(w))$ above every w. The twist $\mathcal{D}$ is the identity for $s \leq 0$ and is $(s, v, w) \mapsto (s, v, qw)$ for $s \geq 1$. Write the twisted form in the coordinates of the image of $\mathcal{D}$ and add $d(J(p)\beta)$, which is defined on the whole model because p and β are. Below the annulus the coefficient is $J(p(w))$. Above it the point w is the image of $z = q^{-1}w$, where the untwisted coefficient is $H^*(p(z)) + J(p(z)) = K(p(z))$. So the coefficient rises from $J(p(w))$ to $K(p(q^{-1}w))$, which is (3). The *layers* are $k + 1$ disjoint subintervals $I_1, \ldots, I_k, I_*$ of $s \in (0, 1)$ (Figure 2). Across I_i the coefficient rises by a smooth cap $h_i(p(w))$, an explicit smoothing of the tent $(a_i - p)_+$, lying below the tent for $2n \geq 6$ and above it by a controlled amount in dimension four; it is the base area needed above w by a ball of capacity slightly less than a_i, and since on I_i the form is toric, such a ball is placed there in product form. Across the last interval I_* the coefficient rises by the remainder $R^*(w)$, which is positive because the gap γ in (4) exceeds the smoothing errors; it is the only layer coefficient that depends on w through more than $p(w)$.

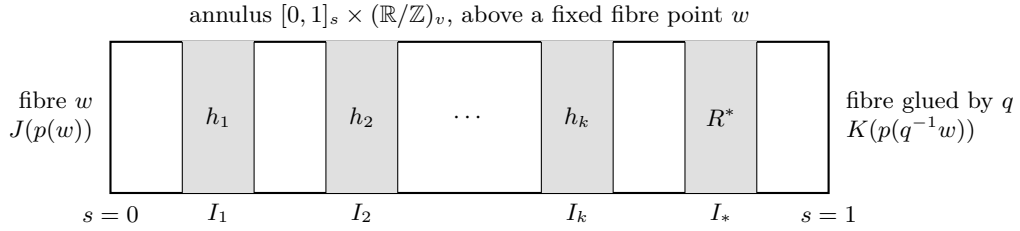


FIGURE 2. The annulus above a fibre point w in dimension four. The β-coefficient rises from $J(p(w))$ to $K(p(q^{-1}w))$; across the layer I_i it rises by the cap $h_i(p(w))$ of the i-th ball and across I_* by the positive remainder $R^*(w)$.

The general comparator is constructed from a concave piecewise affine minorant P with positive mean. For its finite fixed-point construction choose

$$0 < d < \min\bigl(\bar{P}, \beta_P\bigr), \qquad \bar{P} = |\Delta|^{-1} \int_\Delta P, \tag{5}$$

where $P \geq \beta_P > 0$ off the core $\tau_\Delta \Delta$. Section 3 constructs K and q_t and accounts for all approximation errors in the retained gap. The increasing rearrangements act on the moment polytope, and their Hamiltonian realisations act on the fibres. For the equal-ball four-dimensional case with $k \geq 10$ and $\frac{3}{4} < \rho\sqrt{k} < 1$, the linear comparison of Proposition 3.8 gives the datum directly.

2.4. **Model, comparison datum and transfer.**

Definition 2.3 (Model input)**.** A higher-dimensional model input consists of a compact affine core $\Sigma_\ell \subset M$ of an affine curve M, the product $E = M \times \mathbb{C}^m$, a downward polytope Δ, and the physical primitive $\Gamma_{\mathrm{phys}}(p)$ given in the corresponding model section, together with the handle data of Definition 4.1. The physical form is

$$\Omega = \omega_V + d\Gamma_{\mathrm{phys}}, \qquad V_\Delta(E) = \Sigma_\ell \times V_\Delta.$$

The physical curvature $\kappa(p) = d_\Sigma \Gamma_{\mathrm{phys}}(p)$ is positive on the model region and extends smoothly in the moment variables across the coordinate hyperplanes. The four-dimensional input is the periodic cubic model of Definition S8.1; its maps are equivariant for the deck maps given there.

Definition 2.4 (Comparison datum)**.** A comparison datum for (Δ, P) with gap $\gamma > 0$ and reserve $\gamma_* \in (0, \gamma]$ consists of a function K of the moments, smooth on a neighbourhood of Δ in $\mathbb{R}^m$, and a smooth Hamiltonian isotopy $(q_t)_{t\in[0,1]}$ of $\mathbb{C}^m$ with $q_0 = \mathrm{id}$, generated by Q_t with $\iota_{X_{Q_t}}\omega_V = -dQ_t$, such that:

(1) $q_t(V_\Delta) = V_\Delta$ for every t, and there is $\varsigma > 0$ such that for every face ν and every t, Q_t Poisson-commutes with the face moment $\mu_\nu = \langle b_\nu, p\rangle$ on the collar $\{\mu_\nu > \ell_\nu - \varsigma\}$;
(2) $q = q_1$ satisfies the comparison inequality (4).

For the comparator of Theorem 3.5, with gap $\gamma = \frac{11}{16}d$, the models take the reserve $\gamma_* = \frac{11}{64}d$.

Definition 2.5 (Transfer)**.** A transfer to an open target $U \subset \mathbb{C}^n$ assigns to a compact subset $\mathcal{K}$ of the model domain a smooth symplectic embedding $\mathcal{T}_\mathcal{K}$ on a neighbourhood of $\mathcal{K}$, and an explicit compact set $\mathcal{C}_\mathcal{K} \subset U$ containing the image of $\mathcal{K}$. The cubic, quadric and six-dimensional polynomial transfers are fixed maps; in the shell transfer for $n \geq 4$ the slit-shell parameters depend on $\mathcal{K}$.

Theorem 2.6 (Packing in a model)**.** *Let $n \geq 3$ and let $(\Sigma_\ell, \Delta, \Gamma_{\mathrm{phys}})$ be a model input (Definition 2.3) whose handle data satisfy the collar bounds* (12)*. Let $(K, (q_t))$ be a comparison*

datum for (Δ, P) with gap γ and reserve γ_* (Definition 2.4). Let $0 < \rho_i < a_i$ $(1 \le i \le k)$ with $\{|p| \le a_i\} \subset \tau_\Delta \Delta$, and suppose

$$P(p) \le H_{\mathrm{b}}(p) - \sum_i (a_i - |p|)_+ \qquad (p \in \Delta).$$

Then the maps ψ_i of Construction S12.4 are smooth symplectic embeddings of neighbourhoods of the closed balls $B(\rho_i)$ into (E, Ω), and the sets $\psi_i(B(\rho_i))$ are pairwise disjoint and contained in $\operatorname{int} V_\Delta(E)$. *For $n = 2$ the same conclusion holds for the cubic model of Definition S8.1 with the handle data of Proposition S8.3 and the equivariant maps of Section S8, when the comparison datum is replaced by valid cubic data (Definition S10.13): the full-disc comparison for the profile itself and the collar condition $P_{\mathrm{supp}} < h$.*

The model sections verify the density floor, the collar bounds and the profile minorant before applying this theorem; the proof is in §4.5.

Proposition 2.7 (Assembly)**.** *Let ψ_i be the model embeddings of Theorem 2.6 and let $\mathcal{T}$ be its transfer. Then, with $\mathcal{K} = \bigcup_i \psi_i(B(\rho_i))$, the maps $f_i = \mathcal{T}_{\mathcal{K}} \circ \psi_i$ are smooth symplectic embeddings on neighbourhoods of $B(\rho_i)$, have pairwise disjoint images in U, and have images contained in $\mathcal{C}_{\mathcal{K}}$.*

Proof. The set $\mathcal{K}$ is compact, and the transfer is one injective symplectic map on a neighbourhood of it, so composition preserves the pullback identity and disjointness. The inverse images of that neighbourhood under the ψ_i are open neighbourhoods of the source balls. □

input	base and physical primitive	base profile	target
$n = 2$	disc bundle over the cubic, on its periodic cover	$3(1 - 3p)$	$B^\circ(1)$
$n \ge 3$, all $c_i < C/2$	$y_j^2 = t^2 - j$, $m = n - 1$; $(1 - 2\lvert p\rvert)\lambda$	$A_\ell(1 - 2\lvert p\rvert)$; limit $2^m(1 - 2\lvert p\rvert)$	$B^\circ(1)$
$n \ge 4$, one large ball	same affine curve with $q = n - 2$; $(1 - u - 2w)\lambda$	$A_\ell(1 - u - 2w)$; limit $2^q(1 - u - 2w)$	a shell about the reserved ball
$n = 3$, one large ball	an explicit polynomial primitive on a compact affine core	H_ℓ, limit $9(b - \lvert p\rvert) - 2(s - \lvert p\rvert)_+$	common polynomial form, then $B^\circ(1)$

TABLE 2. The four models. A_ℓ is the reference area of the compact affine core; each higher-dimensional total space is a product over that core. The model sections give the heights, truncation level and transfer parameters.

2.5. **The scalar budget.** For a small-ball input define $\mathcal{B}_\Delta$ by (1). The cap identity gives

$$\frac{1 - \sum_i \rho_i^n}{n!} = \left(\frac{1}{n!} - \mathcal{B}_\Delta\right) + \int_\Delta P_{\mathrm{b}} + \sum_i \frac{a_i^n - \rho_i^n}{n!}. \tag{6}$$

The model parameters make the sum of the first and last terms on the right smaller than the left side, so that $\int_\Delta P_{\mathrm{b}} > 0$. In the quadric input the reference-area tail controls the difference from the limiting affine profile; in the six-dimensional input an explicit area modulus controls $H_\ell - H_0$, including the additional two-level collar margin. The comparison operates on a minorant with positive mean and positive values near the outer faces. Its reserve pays for the background and collar mass, and $a_i - \rho_i$ pays for the strip and the source neighbourhood.

For a large-ball input the limiting budget for the remaining balls is $(1-a^n)/n!$, with a the reserved capacity; the shell parameters ($n \geq 4$) or the polynomial profile ($n = 3$) provide it. In all cases the transfer is applied to the already constructed compact family of model images.

symbol	meaning
n, $m = n-1$	phase dimension and fibre rank
p, $\lvert p\rvert$, Δ, V_Δ	fibre moments, their sum, moment polytope and toric domain
κ, H_{b}, $\mathcal{B}_\Delta$	physical base curvature, its intrinsic area and its moment budget
P, K, q_t, γ, γ_*	profile minorant, comparison function, isotopy, gap and reserve
H^*, J, h_i, R^*	normalised handle mass, $K - H^*$, smooth caps and residual
$\mathcal{D}$, Φ_j, $\mathcal{E}_i$	suspension, three Moser endpoints and layer maps
$\mathcal{T}_\mathcal{K}$	transfer on the compact model images

TABLE 3. Notation for the model and handle construction.

3. Hamiltonian comparison

This section produces the comparison data of Definition 2.4. Let Δ be a compact full-dimensional downward polytope with outer faces $\langle b_\nu, p\rangle \leq \ell_\nu$ (§2.4), let $V_\Delta = \{w \in \mathbb{C}^m : p(w) \in \Delta\}$ be its toric domain, with $p_j = \pi|w_j|^2$, $\theta_j = \arg w_j/2\pi \in \mathbb{R}/\mathbb{Z}$ and $\omega_V = \sum_j dp_j \wedge d\theta_j$ as in §2, and let P be a continuous concave function on Δ with positive mean; in the general construction P is rational concave piecewise affine and positive off a core $\tau_\Delta\Delta$. We seek a smooth toric function K and a Hamiltonian isotopy q_t of V_Δ with $(K-P)\circ p \circ q_1 \leq K \circ p - \gamma$ for some $\gamma > 0$. Since q_1 preserves the measure $\omega_V^m/m!$ and, the angles having period 1, $p_*(\omega_V^m/m!)$ is Lebesgue measure on Δ, the map q_1 can only redistribute the values of $K - P$; the inequality asks that after redistribution they lie below $K-\gamma$. The construction has two halves: a scalar half (rearrangement operators on convex functions on Δ and a contraction whose approximate fixed point is K, §§3.1–3.2) and a symplectic half (isotopies realising one coordinate rearrangement up to a prescribed error, §3.3).

The rearrangement operator and the contraction are those of [Ope26, §2]. We replace the fixed point by a finite rational iterate with explicit iteration count and gap (Proposition 3.3), and the existence of realising isotopies by explicit Hamiltonians (Theorem 3.4). For $k \geq 10$ equal balls in dimension four with $\frac{3}{4} < \rho\sqrt{k} < 1$ no fixed point is needed: K is linear, the rearranged inequality is an identity, and the inner planar map costs at most $g/32$ of the gap (§3.4). Every object defined in this section is explicit at level L1 (Definition 1.3).

The polyhedral iterate with log-sum-exp smoothing (Theorem 3.5) is used for all higher-dimensional models. The general four-dimensional case uses the scalar version of §S6, in which the finite iterate is extended by an infimal formula and smoothed by convolution with a fixed bump; its gap and its planar Hamiltonian are Proposition S6.2 and Proposition S6.7. The four-dimensional equal-ball case with $k \geq 10$ uses the linear datum of §3.4.

Throughout, fibre Hamiltonians follow the convention $\iota_{X_Q}\omega_V = -dQ$ of Definition 2.4. Norms $\|\cdot\|$ are uniform norms on Δ. For a coordinate j we write $p = (y, x)$ with $x = p_j$ and $y = (p_i)_{i\neq j}$. The polytope Δ is *rational* if all b_ν, ℓ_ν are rational. A function on Δ is *rational convex polyhedral* if it is the restriction to Δ of $\max_{\iota\in I}\phi_\iota$ for a finite family of affine functions ϕ_ι with rational coefficients, and *rational concave piecewise affine* if it is $\min_{\iota\in I}\phi_\iota$ for such a family. All cutoffs and clocks below use the flat step S of Definition S2.3, all of whose derivatives vanish at 0 and 1.

3.1. **Coordinate rearrangement.** Fix j. Because every b_ν is nonnegative and Δ is compact, at least one ν has $b_{\nu j} > 0$, and

$$L(y) = \min_{\nu:\, b_{\nu j}>0} \frac{\ell_\nu - \sum_{i\neq j} b_{\nu i} y_i}{b_{\nu j}}$$

is a continuous concave piecewise affine function on the projection $\Delta_j = \{y : (y,0) \in \Delta\}$, with $\Delta = \{(y,x) : y \in \Delta_j,\ 0 \le x \le L(y)\}$. The slice of Δ through y is the interval $[0, L(y)]$.

Definition 3.1 (Coordinate rearrangement)**.** For a continuous convex function F on Δ and a coordinate j, put

$$(R_jF)(y,x) = \min_{0\le a,\ a+x\le L(y)} \max\big(F(y,a),\, F(y,a+x)\big), \qquad (y,x) \in \Delta,$$

and $\mathcal{R} = R_m \circ \cdots \circ R_1$.

The minimum exists because the admissible a form the compact interval $[0, L(y) - x]$. Since the sublevel sets of a convex function of one variable are intervals, $x \mapsto (R_jF)(y,x)$ is the increasing rearrangement of $F(y,\cdot)$ on $[0,L(y)]$ with respect to Lebesgue measure: for $x > 0$ it is the smallest value b for which $\{F(y,\cdot) \le b\}$ has length at least x; at $x = 0$ it equals $\min F(y,\cdot)$.

Lemma 3.2 (Properties of R_j)**.** *Let F, G be continuous convex functions on Δ and $j \in \{1,\dots,m\}$.*

- *(i) R_jF is continuous and convex on Δ, and $\int_\Delta R_jF = \int_\Delta F$.*
- *(ii) If $F \le G$ then $R_jF \le R_jG$; $R_j(F+c) = R_jF + c$ and $R_j(\alpha F) = \alpha R_jF$ for constants $c \in \mathbb{R}$, $\alpha \ge 0$; $\|R_jF - R_jG\| \le \|F - G\|$.*
- *(iii) For each y, $x \mapsto (R_jF)(y,x)$ is nondecreasing, $(R_jF)(y,0) = \min_{[0,L(y)]} F(y,\cdot)$ and $(R_jF)(y,L(y)) = \max_{[0,L(y)]} F(y,\cdot)$; in particular $\max_\Delta R_jF = \max_\Delta F$.*
- *(iv) If F is nondecreasing in p_i for some $i \neq j$, so is R_jF. Consequently $\mathcal{R}F$ is nondecreasing in every coordinate, and $\mathcal{R}F(0) = \min_\Delta \mathcal{R}F$.*
- *(v) Let Δ be rational. If F is rational convex polyhedral, so is R_jF. If $F = \max_{\iota\in I} \phi_\iota$, the epigraph of R_jF over Δ is the projection to (y,x,t) of the rational polyhedron in the variables (y,x,t,a) defined by $p = (y,x) \in \Delta$, $a \ge 0$, $b_{\nu j}(a+x) + \sum_{i\neq j} b_{\nu i} y_i \le \ell_\nu$ for every ν with $b_{\nu j} > 0$, and $t \ge \phi_\iota(y,a)$, $t \ge \phi_\iota(y,a+x)$ for every $\iota \in I$; eliminating a by Fourier–Motzkin elimination gives finitely many rational affine functions whose maximum is R_jF.*

Proof. The proofs are in Lemma S3.2 and Lemma S3.3; we give the arguments for convexity, monotonicity and (v). Convexity: if $[a_0, a_0+x_0]$ and $[a_1, a_1+x_1]$ are minimising intervals on the slices through y_0 and y_1, their convex combination with weights $1-\vartheta, \vartheta$ is admissible on the slice through $(1-\vartheta)y_0 + \vartheta y_1$ (concavity of L), and convexity of F bounds its endpoint values by the corresponding convex combinations. Monotonicity in p_i: lowering y_i does not shrink the slice, because $b_{\nu i} \ge 0$, so every admissible interval stays admissible while its endpoint values do not increase. Item (v): the minimum in Definition 3.1 is attained, so $t \ge R_jF(y,x)$ if and only if some a satisfies the displayed system, and the epigraph is the projection. The variable t occurs only in lower bounds $t \ge \dots$, and Fourier–Motzkin elimination of a forms nonnegative combinations of the given inequalities; so it produces only inequalities of the form $t \ge$ (affine function of (y,x)) and inequalities free of t. The latter hold on Δ, because every $(y,x) \in \Delta$ has a feasible a (for instance $a = 0$, with t large), and hence R_jF is the maximum of the former. □

3.2. **A finite comparator iteration.** For $0 < \lambda < 1$ consider $T_\lambda(F) = \mathcal{R}(\lambda F - P)$ on the space of continuous convex functions on Δ. If P is concave, $\lambda F - P$ is convex, and by Lemma 3.2(ii) T_λ is a contraction with constant λ.

Proposition 3.3 (Finite comparator)**.** *Let Δ be rational, $\tau_\Delta \in (0,1)$ rational, and P a rational concave piecewise affine function on Δ with mean $\mu = \mathrm{vol}(\Delta)^{-1}\int_\Delta P > 0$ and $P \ge \beta_P$ on $\Delta \setminus \tau_\Delta\Delta$ for some $\beta_P > 0$. Let d be rational with $0 < d < \min(\mu, \beta_P)$ and put*

$$\eta_\Delta = (1-\tau_\Delta)(\mu - d), \qquad M_P = \max\bigl(1,\ d - \min_\Delta P + 1\bigr), \qquad \lambda = 1 - \frac{\eta_\Delta}{2(M_P + \eta_\Delta)}.$$

Define $K_0 = 0$, $K_{r+1} = \mathcal{R}(\lambda K_r - P)$ and

$$N = \left\lceil \frac{16\|P\|}{d(1-\lambda)(\lambda^{-1}-1)} \right\rceil + 1.$$

Then every K_r is rational convex polyhedral, $\lambda^N\|P\|/(1-\lambda) < d/16$, and

$$\mathcal{R}(K_N - P) \le K_N - \tfrac{7}{8}d \qquad \text{on } \Delta.$$

All quantities in Proposition 3.3 are computed by finite rational arithmetic. For a rational convex polyhedral G, Lemma 3.2(iii),(iv) give $\min_\Delta G = (\mathcal{R}G)(0)$, and the maximum of an affine function ϕ is $-\min_\Delta(-\phi)$; this gives $\min_\Delta P$ and $\|P\|$ by Fourier–Motzkin elimination. The integral $\int_\Delta P$, hence μ, is computed by iterated integration of rational polynomials over rational polyhedral cells, one coordinate at a time (Proposition S3.4(a)). Hence N is obtained by rational arithmetic and a ceiling, and K_N by mN Fourier–Motzkin eliminations (Lemma 3.2(v)).

Proof. Rationality. $K_0 = 0$, and if $K_r = \max_\iota \phi_\iota$ and $P = \min_u \psi_u$ with rational affine ϕ_ι, ψ_u, then $\lambda K_r - P = \max_{\iota,u}(\lambda\phi_\iota - \psi_u)$ is rational convex polyhedral, so K_{r+1} is by Lemma 3.2(v) applied m times.

The fixed point. The continuous convex functions on Δ form a closed subset of $C(\Delta)$, so the λ-contraction T_λ has a unique fixed point K_λ (the comparison function of [Ope26, §2.2]), and $\|K_r - K_\lambda\| \le \lambda^r\|K_\lambda\|$. Since $\mathcal{R}0 = 0$ and $\mathcal{R}$ is 1-Lipschitz, $\|K_\lambda\| \le \lambda\|K_\lambda\| + \|P\|$, so $\|K_\lambda\| \le \|P\|/(1-\lambda)$. By Lemma 3.2(i),(iv), K_λ is convex, nondecreasing in every coordinate, $K_\lambda(0) = \min K_\lambda$, and integrating the fixed-point equation gives $\int K_\lambda = \lambda\int K_\lambda - \int P$, that is, the mean of K_λ is $-\mu/(1-\lambda)$.

The gap of the fixed point. Note $\frac12 < \lambda < 1$, $1-\lambda = \eta_\Delta/(2(M_P+\eta_\Delta))$ and

$$\frac{\lambda\eta_\Delta}{1-\lambda} = 2M_P + \eta_\Delta > d - \min_\Delta P. \tag{7}$$

The idea is that the mean of K_λ is very negative, so by convexity its values on the core $\tau_\Delta\Delta$ lie well below its maximum, while off the core $P \ge \beta_P$ lowers $\lambda K_\lambda - P$; the fixed-point equation then forces the maximum of K_λ to be negative. In detail, let $M_\lambda = \max K_\lambda$ and suppose $M_\lambda \ge -d/(1-\lambda)$, so that $\lambda M_\lambda \le M_\lambda + d$. If $p \in \Delta\setminus\tau_\Delta\Delta$, then $\lambda K_\lambda(p) - P(p) \le \lambda M_\lambda - \beta_P < M_\lambda$, because $(1-\lambda)M_\lambda \ge -d > -\beta_P$. If $p = \tau_\Delta v$ with $v \in \Delta$, convexity and $K_\lambda(0) = \min K_\lambda \le -\mu/(1-\lambda)$ give

$$K_\lambda(p) \le \tau_\Delta M_\lambda + (1-\tau_\Delta)K_\lambda(0) \le M_\lambda - (1-\tau_\Delta)\Bigl(M_\lambda + \frac{\mu}{1-\lambda}\Bigr) \le M_\lambda - \frac{\eta_\Delta}{1-\lambda},$$

the last step by $M_\lambda \ge -d/(1-\lambda)$. Hence, by (7),

$$\lambda K_\lambda(p) - P(p) \le M_\lambda + d - \frac{\lambda\eta_\Delta}{1-\lambda} - \min_\Delta P < M_\lambda.$$

Thus $\max_\Delta(\lambda K_\lambda - P) < M_\lambda$, while Lemma 3.2(iii) and the fixed-point equation give $M_\lambda = \max\mathcal{R}(\lambda K_\lambda - P) = \max(\lambda K_\lambda - P)$, a contradiction. Therefore $M_\lambda < -d/(1-\lambda)$, i.e. $(1-\lambda)K_\lambda < -d$ on Δ, and by Lemma 3.2(ii)

$$\mathcal{R}(K_\lambda - P) = \mathcal{R}\bigl(\lambda K_\lambda - P + (1-\lambda)K_\lambda\bigr) \le \mathcal{R}(\lambda K_\lambda - P) - d = K_\lambda - d.$$

The finite iterate. Let $r_N = \|K_N - K_\lambda\| \le \lambda^N \|P\|/(1-\lambda) < d/16$. By Lemma 3.2(ii), $\mathcal{R}(K_N - P) \le \mathcal{R}(K_\lambda - P) + r_N \le K_\lambda - d + r_N \le K_N - d + 2r_N < K_N - \frac{7}{8}d$. □

The construction applies verbatim when the polytope and the affine pieces have real coefficients: Fourier–Motzkin elimination uses finitely many arithmetic operations and comparisons, and the estimates are unchanged, with "rational convex polyhedral" replaced by "convex polyhedral". The same holds for the smoothing and realisation statements below. This real version is the one used for real capacities.

3.3. **Realisation and the comparison datum.** The symplectic half realises one coordinate rearrangement by an isotopy which changes only the j-th moment; the other fibre coordinates may rotate.

Theorem 3.4 (Realisation of one coordinate rearrangement)**.** [Lean: needed consequence verified.]

Let Δ be rational, F rational convex polyhedral on Δ, $j \in \{1, \dots, m\}$ and $e > 0$ rational. Let $Q_t^{(j)}$ be the Hamiltonian defined from (Δ, F, j, e) in Definition S7.10, and let $(q_t^{(j)})_{t\in[0,1]}$ be its flow ($\iota_X \omega_V = -dQ_t^{(j)}$). Then $Q^{(j)}$ is smooth with compact support in w, $(q_t^{(j)})$ is a smooth Hamiltonian isotopy of $\mathbb{C}^m$, and:

- *(a) $q_0^{(j)} = \mathrm{id}$; every $q_t^{(j)}$ maps V_Δ onto itself and preserves the moments p_i, $i \ne j$.*
- *(b) There is a rational $\varsigma_j > 0$ such that for every ν with $b_{\nu j} > 0$ and every t, $Q_t^{(j)}$ vanishes on $\{\langle b_\nu, p\rangle > \ell_\nu - \varsigma_j\}$; for every ν with $b_{\nu j} = 0$, $Q_t^{(j)}$ Poisson-commutes with $\langle b_\nu, p\rangle$.*
- *(c) $F\big(p(q_1^{(j)}(w))\big) \le (R_j F)(p(w)) + e$ for every $w \in V_\Delta$.*

The proof is Theorem S7.13. The theorem enters through Theorem 3.5, which is used for $2n \ge 6$; the general four-dimensional branch realises its single rearrangement by the planar path of Proposition S6.7 instead. The construction is planar and rests on the following fact: if f is a convex function on $[0,1]$ and $f^\sharp$ its increasing rearrangement, there are area-preserving maps q of the disc of area 1 with $f(\pi|q(z)|^2) \le f^\sharp(\pi|z|^2) + \xi$ for any $\xi > 0$. Such a map carries each disc $\{\pi|z|^2 \le I\}$ onto a region of area I on which $f(\pi|\cdot|^2) \le f^\sharp(I) + \xi$. In the moment–angle rectangle $(0,1) \times (-\frac{1}{2}, \frac{1}{2})$, with area form $dv \wedge d\theta$, we take these regions to be the sublevel sets of one convex potential that approximates $f(v)$ and is proper in the angle, and build the map from its level curves.

In detail, fix y and rescale the inner slice $[0, \ell_{\mathrm{in}}(y)]$ to $[0,1]$, where $\ell_{\mathrm{in}}(y) < L(y)$ is a smooth inner slice length. Then $f(v) = F(y, \ell_{\mathrm{in}}(y)v)$ is convex on $[0,1]$ with increasing rearrangement $f^\sharp$, and we need q_{in} with the displayed inequality on a core disc, with planar error $\xi = e/4$. Smooth f by log-sum-exp and add the barrier $\kappa_0(1/v + 1/(1-v))$ with a small positive κ_0; this gives a real-analytic, strictly convex, proper potential U on $(0,1)$. Add to it a small multiple of the proper even wall $\theta^2/(\frac{1}{4} - \theta^2)$ in the angle. The sublevel sets of the sum are nested ovals in the rectangle, and the level enclosing area I lies below $f^\sharp(I)$ plus a small error (Lemma S7.5). To compare these ovals with round circles we make the potential exactly quadratic near its minimum: the two one-variable wells are modified on small intervals, with error at most $\xi/64$ each (Lemma S7.6). In the signed square roots of the two wells the sum is then the squared radius, so after a rotation and a normalisation of area the centred circle enclosing area I is carried onto the level of the modified potential enclosing area I. The chart $(v, \theta) \mapsto -\sqrt{v/\pi}\, e^{2\pi i\theta}$, in which $dv \wedge d\theta$ is the area form, returns to the disc (Theorem S4.7).

A compactly supported Hamiltonian isotopy of the disc ends at q_{in} on a core disc whose complement has area exactly 8η, with $\eta = \frac{1}{2}\min(1/16, \xi/[32(\Lambda_f + 1)])$, where $\Lambda_f \ge 1$ bounds the slopes of f (Theorem S4.21); outside the core, Lipschitz continuity of $f^\sharp$ gives

$$f \le \max f = f^\sharp(1) \le f^\sharp(\pi|z|^2) + 8\Lambda_f \eta < f^\sharp(\pi|z|^2) + \xi.$$

All constants are chosen uniformly in y over an open parameter domain (Theorem S7.9). The planar generator is lifted as

$$Q_t^{(j)}(w) = \chi(y)S'(t)\ell_{\mathrm{in}}(y)\, Q^{\mathrm{pl}}\big(y, \chi(y)S(t), w_j/\sqrt{\ell_{\mathrm{in}}(y)}\big),$$

with a cutoff $\chi(y)$ vanishing near degenerate slices. The planar Hamiltonian is supported in $\{\pi|u|^2 < r_B\}$ with $r_B < 1$, and $\ell_{\mathrm{in}} > \eta_1$ wherever $\chi \neq 0$; this gives the positive real collar $c_j = \frac{1}{2}(1-r_B)\eta_1 \min_{\nu:\, b_{\nu j}>0} b_{\nu j}$. The rational collar in (b) is $\varsigma_j = 2^{-N_j}$ with N_j the least natural number such that $2^{-N_j} < c_j$; this choice does not affect the Hamiltonian. The lifted motion fixes every point with $p_j \geq r_B\ell_{\mathrm{in}}(y)$, and $\ell_{\mathrm{in}} < L$, so every slice $\{y\} \times [0, L(y)]$ is preserved. The general convex construction is §S7; the equal-ball instance, in which the potential is the tent of §3.4, is §S5; the common planar machinery is §S4.

Theorem 3.5 (Comparison datum for concave profiles)**.** *Assume the hypotheses of Proposition 3.3, and let d, N and K_N be as there. Let $K_N = \max_{\iota\in I} \phi_\iota$ be the representation by rational affine functions produced by the iteration, each step R_j being the Fourier–Motzkin elimination of Lemma 3.2(v), set $c_K = \lceil 32\,\#I/d\rceil + 1$ (so that $\log(\#I)/c_K < d/32$ by Lemma S3.5), and put*

$$K = \frac{1}{c_K} \log \sum_{\iota\in I} e^{c_K\phi_\iota}.$$

Let $F_0 = K_N - P$ and $F_j = R_jF_{j-1}$ $(1 \leq j \leq m)$; let $q_t^{(j)}$ be the isotopy of Theorem 3.4 for F_{j-1}, the coordinate j and $e_j = d/(8(m+1))$. Define q_t by running $q^{(m)}, q^{(m-1)}, \dots, q^{(1)}$ in turn with flat clocks: for $t \in [\frac{r-1}{m}, \frac{r}{m}]$, $r = 1, \dots, m$, and $j = m+1-r$,

$$q_t = q^{(j)}_{S(mt-r+1)} \circ q_1^{(j+1)} \circ \cdots \circ q_1^{(m)}.$$

Then $(K, (q_t))$ is a comparison datum for (Δ, P) with gap $\gamma = \frac{11}{16}d$.

Proof. The two expressions for q_t at $t = r/m$ agree, so q_t is well defined, $q_0 = \mathrm{id}$ and $q_1 = q_1^{(1)} \circ \cdots \circ q_1^{(m)}$. On $[\frac{r-1}{m}, \frac{r}{m}]$ the isotopy is generated by $Q_t = mS'(mt-r+1)\, Q^{(j)}_{S(mt-r+1)}$, since precomposition with the fixed map $q_1^{(j+1)} \circ \cdots \circ q_1^{(m)}$ does not change the generating vector field. All derivatives of S vanish at 0 and 1, so Q_t and q_t are smooth in t across the seams. Each factor maps V_Δ onto itself, so q_t preserves V_Δ. With $\varsigma = \min_j \varsigma_j$, Theorem 3.4(b) shows that every Q_t Poisson-commutes with $\langle b_\nu, p\rangle$ on $\{\langle b_\nu, p\rangle > \ell_\nu - \varsigma\}$ for every ν. This is condition (1) of Definition 2.4.

For condition (2), each F_j is rational convex polyhedral by Lemma 3.2(v), so Theorem 3.4 applies, and $F_m = \mathcal{R}(K_N - P) \leq K_N - \frac{7}{8}d$ by Proposition 3.3. Fix $w \in V_\Delta$ and put $u_m = w$, $u_{j-1} = q_1^{(j)}(u_j)$, so that $u_0 = q_1(w)$ and all $u_j \in V_\Delta$. Theorem 3.4(c) gives $F_{j-1}(p(u_{j-1})) \leq F_j(p(u_j)) + e_j$, and summing over j,

$$(K_N - P)\big(p(q_1(w))\big) \leq F_m(p(w)) + \sum_j e_j \leq K_N(p(w)) - \tfrac{7}{8}d + \tfrac{1}{8}d = K_N(p(w)) - \tfrac{3}{4}d.$$

For log-sum-exp, $\max_\iota \phi_\iota \leq K \leq \max_\iota \phi_\iota + \log(\#I)/c_K$, so $K_N \leq K < K_N + d/32$ on $\mathbb{R}^m$ (Lemma S3.5). Therefore

$$(K - P)\big(p(q_1(w))\big) < (K_N - P)\big(p(q_1(w))\big) + \tfrac{d}{32} \leq K(p(w)) - \tfrac{3}{4}d + \tfrac{1}{32}d < K(p(w)) - \tfrac{11}{16}d.$$

K is real-analytic on $\mathbb{R}^m$, hence smooth near Δ, as Definition 2.4 requires. □

3.4. **The closed-form datum for equal balls.** For $k \geq 10$ equal capacities in dimension four the profile has a single kink, and the fixed point can be written down: K is linear and $K - P$ is, up to a constant, a tent whose increasing rearrangement is linear. This datum is used in the equal-capacity branch of Theorem S13.23; Remark 3.9 treats $k \leq 9$. Here $m = 1$, $\Delta = [0, h]$ with h real (rationality is not used in this subsection), and V_Δ is the closed disc $\bar{D}(h)$ of area h. The symbols s, a, c_s, g, A, B below are local to this subsection and to §S3.7.

Let $k \geq 10$ be an integer and $\eta_\star \in (0, \frac{1}{4})$, set $t_0 = 1 - \eta_\star/2$, and put

$$s = \sqrt{k}, \quad a = \frac{t_0}{s}, \quad h = \frac{t_0}{3}, \quad c_s = 3s - 9, \quad g = 3(1 - t_0), \quad A = \frac{h}{a}, \quad B = \frac{h}{h - a}.$$

Since $s > 3$ we have $0 < a < h$, $c_s > 0$, $A, B > 0$ and

$$\frac{1}{A} + \frac{1}{B} = \frac{a}{h} + \frac{h - a}{h} = 1. \tag{8}$$

Define $\Lambda(x) = A(a - x)$ for $x \leq a$ and $\Lambda(x) = B(x - a)$ for $x \geq a$. On $[0, h]$, Λ is a tent with $\Lambda(0) = \Lambda(h) = h$ and $\Lambda(a) = 0$.

Lemma 3.6 (Sublevel sets of the tent)**.** *For $b \geq 0$, $\{x \in \mathbb{R} : \Lambda(x) \leq b\} = [a - b/A,\, a + b/B]$, an interval of length b. If $b \leq h$, it is contained in $[0, h]$.*

Proof. The interval is read off from the two affine pieces; its length is $b(1/A + 1/B) = b$ by (8). If $b \leq h$, then $a - b/A \geq a - h/A = 0$ and $a + b/B \leq a + h/B = h$. □

Corollary 3.7 (Rearrangement of the tent)**.** *With $m = 1$ and $\Delta = [0, h]$, $R_1\Lambda(x) = x$ for $0 \leq x \leq h$. Consequently $R_1(-g + c_s\Lambda)(x) = -g + c_s x$.*

Proof. For $\alpha = a - x/A$ we have $\alpha \geq 0$, $\alpha + x = a + x/B \leq h$ and $\Lambda(\alpha) = \Lambda(\alpha + x) = x$, so $R_1\Lambda(x) \leq x$. Conversely, if an admissible interval $[\alpha, \alpha + x]$ had $b' = \max(\Lambda(\alpha), \Lambda(\alpha + x)) < x$, then by convexity $[\alpha, \alpha + x] \subset \{\Lambda \leq b'\}$, an interval of length $b' < x$ by Lemma 3.6; this is impossible. The second statement follows from Lemma 3.2(ii), since $c_s > 0$. □

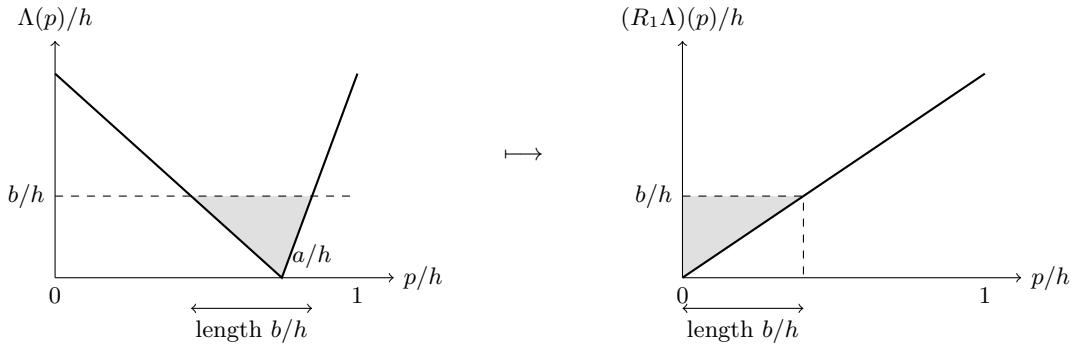


FIGURE 3. The linear comparison: the sublevel interval of Λ has length b, so its increasing rearrangement is p. The drawing uses $a/h = 3/4$ ($k = 16$); the identity holds for every $k \geq 10$. Consequently $K - R_1(K - P) \equiv g$, before the planar realisation error is deducted.

Proposition 3.8 (Linear comparison datum for equal balls)**.** *With the notation above let $\Delta = [0, h]$, $P(p) = 3(1 - 3p) - k(a - p)_+$, $K(p) = c_s p$ and $F = K - P$. Then*

$$F(x) = -g + c_s\Lambda(x) \qquad (x \in \mathbb{R}). \tag{9}$$

Let $e_q = \min\big(h/100,\ g/(128c_s)\big)$ (Definition S3.10) and let (q_t) be the isotopy of Theorem S5.34, whose endpoint $q = q_1$ satisfies $\Lambda(p(q(z))) \le \pi|z|^2 + 4e_q$ on $\bar D(h)$. Then $(K, (q_t))$ is a comparison datum for (Δ, P) with gap $\frac{31}{32}g$.

Proof. By Lemma S3.9, $c_sA = k-3s$, $c_sB = 3s$ and $3sa = 3-g$. For $x \le a$: $F(x) = c_sx-3+9x+k(a-x) = (3s-k)x+ka-3$, and $-g+c_sA(a-x) = (3s-k)x+ka-(g+3sa) = (3s-k)x+ka-3$. For $x \ge a$: $F(x) = c_sx - 3 + 9x = 3sx - 3$, and $-g + c_sB(x-a) = 3sx - g - 3sa = 3sx - 3$. This proves (9).

By Theorem S5.34, $q_0 = \mathrm{id}$, every q_t maps $\bar D(h)$ onto itself and is the identity on $\{\pi|z|^2 > P_{\mathrm{supp}}\}$ for a constant $P_{\mathrm{supp}} < h$ defined there, and the generator Q_t vanishes there; this is condition (1) of Definition 2.4 for the single outer face $p = h$. For condition (2), let $w \in \bar D(h)$ and $I = p(w)$. By (9) and $4c_se_q \le g/32$,

$$K(I) - (K-P)\big(p(q(w))\big) = c_sI + g - c_s\Lambda\big(p(q(w))\big) \ge g - 4c_se_q \ge \tfrac{31}{32}g. \qquad \square$$

Thus $K - R_1F \equiv g$ exactly (Corollary 3.7), and the planar map costs at most $g/32$. As $t_0 \to 1$ the gap $g \to 0$ while the packed volume $ka^2 = t_0^2 \to 1$. The equal-capacity branch uses $\eta_\star = 1 - \rho\sqrt{k}$, that is $t_0 = (1+\rho\sqrt{k})/2$ and $g = 3\eta_\star/2$.

Remark 3.9 (The two four-dimensional branches). The linear comparison is used when the capacities are equal, $k \ge 10$ and $\eta_\star = 1 - \rho\sqrt{k}$ lies in $(0, \frac14)$, that is $\frac34 < \rho\sqrt{k} < 1$. All other four-dimensional inputs, including $k \le 9$, use the finite scalar comparator of §S6.

Remark 3.10 (An unequal input without linear comparison function). In the cubic model (§S8), given capacities ρ_i are realised through auxiliary capacities $a_i > \rho_i$ and a height h with $a_i < h < \frac13$ (Lemma S8.5); the profile is $P(p) = 3(1-3p) - \sum_i (a_i - p)_+$ on $\Delta = [0, h]$, and by Theorem 2.6 any continuous minorant of P may replace it. For $K(p) = \kappa p$ put $F = K - P = (\kappa+9)p - 3 + \sum_i (a_i - p)_+$. If $(K, (q_t))$ is a comparison datum for (Δ, P) with gap $\gamma > 0$, then, evaluating condition (2) of Definition 2.4 at $w = 0$ and at a preimage under q_1 of a maximum point of F (the map q_1 sends $\bar D(h)$ onto itself),

$$\min_{[0,h]} F < K(0) = 0, \qquad \max_{[0,h]} F < \max_{[0,h]} K. \tag{10}$$

Replacing P by a smaller function increases F, so both inequalities only become harder to satisfy, and what follows applies to every minorant.

Consider first the limiting data a^*: nine capacities $8/25$ and one capacity $7/25$, so that $\sum_i a_i^2 = 1$, and $h = \frac13$, where $\int_0^h P = \frac12[1 - \sum_i a_i^2 - (1-3h)^2] = 0$. Then $F(0) = \sum_i a_i - 3 = \frac{4}{25}$, and the slopes of F on $[0, \frac{7}{25}]$, $[\frac{7}{25}, \frac{8}{25}]$, $[\frac{8}{25}, \frac13]$ are $\kappa - 1$, κ, $\kappa + 9$. If $\kappa < 0$, then $\max F \ge F(0) > 0 = \max K$. If $\kappa \ge 1$, then $\min F = F(0) = \frac{4}{25} > 0$. If $0 \le \kappa < 1$, then $\min F = F(\frac{7}{25}) = \frac{7\kappa - 3}{25}$ and $\max F - \max K \ge \frac{4}{25} - \frac{\kappa}{3}$; the larger of these two numbers is at least $\frac{9}{1150}$ (equality at $\kappa = \frac{21}{46}$). So for every $\kappa \in \mathbb{R}$ one of the inequalities (10) fails by at least $\frac{9}{1150}$.

The bound survives perturbation. Let $\eta_0 = 1/2000$, $|a_i - a_i^*| \le \eta_0$ and $\frac13 - \eta_0 \le h < \frac13$. Then F differs from its limiting value by at most $10\eta_0$ at every $p \in [0, h]$, uniformly in κ; minima over $[0, h] \subset [0, \frac13]$ are at least the limiting minima, $F(0)$ is still among the values, and $\max_{[0,h]} K = \kappa h \le \kappa/3$ for $\kappa \ge 0$. The three cases therefore hold with every bound lowered by $10\eta_0 = \frac{1}{200} < \frac{9}{1150}$, and (10) fails again. Such auxiliary data are forced by the given capacities. A comparison datum forces $\int_0^h P > 0$ (integrate condition (2) over $\bar D(h)$, using that q_1 preserves area), i.e. $\sum_i a_i^2 + (1-3h)^2 < 1$. With $D = 1 - \sum_i \rho_i^2$ and $a_i > \rho_i$ this gives $a_i^2 - \rho_i^2 < D$, hence $a_i - \rho_i < D/(2\rho_i)$, and $(1-3h)^2 < D$, hence $\frac13 - h < \sqrt{D}/3$. Therefore, if $|\rho_i - a_i^*| \le \eta_0/2$ for all i (so $\rho_i > \frac14$) and $D \le 9\eta_0^2$, every admissible choice of (a, h) satisfies $|a_i - a_i^*| < \eta_0$ and

$h > \frac{1}{3} - \eta_0$, and no linear K serves; for instance $\rho_i = (1 - 10^{-7})a_i^*$. The scalar construction of Proposition S6.8 applies to all of these inputs.

4. Packing over a handle

The handle construction turns the comparison inequality into ball embeddings by three exact symplectic paths. It is an explicit version of the surface packing lemma of [Ope26, §3.2], in which the Moser argument [Mos65] is carried out with explicit vector fields. The first path puts the usable base mass on an annulus, the second inserts the comparator isotopy across it, and the third separates the caps into disjoint layers. The higher-dimensional models use the finite coordinate construction of §S12; the four-dimensional periodic model uses the equivariant version of Remark 4.7. The suspension and layer identities below have the same form in both.

In the higher-dimensional construction below, $n \geq 3$ and $m = n-1$. Write $\Omega = \omega_V + d\Gamma_{\mathrm{phys}}$ on the explicit affine product, with compact model region $V_\Delta(E) = \Sigma_\ell \times V_\Delta$, physical base curvature $\kappa(p)$ and profile $H_{\mathrm{b}}(p) = \int_{\Sigma_\ell} \kappa(p)$. The comparison datum has gap γ and reserve $0 < \gamma_* \leq \gamma$. A horizontal one-form annihilates fibre tangent vectors. A family of base one-forms depending smoothly on p gives a toric horizontal form; its differential is $d_\Sigma \alpha(p) + \sum_j dp_j \wedge \partial_j \alpha(p)$. This expression is smooth across the coordinate axes because the moments $p_j = \pi|w_j|^2$ are smooth there.

4.1. **Handle data.**

Definition 4.1 (Handle data)**.** Each higher-dimensional model supplies the following data.

(a) An explicit annular coordinate map on $|s_{\mathrm{a}}| < w$, with working band $|s_{\mathrm{a}}| \leq b < w$. We use $s = (s_{\mathrm{a}} + b)/(2b)$, so the working band is $[0, 1]$. For the diagonal-quadric presentation the model uses $w = 1/8$ and $b = 1/16$ inside a double cylinder of half-width $1/4$ on which the annular coordinate is defined.
(b) A fixed compact support set $\mathcal{B} \subset \operatorname{int} \Sigma_\ell$.
(c) A smooth closed one-form β, supported in $\mathcal{B}$, whose annular expression is $b(v)\, dv$, with $b \geq 0$, $b > 0$ on (v_-, v_+) and $\int_0^1 b(v)\, dv = 1$.
(d) An explicit reference primitive Γ_{ref} with positive curvature $\Lambda = d_\Sigma \Gamma_{\mathrm{ref}}$, a reference-area upper bound $A_{\mathrm{ref}} > 0$, and a positive physical density floor κ_- relative to Λ on the correction region: $\kappa(p) \geq \kappa_- \Lambda$ for $p \in \Delta$.
(e) A finite coordinate-box presentation of that region: explicit chart maps and inverses, partition weights, unit-mass coordinate bumps, and a rooted tree on the boxes in which every box other than the root has an overlapping parent of smaller index. Rectangular integration along this tree defines the operator $\mathcal{P}$ of §S12.
(f) The product fibre coordinates of the affine model. In the four-dimensional instance these are the explicit equivariant coordinates on the cover.
(g) A collar cutoff χ_∂, equal to one near the boundary and zero on the inner region containing the annulus, together with the model's proved tail bound. A second, larger finite atlas covers the whole closed model region for the Moser flows.

Every item is supplied by the model formulas. The fixed annulus and dual form are constructed in §S15; the finite atlas and tree are in §S12. The annular gauge is identically one on $|s_{\mathrm{a}}| < b_1 = (b + w)/2$. We write $\mathcal{A}^+$ for this open band and $\mathcal{A}$ for the smaller open band $|s_{\mathrm{a}}| < (b + b_1)/2$. Both contain the working band $0 \leq s \leq 1$ in their interior. Thus the clock $f(s) = S(s)$ is constant on collars of the boundary of $\mathcal{A}$. Put $\eta = df \wedge \beta$ and write $\Lambda = a_\Lambda(s, v)\, ds \wedge dv$, $a_\Lambda > 0$. For the periodic four-dimensional instance the background $e_\Lambda \Lambda$ is $e_h \nu$ (Remark 4.7).

Lemma 4.2 (Top power)**.** *(a) In local product coordinates let κ' be a base two-form and $a'_1, \dots, a'_m$ base one-forms, with coefficients depending on the base point and on p. Then $(\omega_V + \kappa' + \sum_j dp_j \wedge a'_j)^n = n\,\omega_V^m \wedge \kappa'$.*

(b) On $E|_{\mathcal{A}^+}$, in the explicit product coordinates, let $\sigma_\Sigma = a_\Sigma(s,v)\,ds \wedge dv$ be positive, let $G(s,w)$ be smooth and independent of v, and $e > 0$. Then $\big(\omega_V + e\sigma_\Sigma + d(G\beta)\big)^n = n\,(e a_\Sigma(s,v) + b(v)\,\partial_s G)\,\omega_V^m \wedge ds \wedge dv$.

In particular the forms in (a) are nondegenerate where $\kappa' > 0$; when closed, they are symplectic there. The forms in (b) are symplectic where $\partial_s G \geq 0$.

Proof. Write x_j, y_j for the real fibre coordinate covectors and s, t for the base covectors at a point. Every dp_j has the form $u_j x_j + v_j y_j$ and every a'_j has the form $A_j s + B_j t$. For $B = A_j s + B_j t$, the identity

$$x_j \wedge y_j + (u_j x_j + v_j y_j) \wedge B = (x_j - v_j B) \wedge (y_j + u_j B)$$

factors each block as a decomposable two-form. These blocks commute and square to zero. Their sum therefore has zero $(m+1)$-st power, and its m-th power is $m!$ times their product. Against the base area, every base correction in that product vanishes. Adding κ' now gives (a) by the binomial formula. This is a Cartesian identity, including points at which a moment differential vanishes.

For (b), write $d(G\beta) = \partial_s G\, ds \wedge \beta + d_w G \wedge \beta$. Its mixed part has rank one: it is a fibre covector wedged with the single base covector β. Apply the same block factorization with a common base covector to obtain $n\omega_V^m \wedge (e\sigma_\Sigma + \partial_s G\, ds \wedge \beta)$. The displayed coefficient follows from $\beta = b(v)\,dv$. Positive coefficients make the top power nonzero. Thus a closed form of either displayed type is symplectic. □

4.2. **The first path.** Set

$$e_\Lambda = \min\left(\frac{\gamma_*}{32 A_{\mathrm{ref}}}, \frac{\kappa_-}{2}\right) > 0, \qquad \epsilon_0 = \frac{\gamma_*}{16}. \tag{11}$$

The model tail estimate and the reference-area bound give

$$\int_{\Sigma_\ell} \chi_\partial \kappa(p) \leq \frac{\gamma_*}{32}, \qquad e_\Lambda \int_{\Sigma_\ell} \Lambda \leq \frac{\gamma_*}{32} \qquad (p \in \Delta). \tag{12}$$

Apply the finite primitive operator $\mathcal{P}$ of §S12, a finite sum of rectangular integrals over the boxes corrected along the tree so that $d\mathcal{P}\omega = \omega$ for every two-form ω of total mass zero, to $(1 - \chi_\partial)(\kappa(p) - e_\Lambda \Lambda)$, then subtract its annular period and an exact annular gauge. The resulting primitive Γ_n is the formula in Construction S12.2. It has

$$\begin{aligned} &\kappa_{\mathrm{sm}}(p) = \chi_\partial \kappa(p) + (1 - \chi_\partial) e_\Lambda \Lambda, \qquad \epsilon_{\mathrm{sm}}(p) = \int_{\Sigma_\ell} \kappa_{\mathrm{sm}}(p), \\ &H^*(p) = H_{\mathrm{b}}(p) - \epsilon_{\mathrm{sm}}(p), \qquad d_\Sigma \Gamma_n(p) = \kappa_{\mathrm{sm}}(p) + H^*(p)\eta. \end{aligned} \tag{13}$$

The mass equality is proved by calibration with the root bump (Lemma S12.1); the construction itself uses the ratio of two finite coordinate integrals. Since $H^*(p) = \int_{\Sigma_\ell} (1 - \chi_\partial)(\kappa(p) - e_\Lambda \Lambda)$, the support of $1 - \chi_\partial$ lies in the correction region, and $\kappa(p) \geq \kappa_- \Lambda \geq 2 e_\Lambda \Lambda$ there, we have $H^* \geq 0$ on Δ; and by (12)

$$0 \leq \epsilon_{\mathrm{sm}} \leq \epsilon_0 \leq \gamma/16, \qquad H^* \geq H_{\mathrm{b}} - \gamma_*/16. \tag{14}$$

The background is positive, agrees with the physical curvature near the outer boundary, and equals $e_\Lambda \Lambda$ on the annulus. There the period and gauge corrections give $\Gamma_n = e_\Lambda \Gamma_{\mathrm{ref}} + f(s) H^*(p) \beta$ on an open neighbourhood of the working band. Define

$$\alpha_1 = \Gamma_n - \Gamma_{\mathrm{phys}}. \tag{15}$$

Proposition 4.3 (First path)**.** *The form $\Omega_\lambda = \Omega + \lambda d\alpha_1$, $0 \le \lambda \le 1$, is symplectic on a neighbourhood of the model region. Its base curvature is the positive convex combination of κ and $d_\Sigma\Gamma_n$, and on the annulus its endpoint is*

$$\Omega_1 = \omega_V + e_\Lambda\Lambda + d\big(f(s)H^*(p)\beta\big).$$

The variation primitive is horizontal and toric, is smooth in the moment variables, and vanishes near the outer base boundary.

Proof. The normalisation identities give the base curvature and the annular normal form. A toric mixed term does not affect the top power, so the positive convex combination is nondegenerate. The difference $\Gamma_n - \Gamma_{\text{phys}}$ has compact support in the interior of the base, which gives the boundary statement. All moment derivatives are derivatives of the explicit primitive formulas. □

4.3. **The twist.** The generators Q_t satisfy $\partial_t q_t = X_{Q_t} \circ q_t$. On a collar of each outer face the Q_t Poisson-commute with μ_ν, so μ_ν is constant along the trajectories of the isotopy there. The inverse isotopy q_t^{-1} is generated by $-Q_t \circ q_t$, which Poisson-commutes with μ_ν on a slightly smaller collar because q_t preserves μ_ν there. Hence neither q_t nor q_t^{-1} moves a point across an outer face: $q_t(V_\Delta) = V_\Delta$, $q_t(\operatorname{int} V_\Delta) = \operatorname{int} V_\Delta$, and q_t maps each outer face into itself. We write $q = q_1$.

Over the annulus, in the explicit product coordinates, the coefficient of β in Ω_1 rises from 0 to $H^*(p)$. The twist changes the fibre identification across the annulus from the identity to q. For $\lambda \in [0,1]$ define on $E|_{\mathcal{A}}$

(16)

$$\mathcal{D}_\lambda(s,v,w) = \big(s, v, q_{\lambda f(s)}(w)\big), \qquad Q_s^\lambda = \lambda f'(s)\, Q_{\lambda f(s)}, \qquad G_\lambda(s,w) = f(s)\, H^*\big(p(q_\lambda^{-1}(w))\big),$$

and $\mathcal{D} = \mathcal{D}_1$. Since $\partial_s\big(q_{\lambda f(s)}(w)\big) = \lambda f'(s)\,(\partial_t q_t)_{t=\lambda f(s)}(w)$, the function Q_s^λ generates the s-dependence: $\partial_s q_{\lambda f(s)} = X_{Q_s^\lambda} \circ q_{\lambda f(s)}$.

Lemma 4.4 (Suspension)**.** *On $E|_{\mathcal{A}}$, for $\lambda \in [0,1]$:*

(i) $\mathcal{D}_\lambda^*\omega_V = \omega_V + d\big[(Q_s^\lambda \circ q_{\lambda f(s)})\, ds\big]$;

(ii) with the horizontal one-form

$$\eta_\lambda = (Q_s^\lambda \circ q_{\lambda f(s)})\, ds + f(s)\Big[H^*\big(p(q_\lambda^{-1} q_{\lambda f(s)}(w))\big) - H^*\big(p(w)\big)\Big]\beta,$$

one has $\mathcal{D}_\lambda^*\big(\omega_V + e_\Lambda\Lambda + d(G_\lambda\beta)\big) = \omega_V + e_\Lambda\Lambda + d\big(f(s)H^*(p)\beta\big) + d\eta_\lambda$;

(iii) $\eta_0 = 0$, *η_λ is smooth in (λ, s, v, w), and $\eta_\lambda = 0$ for $s \notin (0,1)$. Hence η_λ extends by zero to a horizontal one-form on a neighbourhood of $V_\Delta(E)$, supported over $\mathcal{A}$, and $\Omega_1 + d\eta_\lambda = \mathcal{D}_\lambda^*\big(\omega_V + e_\Lambda\Lambda + d(G_\lambda\beta)\big)$ on $E|_{\mathcal{A}}$.*

Proof. (i) Put $\Theta(s,w) = Q_s^\lambda(q_{\lambda f(s)}(w))$, independent of v; the right side is $\omega_V + d_w\Theta \wedge ds$. The differential of $\mathcal{D}_\lambda$ fixes ∂_v, maps a vertical vector u to $Dq\,u$ ($q = q_{\lambda f(s)}$) and ∂_s to $\partial_s + X_{Q_s^\lambda}(q(w))$; the form ω_V sees only vertical components. On two vertical vectors both sides give $\omega_V(u_1,u_2)$, because q is symplectic. On (∂_s, u) the left side is $\omega_V(X_{Q_s^\lambda}, Dq\,u) = -dQ_s^\lambda(Dq\,u) = -d_w\Theta(u)$, which is also $(d_w\Theta \wedge ds)(\partial_s, u)$. Pairs containing ∂_v, and (∂_s, ∂_v), give zero on both sides. (ii) $\mathcal{D}_\lambda$ fixes base points, so it preserves Λ and β and $\mathcal{D}_\lambda^*(G_\lambda\beta) = (G_\lambda \circ \mathcal{D}_\lambda)\beta$, with $G_\lambda \circ \mathcal{D}_\lambda = f(s)H^*(p(q_\lambda^{-1}q_{\lambda f(s)}w))$; add (i). (iii) For $s \le 0$, $f = f' = 0$ and both terms vanish. For $s \ge 1$, $f = 1$, $f' = 0$ and $q_\lambda^{-1}q_{\lambda f(s)} = q_\lambda^{-1}q_\lambda = \mathrm{id}$, so the bracket vanishes exactly. The last identity is (ii) together with Proposition 4.3. □

Using the fixed map q_λ^{-1} in G_λ, rather than $q_{\lambda f(s)}^{-1}$, makes the bracket vanish identically for $s \ge 1$ and gives $\partial_s G_\lambda = f'(s)H^*(p(q_\lambda^{-1}w))$, with no derivative of the isotopy.

The second path. Put $\Omega_\lambda^{(1)} = \Omega_1 + d\eta_\lambda$. Off $E|_{\mathcal{A}}$ it equals Ω_1. On $E|_{\mathcal{A}}$ it is $\mathcal{D}_\lambda^*$ of a form whose top power is, by Lemma 4.2(b), $n(e_\Lambda a_\Lambda(s,v) + b\, f' H^*(p(q_\lambda^{-1}w)))\,\omega_V^m \wedge ds \wedge dv$. This is positive on a neighbourhood of $V_\Delta(E)$: the background $e_\Lambda\Lambda$ is positive and $H^* \geq 0$ at every fibre moment in Δ. Smoothness and compactness then give an open nondegenerate neighbourhood (Lemma S10.7); $\mathcal{D}_\lambda$ preserves orientation. So $\Omega_\lambda^{(1)}$, $\lambda \in [0,1]$, is an exact symplectic path (Proposition S10.9). The map $\mathcal{D}_\lambda$ itself is defined only over $\mathcal{A}$; only the forms are glued to E.

4.4. **Layers and the Moser map.** Fix source capacities $\rho_i^\sharp < a_i$ and put $\sigma_i = (a_i - \rho_i^\sharp)/2 > 0$. Define the smooth caps and the collar coefficient by

$$h_i(p) = (a_i - |p|)S\big((a_i - |p|)/\sigma_i\big) \quad (1 \leq i \leq k), \qquad J = K - H^* = K - H_{\mathrm{b}} + \epsilon_{\mathrm{sm}}. \tag{17}$$

Put $N = k+1$ and $D_s = 1/(3N+1)$. For $j = 1, \dots, N$ set $l_j = (3j-2)D_s$, $r_j = (3j-1)D_s$, $I_j = [l_j, r_j]$ and $\varsigma_j(s) = S\big(2(s - l_j - D_s/4)/D_s\big)$; the last index $j = N$ is denoted by $*$. The intervals are ordered and disjoint in $(0,1)$, and each clock is constant on collars of its interval endpoints, with $\varsigma_j' \geq 0$.

The constants enter below only through (14) and the residual bound $\gamma_{\mathrm{res}} = \gamma - \sup_\Delta \epsilon_{\mathrm{sm}} \geq 15\gamma/16$.

Proposition 4.5 (Layers)**.** *Under the hypotheses of Theorem 2.6, with ϵ_{sm} from* (13) *and h_i, J from* (17)*:*

(i) each h_i is smooth on $\mathbb{R}^m$ ($2n \geq 6$; for $n = 2$ see Remark 4.7), and $0 \leq h_i(p) \leq (a_i - |p|)_+$ and $(a_i + \rho_i^\sharp)/2 - |p| \leq h_i(p)$; it vanishes for $|p| \geq a_i$ and equals $a_i - |p|$ for $|p| \leq (a_i + \rho_i^\sharp)/2$;

(ii) the residual $R^*(w) = K\big(p(q^{-1}w)\big) - J\big(p(w)\big) - \sum_{i=1}^k h_i\big(p(w)\big)$ satisfies, for $w \in V_\Delta$,

$$R^*(w) \;\geq\; \gamma - \sup_\Delta \epsilon_{\mathrm{sm}} \;\geq\; \frac{15\gamma}{16} > 0;$$

(iii) the final coefficient $G_{\mathrm{f}}(s,w) = J(p(w)) + \sum_{i=1}^k \varsigma_i(s) h_i(p(w)) + \varsigma_*(s) R^*(w)$ satisfies $\partial_s G_{\mathrm{f}} \geq 0$ on $\mathcal{A} \times V_\Delta$; $G_{\mathrm{f}} = J \circ p$ for $s \leq l_1$; $G_{\mathrm{f}} = K \circ p \circ q^{-1}$ for $s \geq r_*$; and on $I_i \times V_\Delta$, $1 \leq i \leq k$,

$$G_{\mathrm{f}}(s,w) = A_i(p(w)) + \varsigma_i(s) h_i(p(w)), \qquad A_i = J + \sum_{j<i} h_j.$$

Proof. (i) Write $x = a_i - |p|$. For $x \leq 0$ the cap is zero. For $0 < x < \sigma_i$, the bounds $0 \leq S(x/\sigma_i) \leq 1$ give $0 \leq xS(x/\sigma_i) \leq x$ and $x - \sigma_i \leq 0 \leq h_i$. For $x \geq \sigma_i$ the cap equals x. Thus $x - \sigma_i \leq h_i \leq x_+$ in all cases. (ii) Let $w \in V_\Delta$ and $z = q^{-1}(w) \in V_\Delta$, so that $w = q(z)$. Using $J = K - H_{\mathrm{b}} + \epsilon_{\mathrm{sm}}$, the hypothesis $H_{\mathrm{b}} - \sum_i (a_i - |p|)_+ \geq P$ and (i),

$$\begin{aligned} R^*(w) &= K(p(z)) - K(p(qz)) + H_{\mathrm{b}}(p(qz)) - \epsilon_{\mathrm{sm}}(p(qz)) - \sum_i h_i(p(qz)) \\ &\geq K(p(z)) - (K - P)(p(qz)) - \sup_\Delta \epsilon_{\mathrm{sm}} \;\geq\; \gamma - \sup_\Delta \epsilon_{\mathrm{sm}}, \end{aligned}$$

by Definition 2.4 at the point z. Finally $\epsilon_{\mathrm{sm}} \leq \gamma/16$ by (14). (iii) $\partial_s G_{\mathrm{f}} = \sum_i \varsigma_i' h_i + \varsigma_*' R^* \geq 0$ by (i), and because $R^* > 0$ on V_Δ by (ii). For $s \leq l_1$ all ς_j vanish; for $s \geq r_*$ all equal one and $J + \sum_i h_i + R^* = K \circ p \circ q^{-1}$. On I_i, $\varsigma_j = 1$ for $j < i$ and $\varsigma_j = 0$ for $j > i$, including $j = *$. □

The comparison is used at the source point $z = q^{-1}(w)$: the twist transports the budget K of level $p(z)$ to the fibre point w where it is spent. The residual layer comes last, so all non-toric dependence on w is confined to I_* and the ball layers see only functions of p.

The third path. Keep $\mathcal{D}$ fixed and interpolate the coefficient from $G_1 = f(s)H^* \circ p \circ q^{-1}$ to G_{f}. On $E|_{\mathcal{A}}$ put $\xi_{\mathcal{A}} = ((G_{\mathrm{f}} - G_1) \circ \mathcal{D})\,\beta$. On the lower collar $s \leq 0$, $\mathcal{D} = \mathrm{id}$, $G_1 = 0$ and $G_{\mathrm{f}} = J \circ p$;

on the upper collar $s \geq 1$, $G_1 \circ \mathcal{D} = H^* \circ p$ and $G_{\mathrm{f}} \circ \mathcal{D} = K \circ p \circ q^{-1} \circ q = K \circ p$. So $\xi_{\mathcal{A}} = J(p)\beta$ on both collars, and

$$\xi = J(p)\,\mathrm{pr}^*\beta + \mathrm{ext}_0\big(\xi_{\mathcal{A}} - J(p)\beta\big), \tag{18}$$

where ext_0 denotes extension by zero (the form $\xi_{\mathcal{A}} - J(p)\beta$ vanishes over both collars of $\mathcal{A}$), is a horizontal one-form on a neighbourhood of $V_\Delta(E)$; the first term is a global toric form, since p and β are. Put $\Omega_\tau^{(2)} = \Omega_1^{(1)} + \tau\, d\xi$ and $\Omega_{\mathrm{final}} = \Omega_1^{(2)}$. On $E|_{\mathcal{A}}$,

$$\begin{aligned} \Omega_\tau^{(2)} &= \mathcal{D}^*\big(\omega_V + e_\Lambda \Lambda + d(\hat{G}_\tau \beta)\big), \qquad \hat{G}_\tau = (1-\tau)G_1 + \tau G_{\mathrm{f}}, \\ \Omega_{\mathrm{final}} &= \mathcal{D}^*\Omega_{\mathrm{pre}}, \qquad\qquad\qquad\quad \Omega_{\mathrm{pre}} = \omega_V + e_\Lambda \Lambda + d(G_{\mathrm{f}}\beta); \end{aligned} \tag{19}$$

off $E|_{\mathcal{A}}$, $\Omega_\tau^{(2)} = \Omega_1 + \tau\, dJ \wedge \beta$ differs from Ω_1 by a toric mixed term. By Lemma 4.2, Proposition 4.5(iii) and Lemma S10.7, every $\Omega_\tau^{(2)}$ is symplectic on a neighbourhood of $V_\Delta(E)$ (Proposition S10.10). Two different toric coefficients at the two ends of $\mathcal{A}$ would have to be interpolated across the rest of Σ, changing its area; the single form $J(p)\mathrm{pr}^*\beta$ adds only the mixed term $dJ\wedge\beta$, which does not enter the top power.

For each of the three paths write its primitive as $\lambda_{j,t} = \lambda_V + Q_{j,t}$ and its variation as $\nu_{j,t} = \partial_t Q_{j,t}$; these are α_1, $\partial_t \eta_t$ and ξ, respectively. In the larger finite atlas use the whole-path Gram cutoff χ_j of Construction S12.4. It is one near $[0,1] \times V_\Delta(E)$ and has compact spatial support in the nondegenerate locus. Define the cut field by $\iota_{Y_{j,t}} d\lambda_{j,t} = -\chi_j \nu_{j,t}$, computed in the charts of the finite atlas, and let Φ_j be its time-one map. Put $\Phi = \Phi_2 \circ \Phi_1 \circ \Phi_0$ and use the abbreviation $\Phi^{-1} = \Phi_0^{-1} \circ \Phi_1^{-1} \circ \Phi_2^{-1}$.

Proposition 4.6 (Moser maps)**.** *The three cut fields are smooth with common compact spatial support. Their time-one maps are diffeomorphisms, preserve the model region and its interior in both directions, and satisfy $\Phi_j^* \Omega_{j,1} = \Omega_{j,0}$ there. Consequently*

$$\Phi^* \Omega_{\mathrm{final}} = \Omega.$$

Proof. Near each outer face the variation primitives annihilate the Hamiltonian field of the face moment. Every path form is invariant under the flow of that moment there: the toric terms are, and the comparator isotopy commutes with it because Q_t Poisson-commutes with μ_ν on the collar (Definition 2.4(1)), so $\iota_{X_{\mu_\nu}} \Omega_{j,t} = -d\mu_\nu$ on the collar. The comparator's face-collar condition therefore gives $d\mu_\nu(Y_{j,t}) = 0$ whenever that moment is near its boundary value. A trajectory cannot cross the corresponding level in either direction; this argument applies to all faces, including their intersections. The variation primitives vanish near the outer base boundary. The fields have compact support, so their flows are complete and their inverse maps are the backward flows. On the model region $\chi_j = 1$, and Cartan's identity gives

$$\frac{d}{dt} \Phi_{j,t}^* \Omega_{j,t} = \Phi_{j,t}^* d\big(\iota_{Y_{j,t}} \Omega_{j,t} + \nu_{j,t}\big) = 0.$$

The finite coordinate ODEs and their inverse endpoints are exactly the maps used in the construction. □

Layer coordinates. On the layer $\Lambda_i = \{l_i < s < r_i,\ v_- < v < v_+\} \times \mathrm{int}\, V_\Delta$ (product coordinates) let $T(v) = \int_{v_-}^{v} b$, a diffeomorphism $(v_-, v_+) \to (0,1)$ with $dT = \beta$, and

$$\ell(s,T) = \frac{e_\Lambda a_\Lambda(s, v(T))}{b(v(T))}, \qquad x_i(s,T,p) = \int_{l_i}^{s} \ell(u,T)\, du + \varsigma_i(s) h_i(p). \tag{20}$$

Then $\partial_s x_i = \ell + \varsigma_i' h_i > 0$, $x_i = 0$ at $s = l_i$ and $x_i > h_i(p)$ at $s = r_i$, so for $0 < x < h_i(p)$ and $0 < T < 1$ there is a unique root $s = s_i(x,T,p) \in (l_i, r_i)$, smooth by the implicit function theorem.

Since $e_\Lambda\Lambda = \ell(s,T)\,ds\wedge dT$ and, by Proposition 4.5(iii), $d(G_{\mathrm f}\beta) = (\varsigma_i' h_i\,ds + dA_i + \varsigma_i\,dh_i)\wedge dT$ on Λ_i,

$$\Omega_{\mathrm{pre}} = \omega_V + dx_i\wedge dT + dA_i(p)\wedge dT \qquad \text{on } \Lambda_i. \tag{21}$$

Two explicit maps finish the layer. The *multi-angle shear* Sh_A, which rotates each fibre coordinate w_j by the angle $-2\pi T\partial_j A(p(w))$, removes the term $dA_i(p)\wedge dT$ from (21) (Lemma S10.31). The *disc-to-strip map* $\Psi_\delta\colon \mathbb{C}\to(0,\infty)\times(0,1)$ is an explicit symplectomorphism onto its image with $0<x<\pi|z|^2+\delta/2$ (Proposition S10.32); the excess $\delta/2$ is unavoidable, since a closed disc of area a does not embed area-preservingly in $\{0<x<a\}\times(0,1)$, and it is paid once for all concentric discs.

4.5. **Proof of Theorem 2.6.** For a higher-dimensional input, let the model, handle data, capacities a_i, function P and comparison datum (K,q) with gap γ be as in Theorem 2.6, with the bounds (12), and let $\rho_i<a_i$. Write $\mathbb{C}^n=\mathbb{C}\times\mathbb{C}^m\ni(z_{\mathrm b},z_{\mathrm f})$, $\varpi=\pi|z_{\mathrm b}|^2$, $p=p(z_{\mathrm f})$, so that $\omega_0=\omega_{\mathbb{C}}(z_{\mathrm b})+\omega_V(z_{\mathrm f})$ and $B(c)=\{\varpi+|p|\le c\}$.

Parameters. Put

$$\hat\rho_i=\frac{3a_i+\rho_i}{4},\qquad \varepsilon_i=\frac{a_i-\hat\rho_i}{4},$$

so that $\rho_i<\hat\rho_i<a_i$ and $B(\rho_i)\subset B^\circ(\hat\rho_i)$; the open ball $B^\circ(\hat\rho_i)$ is the domain of the embedding. In (17) take $\rho_i^\sharp=\hat\rho_i$; these are the constants of equation S57.

Forms and Moser map. Construct c, κ_{sm}, α_1, the paths Ω_λ, $\Omega_\lambda^{(1)}$, $\Omega_\tau^{(2)}$, the twist $\mathcal D$, the coefficient $G_{\mathrm f}$ and the Moser map Φ as above. By Proposition 4.6, $\Phi^*\Omega_{\mathrm{final}}=\Omega$, and by (19) $\Omega_{\mathrm{final}}=\mathcal D^*\Omega_{\mathrm{pre}}$ on $E|_{\mathcal A}$, where $\mathcal D$ is a diffeomorphism of $\mathcal A\times V_\Delta$ fixing base points, with inverse $(s,v,w)\mapsto(s,v,q_{f(s)}^{-1}(w))$ preserving $\operatorname{int}V_\Delta$.

Layer maps. For $(z_{\mathrm b},z_{\mathrm f})\in B^\circ(\hat\rho_i)$ let $(x,T)=\Psi_{\varepsilon_i}(z_{\mathrm b})$ (Proposition S10.32). Then $|p|<\hat\rho_i<a_i$, so $z_{\mathrm f}\in\operatorname{int}V_\Delta$ because $\{|p|\le a_i\}\subset\tau_\Delta\Delta$, and

$$0<x<\varpi+\frac{\varepsilon_i}{2}<\hat\rho_i-|p|+\frac{a_i-\hat\rho_i}{8}<\frac{a_i+\hat\rho_i}{2}-|p|\le h_i(p)$$

by Proposition 4.5(i). Define

$$\mathcal E_i(z_{\mathrm b},z_{\mathrm f})=\Big(s_i\big(x,T,p\big),\ v(T),\ \mathrm{Sh}_{A_i,T}(z_{\mathrm f})\Big)\in\Lambda_i,$$

with the root s_i of (20) and the shear of Lemma S10.31 ($\mathrm{Sh}_{A_i,T}$ preserves p). $\mathcal E_i$ is the composite of $\Psi_{\varepsilon_i}\times\mathrm{id}$, the shear Sh_{A_i} and the inverse of the layer coordinates $(s,v,w)\mapsto(x_i,T,w)$, each a diffeomorphism onto an open set; by (21), Lemma S10.31 and Proposition S10.32,

$$\mathcal E_i^*\Omega_{\mathrm{pre}}=(\Psi_{\varepsilon_i}\times\mathrm{id})^*\big(\omega_V+dx\wedge dT\big)=\omega_{\mathbb{C}}+\omega_V=\omega_0.$$

Put

$$\psi_i=\Phi_0^{-1}\circ\Phi_1^{-1}\circ\Phi_2^{-1}\circ\mathcal D^{-1}\circ\mathcal E_i=\Phi^{-1}\circ\mathcal D^{-1}\circ\mathcal E_i\colon B^\circ(\hat\rho_i)\to E.$$

In Construction S12.4 the map $\mathcal D^{-1}\circ\mathcal E_i$ is written as a composite whose factors are the planar map, the unique inverse layer chart, the inverse angular clock, the annular covering map and the inverse comparator shear. The endpoint factors are, in order, the inverse flows of $Q_0=(1-t)\Gamma_{\mathrm{phys}}+t\Gamma_n$, $Q_1=\Gamma_n+\eta_t$ and $Q_2=\Gamma_n+\eta_1+t\xi$. Thus ψ_i is the map of Construction S12.4 with $\rho_i^\sharp=\hat\rho_i=(3a_i+\rho_i)/4$.

Conclusions. (a) $\psi_i^*\Omega=\mathcal E_i^*(\mathcal D^{-1})^*(\Phi^{-1})^*\Omega=\mathcal E_i^*(\mathcal D^{-1})^*\Omega_{\mathrm{final}}=\mathcal E_i^*\Omega_{\mathrm{pre}}=\omega_0$, and ψ_i is an injective immersion of an open set, hence a smooth symplectic embedding of the neighbourhood $B^\circ(\hat\rho_i)$ of $B(\rho_i)$. (b) The image of $\mathcal E_i$ lies in $\{l_i<s<r_i\}\times\operatorname{int}V_\Delta$, and $\mathcal D^{-1}$, Φ^{-1} map $\operatorname{int}V_\Delta(E)$ into itself (Proposition 4.6), so ψ_i takes values in $\operatorname{int}V_\Delta(E)$. (c) The intervals (l_i,r_i)

are pairwise disjoint, so the images of the $\mathcal{E}_i$ are pairwise disjoint. The maps $\mathcal{D}^{-1}$ and Φ^{-1} are common injective maps for all i, and therefore preserve this disjointness.

Every ingredient is given by a formula, a fixed linear operator, a definite integral, the unique solution of an explicit ODE (Φ) or a unique root (s_i), so ψ_i is explicit in the sense of Definition 1.3. The verification of these maps is Theorem S10.12, with the strip and layer estimates in Proposition S11.15 and Proposition S11.17. □

The source margin $a_i - \rho_i$ pays for the strip excess and for an open neighbourhood of each given closed ball. For $2n \geq 6$ the caps lie below the original tents and their deficit is retained in the final residual layer, while the background and collar consume at most $\gamma_*/16$; in dimension four the caps lie above the tents by a total of at most $g/32$, which the residual budget absorbs (Lemma S10.15).

Remark 4.7 (Dimension four). For $n = 2$ the base is the closed torus of Section S8, with the handle data of Proposition S8.3; there is no collar and no atlas, and the construction is carried out on the cover $\mathbb{R}^2 \times \mathbb{C}$, equivariantly under the deck maps (§S10.9). The background is $e_h\nu$ with $e_h = g/16$, $\nu = du \wedge dv$, and the first path is $\Omega_0 + t\, d\alpha_1$ with the closed-form primitive α_1 of Construction S9.23. The twist uses the clock $f(u) = S((u - \frac{7}{48})/\frac{1}{24})$, which is $S(2s - \frac{1}{2})$ in the handle coordinate $s = 12u - \frac{3}{2}$, and the $k + 1$ layer intervals lie inside its ramp. The caps are $h_i(p) = \frac{1}{2}\big(a_i - p + \sqrt{(a_i - p)^2 + \mu_h^2}\big)$ with $\mu_h = g/(16k)$; they lie above the tents and exceed them in total by at most $g/32$, so a comparison gap $\frac{31}{32}g$ leaves a residual of at least $\frac{7}{8}g$. The three stages are concatenated into one flow, cut off only in the moment variable, on $\{p < h + 2\delta_m\}$, $\delta_m = \min((d_n - h)/4, g/100)$. All three variation primitives vanish as covectors on the marked fibres, so the flow fixes them pointwise and the images avoid $\{Z_0 = 0\}$ (Lemma S10.26, Propositions S10.28 and S10.29).

5. The four models

Theorem 2.6 reduces the packing problem to three inputs: a toric model with handle data, a comparison datum for its profile, and a transfer to the target. This section describes the four models of Table 2 and states the packing theorem each one gives. The definitions, the handle data, the transfers and the proofs are in §S8 (dimension four), §S14 (dimension $2n \geq 6$, capacities below one half) and §S16 (one large ball). Throughout, the target is $B^\circ(1)$ and $\rho_i = c_i/C$.

5.1. **Dimension four.** The model is the normal disc bundle associated with Biran's decomposition of $\mathbb{CP}^2$ along a smooth cubic [Bir01], written on the cover $\mathbb{R}^2 \times \mathbb{C}$ of the Fermat torus $\Gamma = \mathbb{C}/(\mathbb{Z} + \zeta\mathbb{Z})$, $\zeta = e^{2\pi i/3}$. Theta functions give a smooth map $Z\colon \mathbb{R}^2 \to \mathbb{C}^3$ with $|Z| = 1$ and $Z_0^3 + Z_1^3 + Z_2^3 = 0$, periodic up to phase. With $\eta_\Gamma = \frac{1}{2\pi} \operatorname{Im} \sum_j \overline{Z_j}\, dZ_j$, $\sigma = d\eta_\Gamma$ and the fibre moment $p = \pi|w|^2$, the model form is

$$\Omega_0 = \omega_V + (1 - 3p)\,\sigma - 3\, dp \wedge \eta_\Gamma, \qquad \int_\Gamma \sigma = 3,$$

so the base area at level p is $3(1 - 3p)$ and the fibre interval must stop below $p = \frac{1}{3}$; this is the source of the condition $3c_i < C$. We construct the transfer on truncated subbundles and do not identify its image with the complement of a skeleton. The integrated profile over $[0, h]$ is $3h - \frac{9}{2}h^2$, which tends to $\frac{1}{2} = \operatorname{vol} \mathbb{CP}^2$ as $h \uparrow \frac{1}{3}$ (Lemma S8.2). The transfer is an explicit unit lift into the Fermat cone followed by the affine ball chart of the complement of a line; a cube-root symmetry and injectivity modulo phase make the composite well defined on the deck quotient (Theorem S13.21).

Theorem 5.1 (Packing in dimension four)**.** *Let $k \geq 1$ and $0 < \rho_1, \dots, \rho_k < \frac{1}{3}$ with $\sum_i \rho_i^2 < 1$. Then there are explicit numbers $\rho_i < \rho_i^+ < \rho_i^{\mathrm{dom}}$ and smooth maps $f_i \colon B^\circ(\rho_i^{\mathrm{dom}}) \to \mathbb{C}^2$ with symplectic derivative, injective on $B(\rho_i^+)$, such that the sets $f_i(B(\rho_i^+))$ are pairwise disjoint and contained in $B(1-\mu_{\mathrm{ball}})$ for some $\mu_{\mathrm{ball}} \in (0,1)$.*

This is Theorem S13.23. For $k \geq 10$ equal capacities with $\frac{3}{4} < \rho\sqrt{k} < 1$ the comparison datum is the linear datum of Proposition 3.8; the auxiliary capacity is $a = t_0/\sqrt{k}$ and the height is $h = t_0/3$, with $t_0 = (1+\rho\sqrt{k})/2$, so all parameters are algebraic in $\sqrt{k}$ and ρ (Remark S13.25). For all other inputs, including $k = 9$ and unequal capacities, the comparison datum is the scalar comparator of §S6, and the auxiliary capacities $a_i \in (\rho_i, \frac{1}{3})$ and the height $h < \frac{1}{3}$ are the closed formulas in ρ of Lemma S8.5.

5.2. **The four-dimensional argument in outline.** In both branches the comparison datum and the auxiliary numbers form valid cubic data (Definition S10.13), with $d_n = \frac{1}{2}(h + \frac{1}{3})$, and the embedding of the i-th ball is the composite

$$f_i = b_{\mathrm{ball}} \circ \psi_1^{-1} \circ \mathrm{pr} \circ \tau_1 \circ \iota \circ \mathsf{N} \circ \Phi^{-1} \circ \chi_{\mathcal{A}} \circ \mathcal{D}^{-1} \circ E_i \qquad \text{on } B^\circ(\hat\rho_i). \tag{22}$$

Read from the right: E_i places the ball in the i-th layer of the handle (disc-to-strip map, shear, inverse layer coordinates); $\mathcal{D}^{-1}$ undoes the twist, which applies the comparison isotopy $q_{f(u)}$ fibrewise over the annulus $\mathcal{A}$; $\chi_{\mathcal{A}}$ is the frame chart; Φ^{-1} is the inverse handle Moser map, from the final form back to Ω_0; N passes from the torus cover to unit Fermat representatives; ι, τ_1, pr, ψ_1^{-1} are the weight-three injection, the transport from the normal cone X_0 to the graph X_1, the projection to $\mathbb{CP}^2$ and the inverse Moser return to ω_{FS} (Construction S8.4); b_{ball} is the affine chart $\mathbb{CP}^2 \setminus \{Z_0 = 0\} \to B^\circ(1)$. With $\bar\rho_i = \frac{1}{2}(\rho_i + a_i)$, $\delta_i = a_i - \bar\rho_i$ and $\hat\rho_i = a_i - \frac{3}{4}\delta_i$,

$$\rho_i < \rho_i^+ \leq \bar\rho_i < \hat\rho_i < a_i < h < d_n < \tfrac{1}{3}.$$

Three points of the proof are not visible in Theorem 5.1; the margins they use are collected in Table 4.

(i) Where the vector fields are defined. Four flows occur. Each is the flow of a field that is smooth on the whole ambient space and either compactly supported or of bounded speed there, and that agrees with the geometric field on a set which the relevant trajectories never leave. Completeness is therefore automatic, and the pullback identities are proved on that set.

- The comparison isotopy q_t has a smooth Hamiltonian on $\mathbb{C}$ that vanishes for $t \notin (0,1)$ and outside the disc $D(P_{\mathrm{supp}})$, with $P_{\mathrm{supp}} < h$ (Theorem S5.34, Proposition S6.8).
- The handle Moser field Y_t, defined by $\iota_{Y_t}\Omega_t^{\mathrm{M}} = -\mathfrak{v}_t$, needs nondegeneracy of Ω_t^{M}. By the top-power identity $\Omega_\lambda^2 = 2\,\omega_V \wedge \kappa_\lambda$ this reduces to positivity of the base coefficient, and all stage forms are nondegenerate on $U = \{p < h + 2\delta_m\}$, $\delta_m = \min((d_n - h)/4, g/100)$, where $H^* > \frac{15}{16}g - 18\delta_m > 0$ (Lemmas S9.28 and S10.15(d), Propositions S10.21 and S10.25). The field is multiplied by a cutoff $\mathfrak{c}(p)$, equal to one for $p \leq h + \frac{3}{2}\delta_m$ and to zero for $p \geq h + \frac{7}{4}\delta_m$. Above P_{supp} the cut field preserves p, because $\iota_R\Omega_t^{\mathrm{M}} = -dp$ and $\mathfrak{v}_t(R) = 0$ for the fibre rotation R; its base components are deck invariant, hence bounded by their maximum C_b over a fundamental domain. Trajectories stay in compact sets, and on $\{p \leq h + \delta_m\}$ the flow is that of the uncut field (Proposition S10.28).
- The transport field (equation S72) has denominators $A\,N_{\mathrm{den}}$ and $B\,N_{\mathrm{den}}$ with $A = 1 - 3p_W > 1 - 3d_n > 0$, $B > 0$ and $N_{\mathrm{den}} = G/A + T^2/B$. On the set $\mathcal{C}_*$ of points of X_T, $0 \leq T \leq 1$, with $|W|^2 \leq w_* = h/(d_n - h)$, one has $N_{\mathrm{den}} \geq c_* = \min(\frac{3}{2}, T_0^2/d_n)$, $T_0 = \min\bigl(1, (\sqrt{6}\max(1, \sqrt{w_*}))^{-1}\bigr)$: at $T = 0$ the Fermat equation gives $G \geq 3$, for $T \leq T_0$ the constraint $|\mathcal{F}(Z)| = T|W|$ keeps $G \geq \frac{3}{2}$, and for $T \geq T_0$ the term T^2/B suffices (Lemma S13.9). The field is cut off where $N_{\mathrm{den}} \leq c_*/2$ and outside a compact set.

Along the uncut field $|Z|$ and $\mathcal{F}(Z) - TW$ are constant and $|W|^2$ does not increase, and the injection maps $\{p \le h\}$ into $\{|W|^2 \le w_*\}$; so trajectories that start on $X_0 \cap \{p \le h\}$ stay in $\mathcal{C}_*$ and never meet the cutoff (Lemma S13.5, Proposition S13.10).

- The return field has denominator $(1+\mathfrak{u})(A_s + B_s G)$, and $A_s + B_s G \ge 1 - \frac{3}{2} d_n > \frac{1}{2}$ on the unit sphere for $s \in [0,1]$, since $\mathfrak{u} \le 1$ (Lemma S13.12); its cutoff at half this bound is inactive (Proposition S13.15).

The other factors are closed formulas: ι needs $p < d_n$; the layer root exists because the layer coordinate x_i is strictly increasing in u with values filling $(0, h_i(p))$; b_{ball} needs $Z_0 \neq 0$, which is (ii).

(ii) Injectivity and the deleted set. The model uses only the part $\{p < h\}$ of the disc bundle and the transfer is defined on $\{p < d_n\}$, so the isotropic skeleton of Biran's decomposition never enters. The set to be avoided is the line $L = \{Z_0 = 0\}$, which is not contained in the affine chart. It meets the cubic in the three points b_j, over which lie the marked fibres $\widetilde{\mathcal{F}}$ of the cover. The argument has four steps.

(a) The layer maps E_i are injective with images over the pairwise disjoint intervals $l_i < u < r_i$, and $\chi_{\mathcal{A}} \mathcal{D}^{-1}$ and Φ^{-1} are diffeomorphisms common to all i; so the maps $f_i^N = \Phi^{-1} \chi_{\mathcal{A}} \mathcal{D}^{-1} E_i$ are injective with disjoint images (Proposition S10.29(b)).

(b) The points $\chi_{\mathcal{A}} \mathcal{D}^{-1} E_i(X)$ lie over the square $(\frac{1}{8}, \frac{5}{24})^2$, which meets none of its integral translates, and Φ^{-1} commutes with the deck maps; so no two distinct image points are deck equivalent (Proposition S10.29(c)).

(c) The layers lie over $\mathcal{A}$, which misses the b_j. All three variation primitives vanish as covectors on $\widetilde{\mathcal{F}}$ (Lemma S10.26), so Φ fixes $\widetilde{\mathcal{F}}$ pointwise and the bijection Φ^{-1} maps its complement into itself: f_i^N avoids $\widetilde{\mathcal{F}}$, that is, $Z_0 \neq 0$ on its image.

(d) Every factor of $\mathcal{T}_{\mathrm{u}} = \psi_1^{-1} \circ \mathrm{pr} \circ \tau_1 \circ \iota$ is injective on representatives and gauge equivariant, so $\mathcal{T}_{\mathrm{u}}$ is injective modulo gauge (Lemma S13.18); every factor commutes with the symmetry $\mathsf{r}\colon Z_0 \mapsto \zeta Z_0$ (Lemma S13.17). An output with $Y_0 = 0$ is fixed by r, so its input is fixed modulo gauge, which forces $Z_0 = 0$ (Lemma S13.16). Hence inputs with $Z_0 \neq 0$ have outputs off L (Proposition S13.19), and, by injectivity of the Weierstrass parametrisation modulo the lattice, two points of the cover have the same image under $b_{\mathrm{ball}} \circ \mathcal{T}_{\mathrm{u}} \circ \mathsf{N}$ exactly when they are deck equivalent (Theorem S13.21).

By (a)–(d), the maps f_i are injective with pairwise disjoint images. The avoidance of L is a consequence of symmetry and injectivity, not of an estimate.

(iii) Closed balls. No area-preserving map takes a closed disc of area a into the strip $\{0 < x < a\} \times (0,1)$, and the disc-to-strip map Ψ_δ has excess $\frac{\delta}{2}$ (Proposition S10.32). The domain is therefore the open ball $B^\circ(\hat{\rho}_i)$, on which $0 < x < \hat{\rho}_i - p + \frac{\delta_i}{2} = a_i - p - \frac{\delta_i}{4} < h_i(p)$, so that E_i is defined on all of it; it contains the closed ball $B(\bar{\rho}_i) \supset B(\rho_i^+)$. The images lie in $\{p < a_i\} \subset \{p < h\}$, and the Moser identity holds on the larger closed set $\{p \le h + \delta_m\}$, so nothing happens at the face $p = h$. The image $f_i(B(\rho_i^+))$ is compact in $B^\circ(1)$ and hence lies in $B(1 - \mu_{\mathrm{ball}})$, $\mu_{\mathrm{ball}} = \frac{1}{2} \min_{\mathcal{K}} (1 - \pi |z|^2) > 0$ with $\mathcal{K} = \bigcup_i f_i(B(\rho_i^+))$ (Construction S13.22). In Section 6 the enlarged closed capacity ρ_i^+ and the open capacity ρ_i^{dom} give the radii $r_i < r_{1,i} < r_{2,i} < r_{3,i} < r_{4,i} < r_i' < \bar{r}_i$ of Definition 6.4: the Hamiltonian field equals the velocity of the isotopy on $\mathbb{B}^\circ(r_{1,i}) \supset \mathbb{B}(r_i)$, the cutoff acts between $r_{1,i}$ and $r_{2,i}$, disjointness is kept on the closed balls $\mathbb{B}(r_{3,i})$, and every stage is smooth on $\mathbb{B}^\circ(r_{4,i})$ (Proposition 6.6, Lemma 6.12). So each closed source ball $B(s_i, c_i)$ is carried exactly by the isotopy, onto $f_i(B(c_i)) \subset f_i(B(c_i'))$, a compact subset of $B^\circ(C)$; the room left between this set and $\partial B(C)$ is what the finite form of §7.1 uses.

Margin	Definition; dependence	Used for	Positive by
volume defect	$\eta_\star = 1-\rho\sqrt{k}$ (equal); $D = 1-\sum_i \rho_i^2$ (general); input	equal: $\eta_\star = (1-t_0)+\sqrt{k}(a-\rho)$, $\frac{\eta_\star}{2}$ each; general: the slacks δ, ς	hypothesis
capacity slack $a_i-\rho_i$	$\frac{\eta_\star}{2\sqrt{k}}$ (equal); $\min(\frac{\varpi}{8}, \frac{D}{8(S_\rho+k)}, \frac{1}{8})$ (general)	half to $\bar\rho_i$; strip excess $\frac{\delta_i}{2}$; $\frac{\delta_i}{4}$ left below h_i	Lem. S8.5, Rem. S13.25
ball enlargement $\rho_i<\rho_i^+<\rho_i^{\mathrm{dom}}$	$(\rho r_3^2, \rho r_4^2)$ (equal), $(\bar\rho_i, \hat\rho_i)$ (general); ρ, a_i	closed source $B(\rho_i^+)$, open domain $B^\circ(\rho_i^{\mathrm{dom}})$	Rem. S13.25, Prop. S10.29
fibre height $1-3h$	$1-t_0=\frac{\eta_\star}{2}$ (equal); 3ς (general)	$g\le H_{\mathrm{b}}(h)=3(1-3h)$; transfer margins	Lem. S8.5
comparison gap $\gamma=\frac{31}{32}g$	$g=3(1-t_0)$, planar cost $4c_se_q\le\frac{g}{32}$ (equal); $g=\frac34 d$ (general)	background $e_h=\frac{g}{16}$, cap excess $\le\frac{g}{32}$; residual $\ge\frac78 g$	Prop. 3.8; Prop. S6.8, Lem. S10.15
planar support $h-P_{\mathrm{supp}}$	isotopy data of Def. S4.11; equal case: $P_{\mathrm{supp}}>h-\frac{31}{8}e_q$	all flows preserve p above P_{supp}	Lem. S4.12, positivity
safe region $\{p<h+2\delta_m\}$	$\delta_m=\min(\frac{d_n-h}{4}, \frac{g}{100})$; cutoff on $[h+\frac32\delta_m, h+\frac74\delta_m]$	$H^*>\frac{15}{16}g-18\delta_m$; Moser identity on $\{p\le h+\delta_m\}$	Lem. S10.15(d)
base speed C_b	maximum over a compact fundamental set	completeness of the handle flow	positivity only
transfer margins	$d_n-h=\frac12(\frac13-h)$, $1-3d_n=\frac12(1-3h)$; h	ι needs $p<d_n$; $A>1-3d_n$; $w_*=\frac{h}{d_n-h}$	explicit
transport denominator	$N_{\mathrm{den}}\ge c_*=\min(\frac32, \frac{T_0^2}{d_n})$; d_n, w_*	cutoff at $\frac{c_*}{2}$, inactive on $\mathcal{C}_*$	Lem. S13.9
return denominator	$A_s+B_sG\ge 1-\frac32 d_n>\frac12$; d_n	cutoff at half the bound, inactive	Lem. S13.12
target boundary μ_{ball}	$\frac12\min_{\mathcal{K}}(1-\pi\lvert z\rvert^2)=\frac12\min\lvert Y_0\rvert^2$	passed on as compactness of $\bigcup_i f_i(B(c_i'))\subset B^\circ(C)$	Prop. S13.19, positivity only
radii $r_i<r_{1,i}<\cdots<r_{4,i}<r_i'$	$r_{\ell,i}=r_i+\frac{\ell}{5}\min(\frac14\text{centre gap}, r_i'-r_i)$	cutoff collar, disjoint closed balls, open domain	Lem. 6.5
isotopy scale τ	$\min(\frac12, \frac{\delta_c}{2Lr_{\mathrm{tot}}})$; L a maximum over compact sets	separation $\lvert Y_i-Y_j\rvert\ge 4Lr_{\mathrm{tot}}$	Lem. S21.5, positivity only
compilation b_R	$0<b_R<\frac14(R_C-R_*)$, $R_*=\sqrt{C_0/\pi}$, $C_0<C$ by compactness	Grönwall drift $<b_R$, leaving $R_*+b_R<R_C$	Thm. 7.2; C_0 positivity only

TABLE 4. Margins of the four-dimensional construction and of the common steps. "Explicit" means that the margin is a closed formula in the inputs; "positivity only" means that the proof shows the number is positive without bounding it.

A worked instance. §S22 substitutes $k=10$ and $\varepsilon=\frac{1}{10}$ into Theorem A: $R=(k/(1-\varepsilon/4))^{1/4}$, $C=\pi R^2$, $c_i=\pi$, $s_i=(3i-3,0,0,0)$. The normalised capacity is $\rho=\sqrt{39}/20$, with $\rho\sqrt{k}=\sqrt{39/40}\in(\frac34,1)$, so the equal branch with the linear datum is taken. The supplement lists t_0, a, h, d_n, the comparison function $K(p)=(3\sqrt{10}-9)p$ and its gap, the enlarged capacities, the handle and strip constants and the margins of Table 4 for these data, and describes the composition of the resulting Hamiltonian, by substituting the parameters into the closed formulas.

5.3. **Dimension $2n\ge 6$, capacities below one half.** Let $n\ge 3$ and $m=n-1$. The base is a compact piece of the diagonal affine curve

$$\Sigma=\{(t,y)\in\mathbb{C}^{m+1}:\ y_j^2=t^2-j,\ 1\le j\le m\},\qquad \Sigma_\ell=\Sigma\cap\{\lvert t\rvert\le R_\ell\},$$

with $\lambda = d^c \log(1+|z|^2)$, $\sigma = d\lambda$ and $R_\ell = 2 + \sum_j j + \ell$. The model is $\Sigma_\ell \times \mathbb{C}^m$ with $\Omega_0 = \omega_V + d[(1-2|p|)\lambda]$, $|p| = \sum_j \pi |U_j|^2$, so a ball of capacity a needs $a < \frac{1}{2}$. The base area A_ℓ satisfies $0 \le 2^m - A_\ell \le 2^m(m+2)/(\ell+2)$, and the limiting profile $2^m(1-2|p|)$ has integral $1/n!$ over $\{|p| \le \frac{1}{2}\}$, the volume of $B^\circ(1)$ (Lemmas S14.2 and S14.5). The model is an explicit instance of the complete-intersection model of [Ope26, §5.2], with a fixed diagonal curve in place of a generic one. The truncation loss is kept in the comparison profile. The transfer is one map into $B^\circ(1)$: a radial injection, a horizontal transport, an inverse Moser return and an affine ball chart (Theorem S15.12).

Theorem 5.2 (Packing by small balls in dimension $2n \ge 6$)**.** *Let $n \ge 3$ and $0 < \rho_1, \dots, \rho_k < \frac{1}{2}$ with $\sum_i \rho_i^n < 1$, and let $a_1, \dots, a_k$ be real numbers with $\rho_i < a_i < \frac{1}{2}$ and $\sum_i a_i^n < 1$, and put $\rho_i^{\mathrm{dom}} = (\rho_i + 2a_i)/3$. The comparator construction of Theorem 3.5 defines, from these data, smooth symplectic embeddings f_i of open neighbourhoods of the closed balls $B(\rho_i^{\mathrm{dom}}) \subset \mathbb{C}^n$ into $B^\circ(1)$ and a rational $\zeta > 0$, such that the sets $f_i(B(\rho_i^{\mathrm{dom}}))$ are pairwise disjoint and contained in $B(1-\zeta)$.*

This is Theorem S14.7. Construction S20.1 fixes the a_i by explicit slack increments; the further parameters of the model are those of Lemma S14.6.

5.4. **One large ball.** The quadric model accepts only capacities below one half, and the two-ball inequality allows at most one capacity $A \ge \frac{1}{2}$; every other capacity is then below $1 - A \le \frac{1}{2}$. Normalise so that the input is

$$A \ge \tfrac{1}{2}, \qquad A^n + \sum_{i=1}^{k} \rho_i^n < 1, \qquad A + \rho_i < 1 \quad (1 \le i \le k). \tag{23}$$

Valid auxiliary data are $(a; r_1, \dots, r_k)$ with $\max(A, \frac{1}{2}) < a < 1$, $\rho_i < r_i < 1 - a$ and $a^n + \sum_i r_i^n < 1$; they exist under (23), and rational ones exist as well (Lemma S16.2). The large ball is placed in the reserved ball $B^\circ(a)$ and the others in the shell $S_a = \{a < \pi|y|^2 < 1\}$, of volume $(1-a^n)/n!$; the two-ball condition $A + \rho_i < 1$ is what allows $\rho_i < 1 - a$ with $a > A$.

For $n \ge 4$ the second stage is a product mixed model over a finite bordered piece of the affine quadric curve, with a radial-integral phase lift, and the return to the shell is an explicit slit-shell formula in coupled coordinates (Lemma S17.7). For $n = 3$ that base would be a conic; instead we use the polynomial cubic model of [Ope26], and the large ball is inserted by a polynomial bubble before the final return (§S16.2).

Theorem 5.3 (One large ball)**.** *Let $n \ge 3$, let $A, \rho_1, \dots, \rho_k > 0$, and let $(a; r)$ be valid auxiliary data for $(A; \rho)$. For $n \ge 4$ put $\rho_0^{\mathrm{dom}} = (A + a)/2$ and $\rho_i^{\mathrm{dom}} = (3\rho_i + 5r_i)/8$ $(i \ge 1)$. For $n = 3$ assume moreover $a \in \mathbb{Q}$, let $A < r_0 < a$ and $\rho_i < r_i' < r_i$, and put $\rho_0^{\mathrm{dom}} = r_0$, $\rho_i^{\mathrm{dom}} = r_i'$. Then there are smooth symplectic embeddings f_i of open neighbourhoods of $B(\rho_i^{\mathrm{dom}})$ $(0 \le i \le k)$ into $B^\circ(1)$ and a number $\zeta > 0$, defined explicitly from these data, such that the sets $f_i(B(\rho_i^{\mathrm{dom}}))$ are pairwise disjoint and lie in $B(1-\zeta)$. For $n \ge 4$, f_0 is the inclusion, $f_0(B(\rho_0^{\mathrm{dom}})) \subset B^\circ(a)$ and $f_i(B(\rho_i^{\mathrm{dom}})) \subset S_a$ for $i \ge 1$.*

For $n \ge 4$ this is Theorem S17.9; for $n = 3$ it is Theorem S19.7. Construction S20.1 fixes $(a; r)$, and for $n = 3$ also r_i' and r_0, by explicit formulas in ρ.

6. From embeddings to one Hamiltonian; proof of Theorem C

The models of Section 5 produce, for normalised capacities, explicit symplectic embeddings f_i of slightly enlarged balls centred at the origin into $B^\circ(1)$, with pairwise disjoint images. Theorem C asks for more: the balls start at prescribed centres s_i, and the motion must be the

time-one map of one Hamiltonian on $\mathbb{R}\times\mathbb{R}^{2n}$. This section supplies the missing part. We join the translations $u\mapsto s_i+u$ to the embeddings f_i by one explicit family $F_{i,t}$ of symplectic embeddings whose images are pairwise disjoint at every time (§6.2), and we write down a Hamiltonian whose field is the velocity of this family on the moving balls (§6.3). Both steps are explicit: every stage is a formula, an integral, a matrix function, a maximum over a compact set, a definite description, or the least of finitely many angle indices.

Notation. $\mathbb{B}(y,r)$ and $\mathbb{B}^\circ(y,r)$ denote the closed and open Euclidean balls of radius r about $y\in\mathbb{R}^{2n}$, and $\mathbb{B}(r)=\mathbb{B}(0,r)$; thus $B(s,c)=\mathbb{B}(s,\sqrt{c/\pi})$. We write $\omega_0(v,w)=v^{\mathsf T}J_0w$ with $J_0=\left(\begin{smallmatrix}0 & I\\ -I & 0\end{smallmatrix}\right)$ in the coordinates (x,y), so that $X_H=J_0\nabla H$ and $\iota_{X_H}\omega_0=dH$. Multiplication by i on $\mathbb{C}^n$, $z_j=x_j+iy_j$, is $J_{\mathbb{C}}=-J_0$. $\mathrm{Sp}(2n)$ is the group of matrices A with $A^{\mathsf T}J_0A=J_0$, and $\|\cdot\|$ is the Euclidean operator norm.

The construction rests on two observations. The first is a common Alexander trick: for $0<v\le 1$ the maps $u\mapsto v^{-1}f_i(vu)$ are symplectic, equal f_i at $v=1$, and have pairwise disjoint images as long as all balls use the same v, because the images are those of disjoint sets under one dilation. The second is that the Hamiltonian is computed in the source: on the round source ball the one-form $\omega_0(\dot F, DF\,\cdot)$ is closed, its radial primitive is an explicit integral of forward data, and a fixed radial cutoff in the source, pushed forward, localises it.

6.1. **The embeddings in each case.**

Definition 6.1 (Normalisation)**.** Let $C>0$ and $c\in(0,\infty)^k$. The *normalised capacities* are $\rho_i=c_i/C$. For a map $f^\natural$ defined on a subset of $\mathbb{R}^{2n}$, its *physical rescaling* is $f(u)=\sqrt{C}\,f^\natural(u/\sqrt{C})$.

The dilation $\delta(z)=z/\sqrt{C}$ maps $B(s,c)$ onto $B(s/\sqrt{C},c/C)$ and $B^\circ(C)$ onto $B^\circ(1)$, and (n,k,C,c) is admissible if and only if $(n,k,1,\rho)$ is, since both conditions of Definition 1.2 are homogeneous. The physical rescaling is $\delta^{-1}\circ f^\natural\circ\delta$, so $Df(u)=Df^\natural(u/\sqrt{C})$; it is symplectic, injective or of disjoint images exactly when $f^\natural$ is. The models are built for normalised capacities, and their embeddings enter the construction only through the physical rescaling.

Construction S20.1 defines for admissible normalised data $(n,k,1,\rho)$ the *normalised embeddings* $f_i^\natural$ together with an enlarged closed capacity ρ_i^+ and an open domain capacity ρ_i^{dom}, $\rho_i<\rho_i^+<\rho_i^{\mathrm{dom}}$. It is a case distinction: (i) $n=2$, the cubic model, with the linear datum when $k\ge 10$, the capacities are equal and $\frac34<\rho\sqrt{k}<1$, and the scalar comparator otherwise; (ii) $n\ge 3$ and all $\rho_i<\frac12$, the quadric model; (iii) $n\ge 4$ and one $\rho_j\ge\frac12$, the shell model; (iv) $n=3$ and one $\rho_j\ge\frac12$, the polynomial cubic model with the bubble. In each case the auxiliary capacities of the model theorem are closed formulas in ρ, built from arithmetic, finite minima and maxima, square roots (in case (i)) and, in case (iv), one integer floor and one ceiling; for instance in case (ii)

$$a_i=\rho_i+\min\Big(\frac{1/2-\rho_i}{2},\frac{1-\sum_j\rho_j^n}{2nk}\Big),\qquad(\rho_i^+,\rho_i^{\mathrm{dom}})=\Big(\frac{2\rho_i+a_i}{3},\frac{\rho_i+2a_i}{3}\Big).$$

The supplement checks that these data satisfy the hypotheses of the model theorems.

Definition 6.2 (Embedding datum)**.** Let $n\ge1$, $k\ge1$, $C>0$ and $c\in(0,\infty)^k$. An *embedding datum* for (n,k,C,c) consists of numbers $c_i',\bar c_i$ with $c_i<c_i'<\bar c_i$ and maps $f_i\colon B^\circ(\bar c_i)\to\mathbb{R}^{2n}$ such that

(i) f_i is C^∞ and $Df_i(u)\in\mathrm{Sp}(2n)$ for all $u\in B^\circ(\bar c_i)$;
(ii) f_i is injective on $B(c_i')$;
(iii) the sets $f_i(B(c_i'))$, $1\le i\le k$, are pairwise disjoint and contained in $B^\circ(C)$.

Proposition 6.3 (The embeddings)**.** *Let (n,k,C,c) be admissible, $\rho = c/C$, and let $f_i^\natural$, ρ_i^+ and ρ_i^{dom} be given by Construction S20.1. Put*

$$c_i' = C\rho_i^+, \qquad \bar{c}_i = C\rho_i^{\mathrm{dom}}, \qquad f_i(u) = \sqrt{C}\, f_i^\natural(u/\sqrt{C}).$$

Then $(c_i', \bar{c}_i, f_i)$ is an embedding datum for (n,k,C,c), and $\bigcup_i f_i(B(c_i'))$ is a compact subset of $B^\circ(C)$.

Proof. In each case $\rho_i < \rho_i^+ < \rho_i^{\mathrm{dom}}$, and the model theorem gives exactly the properties required on these balls. In case (i), Theorem S13.23 states them for the displayed pairs. In cases (ii)–(iv) the model theorems (Theorems S14.7, S17.9 and S19.7) give smooth symplectic embeddings of open neighbourhoods of $B(\rho_i^{\mathrm{dom}})$ with pairwise disjoint images of $B(\rho_i^{\mathrm{dom}})$ in $B(1-\zeta)$: in case (ii) $\rho_i^{\mathrm{dom}} = (\rho_i + 2a_i)/3$; in case (iii) $\rho_i^{\mathrm{dom}} = (3\rho_i + 5r_i)/8$ for $i \neq j$, while the distinguished ball is placed by the inclusion, which is symplectic and injective on all of $\mathbb{C}^n$ and maps $B(\rho_j^+) = B((A+a)/2)$ into $B^\circ(a)$, disjoint from the shell S_a containing the other images; in case (iv) $\rho_i^{\mathrm{dom}} = r_i'$ for $i \neq j$ and $\rho_j^{\mathrm{dom}} = r_0$, with r_i' and r_0 of Construction S20.1. Since $B(\rho_i^+) \subset B^\circ(\rho_i^{\mathrm{dom}}) \subset B(\rho_i^{\mathrm{dom}})$, this is smoothness and the symplectic identity on $B^\circ(\rho_i^{\mathrm{dom}})$ and injectivity, pairwise disjointness and target containment on $B(\rho_i^+)$. Physical rescaling preserves these properties, because $Df_i(u) = Df_i^\natural(u/\sqrt{C})$. The union is a finite union of continuous images of compact sets. □

6.2. **A common symplectic isotopy.** In this subsection we fix $n \geq 1$, an embedding datum $(c_i', \bar{c}_i, f_i)$ for (n,k,C,c), and centres $s_1, \dots, s_k$ such that the closed balls $B(s_i, c_i)$ are pairwise disjoint. The isotopy runs through four kinds of stages. The round balls are first translated, along a collision-free route, to far-apart centres Y_i; there the linear map is deformed from I to $A_i = Df_i(0)$ in $\mathrm{Sp}(2n)$; then the shape is deformed from $Y_i + A_i u$ to $\tau^{-1} f_i(\tau u)$ with the centre fixed, for a small scale τ (Definition S21.4); finally the common Alexander parameter v runs from τ to 1. All proofs are in §S21; here we give the definitions and the statements used later.

Definition 6.4 (Radii)**.** Put $r_i = \sqrt{c_i/\pi}$, $r_i' = \sqrt{c_i'/\pi}$, $\bar{r}_i = \sqrt{\bar{c}_i/\pi}$,

$$\delta_{\mathrm{sep}} = \begin{cases} \min_{i \neq j} \big(|s_i - s_j| - r_i - r_j \big), & k \geq 2, \\ 1, & k = 1, \end{cases}$$

$$\delta_i = \min\big(\tfrac{\delta_{\mathrm{sep}}}{4},\ r_i' - r_i\big), \qquad r_{\ell,i} = r_i + \tfrac{\ell}{5}\delta_i \quad (\ell = 1, \dots, 4).$$

Lemma 6.5 (Radii)**.** *$\delta_{\mathrm{sep}} > 0$, $\delta_i > 0$, $r_i < r_{1,i} < r_{2,i} < r_{3,i} < r_{4,i} < r_i' < \bar{r}_i$, and*

$$|s_i - s_j| > r_{3,i} + r_{3,j} \qquad (i \neq j). \tag{24}$$

Consequently f_i is C^∞ with symplectic derivative on $\mathbb{B}^\circ(\bar{r}_i)$, injective on $\mathbb{B}(r_{4,i})$, and the sets $f_i(\mathbb{B}(r_{4,i}))$ are pairwise disjoint and contained in $\mathbb{B}^\circ(R_0)$, $R_0 = \sqrt{C/\pi}$: the data form an isotopy input (Definition S21.1).

Proof. If $|s_i - s_j| \leq r_i + r_j$, the point $s_i + \frac{r_i}{r_i + r_j}(s_j - s_i)$ lies in $\mathbb{B}(s_i, r_i) \cap \mathbb{B}(s_j, r_j)$; so $|s_i - s_j| > r_i + r_j$ for $i \neq j$, and $\delta_{\mathrm{sep}} > 0$. Since $c_i' > c_i$, $\delta_i > 0$, and the $r_{\ell,i}$ increase with ℓ; $r_{4,i} < r_i + \delta_i \leq r_i' < \bar{r}_i$. For $i \neq j$, $r_{3,i} + r_{3,j} = r_i + r_j + \frac{3}{5}(\delta_i + \delta_j) \leq r_i + r_j + \frac{3\delta_{\mathrm{sep}}}{10} < r_i + r_j + \delta_{\mathrm{sep}} \leq |s_i - s_j|$. The rest follows from Definition 6.2, because $\mathbb{B}^\circ(\bar{r}_i) = B^\circ(\bar{c}_i)$, $\mathbb{B}(r_{4,i}) \subset B(c_i')$ and $B^\circ(C) = \mathbb{B}^\circ(R_0)$. □

The ball $\mathbb{B}^\circ(r_{4,i})$ is the open source domain of every stage, the images of the closed balls $\mathbb{B}(r_{3,i})$ are kept pairwise disjoint, and the cutoff of §6.3 interpolates between $r_{1,i}$ and $r_{2,i}$. Table 4 lists these radii with the other margins of the construction, from the capacity slack of the model to the compilation margin b_R of the finite form.

The four kinds of stages are given by explicit formulas in §S21; we describe them only so far as the later arguments use them. The *route* (Definition S21.10) translates the round balls to centres $Y_i = y_i/\tau$, $y_i = f_i(0)$: each ball first rises in the x_1-direction to a level given by the rank of its starting height, then in the y_1-direction to a level given by the rank of its target height, then all balls move straight to their targets, and the two offsets are removed in reverse order; ranks are compatible with heights, so the balls $\mathbb{B}(r_{3,i})$ stay disjoint (Lemma S21.11). The *linear stage* is $Y_i + M_{i,\sigma}u$, where $\sigma \mapsto M_{i,\sigma}$ is an explicit smooth path in $\mathrm{Sp}(2n)$ from I to $A_i = Df_i(0)$, obtained from a polar decomposition, a Cayley logarithm and a shear (Definition S21.2). The *shape stage* is $Y_i + \kappa_{i,\tau\sigma}(u)$ with $\kappa_{i,v}(u) = \int_0^1 Df_i(\vartheta v u)\, u\, d\vartheta$, which runs from $Y_i + A_i u$ to $\tau^{-1} f_i(\tau u)$. The *tail* is $T_{i,v}(u) = v^{-1} f_i(vu)$ for v from τ to 1; a single dilation acts on all images at once, so disjointness is preserved, and $T_{i,1} = f_i$ (Lemma S21.9). The scale $\tau \in (0, \frac{1}{2}]$ is given by a formula in a bound L for $\|Df_i\|$ and $|M_{i,\sigma}u|$ on $\mathbb{B}(r_{3,i})$, a maximum of a continuous function over a compact set, and in the minimal distance of the points y_i, so that the balls $\mathbb{B}(Y_i, Lr_{\mathrm{tot}})$ are pairwise disjoint (Definition S21.4). The stages occupy the eight slots $[\frac{m}{8}, \frac{m+1}{8}]$ of $[0,1]$ and are joined by flat clocks $S(8t-m)$, with S the flat step; on each slot the defining sum $F_{i,t}$ reduces to one stage (Construction S21.12 and Lemma S21.13).

Proposition 6.6 (The common symplectic isotopy)**.** *(a) $(t,u) \mapsto F_{i,t}(u)$ is C^∞ on $\mathbb{R} \times \mathbb{B}^\circ(r_{4,i})$; $F_{i,0}(u) = s_i + u$, $F_{i,1} = f_i$; $F_{i,t} = F_{i,0}$ for $t \le 0$ and $F_{i,t} = F_{i,1}$ for $t \ge 1$;*

(b) $D_u F_{i,t}(u) \in \mathrm{Sp}(2n)$ for all t and $u \in \mathbb{B}^\circ(r_{4,i})$;

(c) each $F_{i,t}$ is injective on $\mathbb{B}(r_{3,i})$, and for every t the sets $F_{i,t}(\mathbb{B}(r_{3,i}))$, $1 \le i \le k$, are pairwise disjoint;

(d) $F_{i,t}(\mathbb{B}(r_{3,i})) \subset \mathbb{B}(R_F)$ for all i, t, for the explicit radius R_F of Proposition S21.14.

Proof. By Lemma 6.5 the data form an isotopy input, and Proposition S21.14 gives (a)–(d). On each slot it suffices to check (b)–(d) for one stage. Route slots are translations, with disjoint images by the rank argument. The linear and shape slots have symplectic derivatives $M_{i,\sigma}$ and $Df_i(\tau\sigma u)$ and images in the pairwise disjoint balls $\mathbb{B}(Y_i, Lr_{\mathrm{tot}})$. The tail slot has derivative $Df_i(vu)$ and disjoint images because one dilation $y \mapsto v^{-1}y$ is applied to the disjoint sets $f_i(v\mathbb{B}(r_{3,i})) \subset f_i(\mathbb{B}(r_{4,i}))$. □

6.3. **The Hamiltonian.** Let F be the isotopy of Construction S21.12. Only properties (a)–(d) of Proposition 6.6, with the radii $r_{1,i} < r_{2,i} < r_{3,i} < r_{4,i}$, are used below.

Definition 6.7 (Source one-form and primitive)**.** For $t \in \mathbb{R}$, $u \in \mathbb{B}^\circ(r_{4,i})$ and $w \in \mathbb{R}^{2n}$,

$$\mathfrak{a}_{i,t}(u)[w] = \omega_0\big(\dot F_{i,t}(u), D_u F_{i,t}(u) w\big), \qquad \mathfrak{h}_{i,t}(u) = \int_0^1 \mathfrak{a}_{i,t}(\vartheta u)[u]\, d\vartheta,$$

where $\dot F = \partial_t F$.

Lemma 6.8 ($\mathfrak{a}_{i,t}$ is closed)**.** *The bilinear form $(w, w') \mapsto D_u \mathfrak{a}_{i,t}(u)[w][w']$ is symmetric; that is, $d_u \mathfrak{a}_{i,t} = 0$.*

Proof. Write $F = F_{i,t}$ and $\partial_l = \partial/\partial u_l$. The coefficient of $du_l \wedge du_m$ in $d_u \mathfrak{a}_{i,t}$ is

$$\partial_l\, \omega_0(\dot F, \partial_m F) - \partial_m\, \omega_0(\dot F, \partial_l F) = \omega_0(\partial_l \dot F, \partial_m F) + \omega_0(\partial_l F, \partial_m \dot F) = \partial_t\big[\omega_0(\partial_l F, \partial_m F)\big],$$

since the terms $\omega_0(\dot F, \partial_l \partial_m F)$ and $\omega_0(\dot F, \partial_m \partial_l F)$ cancel and $\partial_l \dot F = \partial_t \partial_l F$. By Proposition 6.6(b), $\omega_0(\partial_l F, \partial_m F) = \omega_0(e_l, e_m)$ does not depend on t. □

Lemma 6.9 (Radial primitive)**.** *$(t,u) \mapsto \mathfrak{h}_{i,t}(u)$ is C^∞ on $\mathbb{R} \times \mathbb{B}^\circ(r_{4,i})$, $\mathfrak{h}_{i,t}(0) = 0$, $d_u \mathfrak{h}_{i,t} = \mathfrak{a}_{i,t}$, and $\mathfrak{h}_{i,t} \equiv 0$ for $t \notin (0,1)$.*

Proof. The integrand is C^∞ in (ϑ, t, u), so $\mathfrak{h}$ is smooth and may be differentiated under the integral. For $w \in \mathbb{R}^{2n}$, by Lemma 6.8,

$$d_u\mathfrak{h}_{i,t}(u)[w] = \int_0^1 \big(\vartheta\, D\mathfrak{a}_{i,t}(\vartheta u)[u][w] + \mathfrak{a}_{i,t}(\vartheta u)[w]\big)\, d\vartheta = \int_0^1 \frac{d}{d\vartheta}\big(\vartheta\, \mathfrak{a}_{i,t}(\vartheta u)[w]\big)\, d\vartheta = \mathfrak{a}_{i,t}(u)[w].$$

For $t \notin (0,1)$, $F_{i,\cdot}(u)$ is constant on $(-\infty, 0]$ or on $[1, \infty)$ and smooth, so $\dot F_{i,t} = 0$ and $\mathfrak{a}_{i,t} = 0$. □

The primitive is fixed by $\mathfrak{h}_{i,t}(0) = 0$. This matters after the cutoff: adding a time-dependent constant $c(t)$ would contribute $c(t)\, d\chi_i$ to $d(\chi_i\mathfrak{h}_{i,t})$, so the field in the transition collar would depend on that choice.

Definition 6.10 (Source cutoff and the Hamiltonian of an embedding datum)**.** Let $\chi_i = \chi_{r_{1,i}, r_{2,i}}$ be the radial cutoff $\chi_{r,r'}(u) = 1 - S\big((|u|^2 - r^2)/(r'^2 - r^2)\big)$, so $\chi_i = 1$ on $\mathbb{B}(r_{1,i})$ and $\chi_i = 0$ outside $\mathbb{B}^\circ(r_{2,i})$ (Lemma S2.7). Put $H_i(t, y) = \chi_i(u)\, \mathfrak{h}_{i,t}(u)$ if u is the unique point of $\mathbb{B}(r_{3,i})$ with $F_{i,t}(u) = y$, and $H_i(t, y) = 0$ if there is no such unique point; and $H_{\mathfrak{E}} = \sum_{i=1}^k H_i$, where $\mathfrak{E}$ denotes the embedding datum together with the centres. The unique preimage is a definite description (Definition 1.4).

By Proposition 6.6(c) the first case applies exactly when $y \in F_{i,t}(\mathbb{B}(r_{3,i}))$.

Lemma 6.11 (Smoothness and support)**.** [Lean: needed consequence verified.]

Each H_i is C^∞ on $\mathbb{R} \times \mathbb{R}^{2n}$; $H_i(t, \cdot)$ vanishes outside $F_{i,t}(\mathbb{B}(r_{2,i}))$ and vanishes identically for $t \notin (0,1)$. Hence $H_{\mathfrak{E}}$ is C^∞, vanishes for $t \notin (0,1)$, and $H_{\mathfrak{E}}(t, \cdot)$ vanishes outside the compact set $K_H = \bigcup_i \bigcup_{t \in [0,1]} F_{i,t}(\mathbb{B}(r_{2,i})) \subset \mathbb{B}(R_F)$.

Proof. The map $E_i(t, u) = (t, F_{i,t}(u))$ is C^∞ on $\mathbb{R} \times \mathbb{B}^\circ(r_{4,i})$, injective on $\mathbb{R} \times \mathbb{B}(r_{3,i})$, and its derivative $\left(\begin{smallmatrix} 1 & 0 \\ \dot F & D_u F \end{smallmatrix}\right)$ is invertible. By the inverse function theorem $O_1 = E_i(\mathbb{R} \times \mathbb{B}^\circ(r_{3,i}))$ is open and E_i^{-1} is smooth on it, and $H_i = (\chi_i\mathfrak{h}_i) \circ E_i^{-1}$ there. The set $Z = E_i(\mathbb{R} \times \mathbb{B}(r_{2,i}))$ is the union of $(-\infty, 0] \times F_{i,0}(\mathbb{B}(r_{2,i}))$, $E_i([0,1] \times \mathbb{B}(r_{2,i}))$ and $[1, \infty) \times F_{i,1}(\mathbb{B}(r_{2,i}))$ (Proposition 6.6(a)), hence closed, and $Z \subset O_1$. On $\mathbb{R}^{1+2n} \setminus Z$ we have $H_i = 0$: either y has no preimage in $\mathbb{B}(r_{3,i})$, or its preimage lies outside $\mathbb{B}(r_{2,i})$, where $\chi_i = 0$. The two open sets cover, so H_i is smooth (in full: Lemma S21.16). Vanishing for $t \notin (0,1)$ is Lemma 6.9. K_H is compact as a finite union of continuous images of compact sets (F is constant in t outside $[0,1]$), and lies in $\mathbb{B}(R_F)$ by Proposition 6.6(d). □

Lemma 6.12 (The field on the moving balls)**.** *For all $t \in \mathbb{R}$, $u \in \mathbb{B}^\circ(r_{1,i})$ and $w \in \mathbb{R}^{2n}$,*

$$d_y H_{\mathfrak{E}}(t, F_{i,t}(u))[w] = \omega_0\big(\dot F_{i,t}(u), w\big), \qquad \text{hence} \qquad X_{H_{\mathfrak{E}}}(t, F_{i,t}(u)) = \dot F_{i,t}(u).$$

Proof. Since $|u| < r_{1,i}$, we have $H_i(t, F_{i,t}(x)) = \mathfrak{h}_{i,t}(x)$ for x near u. Differentiating and using Lemma 6.9, $d_y H_i(t, F_{i,t}(u)) \circ D_u F_{i,t}(u) = \mathfrak{a}_{i,t}(u) = \omega_0(\dot F_{i,t}(u), D_u F_{i,t}(u)\,\cdot)$, and $D_u F_{i,t}(u)$ is invertible. For $j \neq i$, $H_j(t, \cdot)$ vanishes outside $F_{j,t}(\mathbb{B}(r_{2,j}))$, which is disjoint from the set $F_{i,t}(\mathbb{B}^\circ(r_{3,i}))$ by Proposition 6.6(c); this set is open, since $F_{i,t}$ is an injective local diffeomorphism on the open ball, and contains $F_{i,t}(u)$. So $d_y H_j(t, F_{i,t}(u)) = 0$. Since $\omega_0(X_H, w) = dH[w]$ and ω_0 is nondegenerate, $X_{H_{\mathfrak{E}}} = \dot F_{i,t}(u)$. □

Lemma 6.13 (Integral curves)**.** *Let $H = H_{\mathfrak{E}}$. Through every point of $\mathbb{R}^{2n}$ passes an integral curve $\gamma\colon \mathbb{R} \to \mathbb{R}^{2n}$ of X_H with $\gamma(0)$ that point. If $|u| < r_{1,i}$ and $\gamma\colon [0,1] \to \mathbb{R}^{2n}$ is an integral curve of X_H with $\gamma(0) = s_i + u$, then $\gamma(t) = F_{i,t}(u)$ for all $t \in [0,1]$.*

Proof. $X_H = J_0 \nabla H$ is C^∞ and vanishes outside $[0,1] \times K_H$ (Lemma 6.11), so $D_y X_H$ is bounded and X_H is globally Lipschitz in y, uniformly in t. By the Picard–Lindelöf theorem integral curves exist for all time and are unique. The curve $t \mapsto F_{i,t}(u)$ starts at $s_i + u$ and is an integral curve by Lemma 6.12. □

Theorem 6.14 (The Hamiltonian realises the isotopy)**.** *Let $H = H_{\mathfrak{E}}$. The flow ϕ_H^t is defined for all $t \in \mathbb{R}$ and consists of symplectic diffeomorphisms of $\mathbb{R}^{2n}$, and*

$$\phi_H^t(s_i + u) = F_{i,t}(u) \qquad (u \in \mathbb{B}(r_i),\ t \in \mathbb{R},\ 1 \le i \le k).$$

In particular $\phi_H^1(B(s_i, c_i)) = f_i(B(c_i))$, and these sets are pairwise disjoint and contained in $B^\circ(C)$.

Proof. By the proof of Lemma 6.13 the flow is global; the time-dependent flow of a Hamiltonian field consists of symplectic diffeomorphisms. Since $\mathbb{B}(r_i) \subset \mathbb{B}^\circ(r_{1,i})$, Lemma 6.12 holds for all $t \in \mathbb{R}$ at $u \in \mathbb{B}(r_i)$, so $t \mapsto F_{i,t}(u)$ is the integral curve through $s_i + u$. At $t = 1$, $F_{i,1} = f_i$ on $\mathbb{B}^\circ(r_{4,i}) \supset \mathbb{B}(r_i) = B(c_i)$, and $B(c_i) \subset B(c_i')$. □

On the collar $F_{i,t}(\mathbb{B}(r_{2,i}) \setminus \mathbb{B}(r_{1,i}))$ the field X_H need not equal $\dot{F}_{i,t}$; trajectories starting in $B(s_i, c_i)$ stay in $F_{i,t}(\mathbb{B}(r_i))$ and never meet it.

Proposition 6.15 (Hamiltonian of an embedding datum)**.** *Let $n \ge 1$, let $(c_i', \bar{c}_i, f_i)$ be an embedding datum for (n, k, C, c), and let $s_1, \dots, s_k$ be centres such that the closed balls $B(s_i, c_i)$ are pairwise disjoint. Then the Hamiltonian $H_{\mathfrak{E}}$ of Definition 6.10 satisfies* $\mathrm{Pack}(n, k, C, c, s)$.

Proof. Condition (a) of Definition 1.1 is Lemma 6.11 with $K = \mathbb{B}(R_F)$. For (b), let $z = s_i + u \in B(s_i, c_i)$, so $|u| \le r_i < r_{1,i}$. Lemma 6.13 gives an integral curve on $\mathbb{R}$ through z, and every integral curve γ on $[0, 1]$ with $\gamma(0) = z$ has $\gamma(1) = F_{i,1}(u) = f_i(u) \in f_i(B(c_i')) \subset B^\circ(C)$. □

Definition 6.16 (The Hamiltonian $H[n, k, C, c, s]$)**.** On admissible input with pairwise disjoint closed source balls, let $\mathfrak{E}$ be the embedding datum of Proposition 6.3, together with the centres s_i, and put $H[n, k, C, c, s] = H_{\mathfrak{E}}$ (Definition 6.10).

6.4. **Proof of Theorem C and of the corollaries.**

Proof of Theorem C(i). Let $H = H[n, k, C, c, s]$ with the hypotheses of Theorem C, and let $\mathfrak{E}$ be the datum of Definition 6.16. By Lemma 6.11, H is C^∞, vanishes for $t \notin (0, 1)$, and vanishes outside $\mathbb{R} \times K$ with the explicit compact set $K = \mathbb{B}(R_F)$. By Proposition 6.15, H satisfies $\mathrm{Pack}(n, k, C, c, s)$. By Theorem 6.14, $\phi_H^1(B(s_i, c_i)) = f_i(B(c_i))$; these sets are pairwise disjoint and lie in the compact subset $\bigcup_i f_i(B(c_i'))$ of $B^\circ(C)$ (Proposition 6.3). □

Proof of Theorem C(ii). Every entry of Table 5 is one of the operations of Definition 1.3; the only search in it is the least of $n+1$ angle indices in the linear path. The same definition applies to all (n, k, C, c, s). □

Part (iii) of Theorem C is Theorem 7.4.

Proof of Corollary D. Let $n \ge 3$. Suppose Ψ is a symplectic embedding of a neighbourhood of $\bigsqcup_i B(s_i, c_i)$ into $B^\circ(C)$. The image of the compact set $\bigsqcup_i B(s_i, c_i)$ is compact, hence contained in $B(C')$ for some $C' < C$. Symplectic embeddings preserve ω_0^n, so $\sum_i c_i^n/n! = \mathrm{vol}\,\Psi(\bigsqcup_i B(s_i, c_i)) \le C'^n/n! < C^n/n!$. For $i \ne j$, Ψ restricts to a symplectic embedding of the union of the interiors of $B(s_i, c_i)$ and $B(s_j, c_j)$ into $B^\circ(C'')$ for any $C'' \in (C', C)$, and Gromov's two-ball theorem [Gro85] gives $c_i + c_j \le C'' < C$. Conversely, if both inequalities hold, then (n, k, C, c) is admissible and Theorem C(i) applies to $H = H[n, k, C, c, s]$. The global symplectic diffeomorphism ϕ_H^1 maps each $B(s_i, c_i)$ into $B^\circ(C)$, so its restriction to the open neighbourhood $(\phi_H^1)^{-1}(B^\circ(C))$ of $\bigsqcup_i B(s_i, c_i)$ is a symplectic embedding into $B^\circ(C)$. □

Proof of Corollary E. All $c_i = c$, and $\sum_i c_i^n = kc^n < C^n$. If $n \ge 3$, then $2^n c^n \le kc^n < C^n$, so $c_i + c_j = 2c < C$. If $n = 2$, then $9c^2 \le kc^2 < C^2$, so $3c < C$. Thus (n, k, C, c) is admissible and Theorem C(i) applies. The images are pairwise disjoint and ϕ_H^1 preserves volume, so they fill the fraction $k \,\mathrm{vol}\, B(c)/\,\mathrm{vol}\, B(C) = kc^n/C^n$ of $B(C)$, and $kc^n/C^n > 1 - \varepsilon$ by hypothesis. □

Ingredient	Operation of Definition 1.3	Where shown
hypotheses; branch	case distinctions on propositions ($n = 2$; some $\rho_j \geq \frac{1}{2}$; for $n = 2$, $k \geq 10$, equal $\rho_i = \rho$ and $\frac{3}{4} < \rho\sqrt{k} < 1$)	Def. 6.16, Cons. S20.1
branch margins	arithmetic, finite minima and maxima, square roots, an integer floor and ceiling	Cons. S20.1
model embeddings $f_i^{\natural}$, ρ_i^{+}	model formulas and compact flow parameters; in dimension six, integer weights and explicit polynomial lifts	Thms. 5.1, 5.2, 5.3; Prop. S16.8
c_i', $\bar{c}_i$, f_i	arithmetic; composition with a dilation	Prop. 6.3
radii $r_{\ell,i}$	square roots; finite minima; arithmetic	Lem. 6.5
linear path $M_{i,\sigma}$	positive square root (definite description); inverses by the adjugate formula; least of $n+1$ angle indices; sin, cos	Lem. S21.3
L, δ_c, r_{tot}, τ, Y_i	maxima of continuous functions over compact sets; finite minima; arithmetic	Lem. S21.5
route z_i	ranks (counts of propositions); arithmetic	Lem. S21.11
shape, tail, clocks	integral over $[0,1]$; composition; the flat step	Lem. S21.8, S21.9
$H_{\mathfrak{E}}$	integral over $[0,1]$; radial cutoff; definite description of the unique preimage	Def. 6.10, Lem. 6.11

TABLE 5. The construction of $H[n,k,C,c,s]$ by the operations of level L1.

Proof of Corollary F. Here $n = 2$, $c_i = \pi$, so $B(s_i, \pi)$ is the closed Euclidean unit ball about $s_i = (3i-3, 0, 0, 0)$; distinct centres are at distance at least $3 > 2$, so these balls are pairwise disjoint. Let $C = \pi R^2$ with $R^4 = k/(1-\varepsilon/2)$. Then $\sum_i c_i^2 = k\pi^2 < \pi^2 R^4 = C^2$ since $1-\varepsilon/2 < 1$, and $3c_i < C$ since $R^4 > k \geq 9$. Theorem C(i) applies, and $B^{\circ}(\pi R^2)$ is the open Euclidean ball of radius R about the origin. Finally $k/R^4 = 1 - \varepsilon/2 > 1 - \varepsilon$. □

Proof of Corollary G. Let $V = \sum_i c_i^2$ and $C = C_\varepsilon$, so $V/C^2 = 1 - \varepsilon/2$. Then $V < C^2$, and $3c_i \leq V^{1/2} < C$. So $(2, k, C, c)$ is admissible and Theorem C(i) applies; as in the proof of Corollary E the images fill the fraction $V/C^2 = 1 - \varepsilon/2$ of $B(C)$. □

7. Computability and the finite form

This section proves Theorem C(iii). The output is a rational polynomial multiplied by a fixed smooth spatial cutoff, and it is found by a search over finite certificates of trajectory containment: a candidate polynomial is accepted when Euler trajectories from a grid covering the source balls, with an error bound valid on the whole time interval, stay in the region where the cutoff equals one and end inside the target. Every accepted candidate is correct (Theorem S25.5). Termination needs one smooth compactly supported Hamiltonian satisfying the packing specification with a positive margin. We take $H[n,k,C,c,s]$ and compile it by Bernstein approximation (§7.1); any packing Hamiltonian with a margin would serve equally.

For the Epoch family the termination witness is Corollary F, at density $1 - \varepsilon/2$; the finite verifier is the same in every case.

A name for a real x is a rational stream $q(p)$ with $|q(p) - x| \leq 2^{-p}$; a rational is named by its constant stream. The finite verifier requires names for the source radii $r_i = \sqrt{c_i/\pi}$ and target radius $R = \sqrt{C/\pi}$; these are computed from the supplied capacity names by the programs in §S23. The symbol $|\cdot|$ denotes Euclidean norm, and S is the flat step of Definition S2.3.

Object	Definition	Effective data	Reference
Input radii	$r_i = \sqrt{c_i/\pi}$, $R = \sqrt{C/\pi}$	names with error 2^{-p}; lower bound $1 < \pi$	§S23
Source enclosures	s_i^p, $r_{i,p}^+$	whole closed source covered; retained nodes within $(2d+3)2^{-p} + d\rho$ of a source	Eq. S97
Polynomial field	$J_0 \nabla P_e$, arguments clamped to $[0,1] \times [-\frac{B}{2}, \frac{B}{2}]^d$	exact rational evaluation; integer $\Theta = h_B(e)$ bounding speed and Lipschitz constants	Lem. S25.2
Covering grid	mesh $\rho = 2Q2^{-m}$ in $[-Q,Q]^d$	$(2^m+1)^d$ nodes; maximum-norm radius ρ	Def. S25.4
Euler table	$x_{j+1} = x_j + F_e(j/N, x_j)/N$	error $E = (\rho + (\Theta+\Theta^2)/N)4^\Theta + \Theta/N$ on $[0,1]$	Lem. S25.3
Finite output	$\beta_B P_e$	margins $B/2 - \lvert x_{j,l}\rvert - E$ to the region where $\beta_B = 1$ and $R_p^- - \lvert x_N\rvert - dE$ to the target boundary	Thm. S25.5
Bernstein witness	compilation datum (C1)–(C7)	derivative envelope M_0; field error η; Grönwall factor $(e^{L_X}-1)/L_X$	Thm. 7.2

TABLE 6. Data for the finite verifier and its nonemptiness proof; $d = 2n$.

Search	Nonempty because	Checkable condition
Square-root names	none: a direct total program (integer square root at doubled precision, Lemma S23.6)	—
Candidate (B, e, Q, p, m, N)	smaller-target geometric packing, compilation and source-neighbourhood margins	finite strict rational tests
First passing candidate	Thm. S25.7	zero natural failure count

TABLE 7. Searches of the finite algorithm. The model parameters and their margins are given in the geometric sections.

7.1. **The finite form.** One estimate on the vector fields controls the time-one maps at all starting points at once.

Lemma 7.1 (Grönwall)**.** *Let $G, H\colon \mathbb{R} \times \mathbb{R}^{2n} \to \mathbb{R}$ be C^∞, let $\eta > 0$ and $L_X > 0$, and suppose that*

(i) $\lvert X_G(t,z) - X_H(t,z)\rvert \le \eta$ for all $t \in [0,1]$ and all $z \in \mathbb{R}^{2n}$;
(ii) $X_H(t,\cdot)$ is L_X-Lipschitz on all of $\mathbb{R}^{2n}$ for every $t \in [0,1]$.

If $\gamma_G, \gamma_H\colon [0,1] \to \mathbb{R}^{2n}$ are integral curves of X_G and X_H with $\gamma_G(0) = \gamma_H(0)$, then $\lvert\gamma_G(1) - \gamma_H(1)\rvert \le \eta(e^{L_X}-1)/L_X$. If moreover $\eta = b_R e^{-L_X}/2$ with $b_R > 0$, then $\eta(e^{L_X}-1)/L_X < b_R$.

Proof. The difference $u = \gamma_G - \gamma_H$ satisfies $u(0) = 0$ and $\lvert\dot u\rvert \le \lvert X_G(\gamma_G) - X_H(\gamma_G)\rvert + \lvert X_H(\gamma_G) - X_H(\gamma_H)\rvert \le \eta + L_X\lvert u\rvert$, and Grönwall's inequality gives the first bound. With $\eta = b_R e^{-L_X}/2$ it equals $\frac{b_R}{2}(1 - e^{-L_X})/L_X < b_R/2$, since $1 - e^{-x} < x$ for $x > 0$. (Lemma S24.14 gives the same argument with all constants.) □

Both hypotheses must hold on the whole space: before the estimate is proved, the approximate trajectory is not known to stay near the exact one, so a bound valid only near the exact trajectories could not be applied to it. For smooth compactly supported H such global

bounds are finite: since $X_H = J_0\nabla_z H$ with J_0 orthogonal, (ii) holds for every $L_X > 0$ with $L_X \geq \max_{[0,1]\times\mathbb{R}^{2n}} \|D_z^2 H\|$.

We now fix the compilation. Let $n \geq 2$, $k \geq 1$, $C > 0$, $c \in (0,\infty)^k$ and $s \in (\mathbb{R}^{2n})^k$, and let H satisfy $\mathrm{Pack}(n,k,C,c,s)$ (Definition 1.1); thus H is C^∞ and vanishes outside $\mathbb{R}\times K_H$ for a compact set $K_H \subset \mathbb{R}^{2n}$, its flow ϕ_H^t is global, and $\phi_H^1(B(s_i,c_i)) \subset B^\circ(C)$ for every i. Write $\zeta_1,\dots,\zeta_{2n}$ for the real coordinates of $z \in \mathbb{R}^{2n}$. A *compilation datum* for H consists of the following numbers.

(C1) A capacity $C_0 \in (0,C)$ with $\phi_H^1(B(s_i,c_i)) \subset B(C_0)$ for all i. Put $R_* = \sqrt{C_0/\pi} < R_C = \sqrt{C/\pi}$.

(C2) A rational $a_{\mathrm{box}} > 0$ with $H(t,z) = 0$ whenever $\max_l |\zeta_l| \geq a_{\mathrm{box}}$, and an integer $B > \max(2n, 2a_{\mathrm{box}})$. The box cutoff is

$$\beta_B(z) = \prod_{l=1}^{2n} \beta_{B,1}(\zeta_l), \qquad \beta_{B,1}(x) = 1 - S\Big(\frac{x^2 - (B/2)^2}{(3B/4)^2 - (B/2)^2}\Big),$$

where S is the flat step, so that $0 \leq \beta_B \leq 1$, $\beta_B = 1$ on $[-B/2,B/2]^{2n}$, $\beta_B = 0$ outside $(-3B/4,3B/4)^{2n}$, and $\beta_B H = H$.

(C3) Reals $L_X \geq 1 + \sup_{[0,1]\times\mathbb{R}^{2n}} \|D_z^2 H\|$ and $C_\beta \geq 1 + \sup_{\mathbb{R}^{2n}} |\nabla\beta_B|$.

(C4) A rational b_R with $0 < b_R < (R_C - R_*)/4$; put $\eta = b_R e^{-L_X}/2$.

(C5) A real $M_0 \geq 1$ bounding all first and second partial derivatives on $[0,1]^{2n+1}$ of $\tilde H(\xi) = H(\xi_0, 2B\xi_1 - B, \dots, 2B\xi_{2n} - B)$.

(C6) A rational δ with $0 < \delta < \min\big(\frac12, \eta/(16(1+C_\beta))\big)$ and an integer $N_{\mathrm{B}} > \max\big(1, ((2n+1)M_0/\delta)^2, 4M_0/\delta\big)$.

(C7) For every $m \in \{0,\dots,N_{\mathrm{B}}\}^{2n+1}$ a rational $\tilde H_m$ with $|\tilde H_m - \tilde H(m/N_{\mathrm{B}})| < \delta/(2N_{\mathrm{B}}+1)$.

With $b_{N,j}(x) = \binom{N}{j} x^j (1-x)^{N-j}$ put

$$P(t,z) = \sum_{m\in\{0,\dots,N_{\mathrm{B}}\}^{2n+1}} \tilde H_m \, b_{N_{\mathrm{B}},m_0}(t) \prod_{l=1}^{2n} b_{N_{\mathrm{B}},m_l}\Big(\frac{\zeta_l + B}{2B}\Big), \qquad H_{\mathrm{fin}} = \beta_B \cdot P.$$

Since $B \in \mathbb{N}$ and $\tilde H_m \in \mathbb{Q}$, $P \in \mathbb{Q}[t,z]$, of degree at most N_{B} in each variable. Compilation data exist: C_0 because $\bigcup_i \phi_H^1(B(s_i,c_i))$ is a compact subset of $B^\circ(C)$, a_{box} because H vanishes outside $\mathbb{R}\times K_H$, the bounds of (C3) and (C5) because they are maxima of continuous functions on compact sets, and the rest because each is a strict lower or upper bound on a single number.

Theorem 7.2 (Finite output)**.** *Let H satisfy* $\mathrm{Pack}(n,k,C,c,s)$. *For every compilation datum (C1)–(C7) for H, the function $H_{\mathrm{fin}} = \beta_B \cdot P$ satisfies* $\mathrm{Pack}(n,k,C,c,s)$. *More precisely: H_{fin} is C^∞ and $H_{\mathrm{fin}}(t,z) = 0$ for all t whenever $z \notin (-3B/4,3B/4)^{2n}$; $|X_{H_{\mathrm{fin}}} - X_H| < \eta$ on $[0,1]\times\mathbb{R}^{2n}$; and for every $z \in \bigcup_i B(s_i,c_i)$,*

$$|\phi^1_{H_{\mathrm{fin}}}(z) - \phi^1_H(z)| < b_R, \qquad |\phi^1_{H_{\mathrm{fin}}}(z)| < R_* + b_R < R_C,$$

so that $\phi^1_{H_{\mathrm{fin}}}$ maps every $B(s_i,c_i)$ into $B^\circ(C)$. If the source balls are pairwise disjoint, their images are pairwise disjoint.

Proof (complete version: Theorem S24.15). Write $\mathcal{B}_{N_{\mathrm{B}}}$ for the tensor Bernstein operator on $[0,1]^{2n+1}$. It satisfies $\|\mathcal{B}_{N_{\mathrm{B}}}\tilde H - \tilde H\|_\infty \leq (2n+1)M_0/(2\sqrt{N_{\mathrm{B}}}) < \delta/2$ and $\|\partial_j\mathcal{B}_{N_{\mathrm{B}}}\tilde H - \partial_j\tilde H\|_\infty \leq M_0\big((2n+1)/(2\sqrt{N_{\mathrm{B}}}) + 2/N_{\mathrm{B}}\big) < \delta$, by the binomial variance identity and summation by parts (Proposition S24.7). Because the Bernstein weights are nonnegative with sum 1, rounding changes values by less than $\delta/(2N_{\mathrm{B}}+1)$ and first derivatives by less than $2N_{\mathrm{B}}\delta/(2N_{\mathrm{B}}+1) < \delta$.

Hence on $[0,1]\times[-B,B]^{2n}$ we have $|P-H|<\delta$ and, since $\partial_{\zeta_l}=(2B)^{-1}\partial_{\xi_l}$ and $B>2n$, $|\nabla_z P-\nabla_z H|\le\sqrt{2n}\,\delta/B<\delta$ (Lemma S24.9). There, differentiating $\beta_B H=H$ gives

$$\nabla_z H_{\mathrm{fin}}-\nabla_z H=\beta_B\nabla_z(P-H)+(P-H)\nabla\beta_B,$$

whose norm is less than $\delta(1+C_\beta)<\eta/16$ by (C6); outside $(-B,B)^{2n}$ both H_{fin} and H vanish on a neighbourhood. As $X_G=J_0\nabla_z G$ with J_0 orthogonal, $|X_{H_{\mathrm{fin}}}-X_H|<\eta$ everywhere on $[0,1]\times\mathbb{R}^{2n}$, and $X_H(t,\cdot)$ is L_X-Lipschitz by (C3). The field $X_{H_{\mathrm{fin}}}$ is smooth and vanishes outside $\mathbb{R}\times[-3B/4,3B/4]^{2n}$, so its integral curves exist on $\mathbb{R}$ and are unique; the same holds for X_H, which vanishes outside $\mathbb{R}\times K_H$. Lemma 7.1 gives $|\phi^1_{H_{\mathrm{fin}}}(z)-\phi^1_H(z)|<b_R$ for every z. For $z\in B(s_i,c_i)$ we have $|\phi^1_H(z)|\le R_*$ by (C1), hence $|\phi^1_{H_{\mathrm{fin}}}(z)|<R_*+b_R<R_C$, that is $\pi|\phi^1_{H_{\mathrm{fin}}}(z)|^2<C$. Hence H_{fin} satisfies both conditions of Definition 1.1: it is C^∞ and vanishes for $z\notin(-3B/4,3B/4)^{2n}$ and all t, because β_B does not depend on t; and every integral curve of $X_{H_{\mathrm{fin}}}$ on $[0,1]$ starting at $z\in B(s_i,c_i)$ is the flow line, which ends in $B^\circ(C)$. The images are disjoint because $\phi^1_{H_{\mathrm{fin}}}$ is injective. □

The theorem applies to every compilation datum; the existence of one is used in the proof that the search below terminates.

7.2. **The algorithm.**

Remark 7.3 (Accepted outputs). The polynomial returned by the algorithm below need not be the Bernstein approximation of $H[n,k,C,c,s]$ for any compilation datum. Bernstein compilation shows that some candidate is accepted; every accepted candidate is correct by the soundness theorem.

Put $d=2n$. A candidate consists of integers $B,Q,N\ge1$, precisions $p,m\ge0$, and a finite rational arithmetic expression e for a polynomial $P_e(t,z)$. Let $\Theta=h_B(e)$ be the integer of Definition S25.1. It bounds the speed and the temporal and spatial Lipschitz constants, in maximum norm, of the field with clamped arguments

$$F_e(t,z)=J_0\nabla_z P_e(\mathrm{clip}_{[0,1]}\,t,\mathrm{clip}_{[-B/2,B/2]^d}\,z),$$

where clip_X is the nearest-point projection to the box X. Read the input names to error 2^{-p}. From them form rational outer balls $B(s_i^p,r_{i,p}^+)$ containing the original closed sources, and a rational $R_p^-\le\sqrt{C/\pi}$, as in Equation S97. First test $|s_{i,l}^p|+r_{i,p}^+<Q$ for every i,l.

Use the grid of mesh $\rho=2Q2^{-m}$ in $[-Q,Q]^d$. For each node a and source i, discard the node if $|a-s_i^p|^2>(r_{i,p}^++d\rho)^2$. Otherwise compute the exact rational Euler table

$$x_0=a,\quad x_{j+1}=x_j+N^{-1}F_e(j/N,x_j),\quad E=\left(\rho+\frac{\Theta+\Theta^2}{N}\right)4^\Theta+\frac{\Theta}{N}.$$

Require $|x_{j,l}|+E<B/2$ for every $j\le N,l$, and

$$R_p^--dE>0,\qquad |x_N|^2<(R_p^--dE)^2.$$

These are finite strict rational tests. Enumerate all candidate tuples and return the first one passing every test; the output is B and the program e, which determines $P_e\in\mathbb{Q}[t,z]$.

The real inputs are supplied as oracles and n,k as ordinary input; §S26 gives the encoding.

Theorem 7.4 (Termination and correctness)**.** *There is one oracle machine with the following property. Let $n\ge2$, $k\ge1$, C, c and s satisfy the hypotheses of Theorem C, and supply as oracles names of $C,c_1,\dots,c_k$ and of the $2nk$ coordinates of $s_1,\dots,s_k$. On input (n,k) the machine halts and outputs an integer B and a polynomial $P\in\mathbb{Q}[t,z_1,\dots,z_{2n}]$ such that $\beta_B P$ satisfies* $\mathrm{Pack}(n,k,C,c,s)$. *The output may depend on the names, not only on the numbers they name.*

If the names are computable, for instance the constant names of rational inputs, replacing the oracles by programs for them gives an ordinary algorithm.

There is also one oracle machine which, given a name of $\varepsilon \in (0,1)$ *as oracle and an integer* $k \geq 9$ *as input, computes names of the source radius* 1 *and of the target radius* $R = (k/(1-\varepsilon/2))^{1/4}$ *for the data*

$$n = 2, \quad C = \pi R^2, \quad c_i = \pi, \quad s_i = (3i-3, 0, 0, 0),$$

runs the same certificate search on them, and halts with (B, P) *such that* $\beta_B P$ *satisfies* $\mathrm{Pack}(2, k, C, c, s)$*; here* $k/R^4 = 1 - \varepsilon/2 > 1 - \varepsilon$.
[Lean: full statement verified; `original_name_algorithm, epoch_name_algorithm, epoch_name_algorithm_packing`]

Proof. For an accepted candidate, each source point lies within ρ in maximum norm of a grid node that cannot be discarded. The Euler error is at most E throughout $[0,1]$. The plateau tests therefore keep the trajectory of F_e from that point in the open box where $F_e = X_{\beta_B P_e}$, and the terminal test places its endpoint in the open target. Smoothness, spatial compact support and uniqueness of integral curves give the packing specification (Theorem S25.5).

For termination, $H = H[n, k, C, c, s]$ maps the compact source balls into $B^\circ(C)$, so their images lie in $B^\circ(C_0)$ for some $C_0 < C$; put $C_1 = (C_0 + C)/2$. Then H satisfies $\mathrm{Pack}(n, k, C_1, c, s)$, and Theorem 7.2 with target C_1 and B large gives a rational polynomial whose trajectories from the sources stay a positive distance inside the region where $\beta_B = 1$ and end in $B^\circ(C_1)$, at a positive distance from the boundary of $B^\circ(C)$. These margins persist for starting points in a neighbourhood of the sources. A node not discarded lies at maximum-norm distance at most $(2d+3)2^{-p} + d\rho$ from an original source ball. Taking p, m, N sufficiently large therefore makes all tests pass (Theorem S25.7). Each candidate needs finitely many oracle queries and rational operations, so the enumeration reaches an accepted candidate in finite time. The encodings of §S26 treat n and k as inputs of the same machine. The Epoch input is admissible with disjoint source balls by Corollary F; the Epoch machine forms the radius names from the name of ε by total programs (the reciprocal of $1 - \varepsilon/2 > \frac{1}{2}$ and two square roots, Lemma S23.6) and runs the same search. □

Remark 7.5 (The four cases). The exact construction distinguishes the four cases of Construction S20.1. The finite verifier has the same form in every case: it tests rational trajectories of the proposed output. No equality test between named capacities or decision at the threshold $c_i = C/2$ is needed.

Remark 7.6 (Size). A certificate with parameters (B, e, Q, p, m, N) inspects $k(2^m+1)^{2n}$ node-source pairs and up to $N+1$ Euler rows for each retained pair. The factor 4^Θ in the Euler bound can require very large spatial grids and time tables; the theorem concerns computability, not running time.

Remark 7.7 (The Epoch submission format). The Epoch problem [Epo26] asks for submissions as a Python function returning a PyTorch implementation of the Hamiltonian, which a numerical verifier integrates from sampled boundary points of the source balls within fixed time limits. We read its variables as $(Q, P) = (x, y)$, with $X_H = (\partial_P H, -\partial_Q H)$, and its target as the time-one map. The results here answer the mathematical part of the problem: Theorem A gives the Hamiltonian, Proposition B the vector-field tolerance that a numerical implementation must meet, and Theorem 7.4 a uniform algorithm with finite output. A program within the verifier's limits of ten minutes per solution and ten seconds per evaluation is not part of this paper.

8. Formal verification

The Lean 4 development [Son26, dMU21] uses the toolchain `v4.34.1`, Mathlib [The20] at the tag of the same name, and the unmodified Lean library accompanying [Ope26]. That library contributes lemmas and a few generic differential-geometric operators (exterior derivative, chart differential); the definitions of the Hamiltonian and of the algorithms belong to the development, and none of the library's packing constructions or existence results is used. The entry point is five short files in `verification/`: `Problem.lean` defines the problem using Mathlib alone; `Algorithm.lean` exposes the top level of the construction; `Statement.lean` states correctness of the construction; `Computability.lean` and `EpochComputability.lean` state correctness and termination of the finite-output algorithms. Theorems in `verification/` have no namespace; other names are relative to `SymplecticPacking`.

Statement	Lean
Theorem A	`four_dim_theorem`, with `General.epochRadiusQ`, `General.epochHamiltonianQ`
Proposition B	`four_dim_stability`
Theorem C(i)	`main_theorem`, `main_time_support`, `main_support_radius`; for the flow, `flow_is_integral_curve`, `flow_symplectic_diffeomorphism`, `flow_images_disjoint`
Theorem C(ii)	`#check_explicit General.packingHamiltonian` (a metaprogram, see below)
Theorem C(iii)	`original_name_algorithm`, with `#check_constructive` on the machine; finite form: `General.Compile.main_finite`
Corollaries D–G	`neighbourhood_criterion`, `flow_neighbourhood_embedding`; `equal_balls`; `epoch_packing`, `epoch_theorem_ten`, `epoch_name_algorithm`, `epoch_name_algorithm_packing`; `unequal_cone`
embeddings, by case	$n = 2$: `Branch2.embeddingsEq_spec` (linear datum), `Branch2.embeddingsGen_spec` (scalar comparator); $n \geq 3$: `BranchSmall.embeddings_spec` (all $c_i < \frac{C}{2}$), `BranchLarge3.embeddings_spec` ($n = 3$, one $c_j \geq \frac{C}{2}$), `BranchLargeHigh.embeddings_spec` ($n \geq 4$, one $c_j \geq \frac{C}{2}$)

Table 8. Formal counterparts of the main statements. Every case of Construction S20.1 is covered by `main_theorem` and by the same algorithm theorems.

The statement. Phase space is `Problem.E` n = `EuclideanSpace` $\mathbb{R}$ (`Fin` $(2n)$), coordinates $j < n$ and $n + j$ being x_{j+1} and y_{j+1} (named q and p in the file); `omega0` and `hamVF` give $\omega_0 = \sum_j dx_j \wedge dy_j$ and $X_H = (\partial_y H, -\partial_x H)$, so $\iota_{X_H}\omega_0 = dH$. `capBall` $s\ c = \{z : \pi\|z - s\|^2 \leq c\}$ and `capBallOpen` is its strict version. `Admissible`, `BallsDisjoint` and `PackingSpec` are Definitions 1.2 and 1.1 and the disjointness hypothesis, field by field; `PackingSpec` quantifies over every integral curve on $[0, 1]$, so it does not depend on a chosen solution. `EpochSpecR` is the problem as posed, with the closed Euclidean unit balls about `epochCenter` $i = (3i, 0, 0, 0)$, $0 \leq i < k$ (the centre s_{i+1}), and the closed target of radius R.

Each theorem is a statement about the defined term. `Algorithm.lean` shows by `rfl` that `General.packingHamiltonian` is `HamEnd.hamiltonianOfEmbeddings` applied to `General.selectEmbeddings`, the case distinction of Construction S20.1, and that the Epoch Hamiltonian is its instance with $C = \pi R^2$. The Lean statements agree with the paper, with these differences of form: `equal_balls` does not assume $\varepsilon > 0$; `four_dim_stability` also records $R < R'$; `four_dim_theorem` contains both `EpochSpecR` and the packing specification; the algorithm theorems assert `Nat.RecursiveIn` relative to the oracles given by the input names, and that one program serves every input is certified by `#check_constructive`;

`flow_neighbourhood_embedding` uses the neighbourhood $(\phi_H^1)^{-1}(B^\circ(C))$ and holds for every $n \geq 2$.

What is verified. The kernel checks the theorems of `Statement.lean`, `Computability.lean` and `EpochComputability.lean` and all their dependencies in the development, the vendored library and Mathlib. The trusted base is the kernel, the definitions in `Problem.lean` and the axioms `propext`, `Classical.choice` and `Quot.sound`; a command fails the build if the transitive axiom set of any of these theorems contains anything else, in particular `sorryAx`.

Since Mathlib's real numbers use classical choice, axioms cannot distinguish a formula from an arbitrary existential witness. Explicitness (level L1) and uniformity (level L3) are therefore checked by two metaprograms, which are not themselves verified. `#check_explicit` traverses the non-proof dependency closure of a definition, through project, vendored and Mathlib definitions, and accepts case distinctions on propositions, least-element searches over integers, rationals and finite data with a non-classical decision procedure, uses of choice inside an explicit allow-list of Mathlib definitions (`SymplecticPacking/Explicit/Whitelist.lean`), and definite descriptions after unique existence over types without a discrete factor. The 26 definite descriptions reached from `General.packingHamiltonian` (inverses, roots and critical points, linear solves, matrix square roots, solutions of ordinary differential equations, extrema and measures) are listed in Appendix SA. The command is applied to `General.packingHamiltonian`, the Epoch Hamiltonians, the support radius, the case embeddings and the algorithm terms. `#check_constructive` is stricter: the computational closure may contain no classical constant at all, including classical decidability. It is applied to the two oracle machines, so each is one fixed program in its oracle streams and cannot branch on an undecidable property of the reals they name. `test/ExplicitTest.lean` contains terms both commands must reject, among them a search for the least index whose enumerated Hamiltonian satisfies the specification, the same search wrapped as a unique pair in $\mathbb{N} \times \mathbb{R}$, and a machine branching on $x < \frac{1}{3}$ for a real x.

Independence. The file `verification/Audit/Independence.lean` runs the command `#assert_construction_independence`, which traverses types, definitions and proof bodies of the packing, stability, flow, corollary and algorithm theorems and fails if they reach the vendored packing predicate or existence theorems, the holomorphic-line and two-ball obstruction proofs, or the curve-analysis modules of the vendored library. The converse in Corollary D is checked separately: it may use the classical obstructions, but no sufficiency theorem. A positive control detects its dependence on `adapted_affine_line`, and a regression test detects a two-ball obstruction hidden inside a wrapper. The vendored declarations used are listed by `verification/Audit/VendoredUsage.lean` (Appendix SA.4): explicit charts, forms, primitives, Moser fields and model coefficients.

Further statements carry source tags naming related Lean declarations; `verification/PAPER-INDEX.md` indexes them, and the build checks that every tagged name exists and depends only on the three axioms. The visible label *full statement verified* denotes the full mathematical statement, up to the presentation differences described above. The label *needed consequence verified* denotes only the version used by the construction, not every assertion or the entire domain of the printed statement. Appendix SA.5 records these restrictions, including return identities along smooth unit-vector families, transfer properties along the constructed embeddings, and the coarser Hamiltonian support bound. An unqualified source tag is a reference, without a full-coverage claim. The scope labels are editorial correspondences for readers to review; the kernel checks the Lean statements. The main correctness theorems concern the defined Hamiltonian and depend on every step of the construction.

Reproduction. With Elan, Node.js and Tectonic installed, `lake exe cache get` followed by `bash scripts/verify.sh` builds or incrementally checks the development with all axiom, explicitness, uniformity and independence gates, runs the regression tests and the paper-tag check, and compiles both papers; every stage is fatal. The development is released under Apache-2.0; the vendored library keeps its licence and notices. The repository includes a short reviewer guide, `REVIEW.md`, and an optional project cache. After building, `bash scripts/check_kernel.sh` replays the complete proof closure of `verification/Kernel.lean` into a fresh environment using the pinned toolchain's `leanchecker`; this checks stored proof terms with the same Lean kernel. A source rebuild from an empty project build directory checks elaboration without the project cache.

Authorship and the role of AI

Liang Song is the sole author and is responsible for the manuscript and its release. The mathematics, the Lean formalisation and the text were produced by AI systems, including GPT-6.1 Sol (OpenAI), Claude Opus 5.5 (Anthropic) and other models, working in a research environment designed and directed by Liang Song, who chose the problem, assembled the literature and context given to the systems, designed the verification pipeline and made the editorial decisions. Every statement marked [Lean: …] is checked by the Lean kernel, for the problem as stated in `verification/Problem.lean` from Mathlib alone, and the build fails if any axiom beyond `propext`, `Classical.choice` and `Quot.sound` is used. Explicitness and the uniformity of the algorithm are checked by two metaprograms, which are not themselves verified (Section 8).

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